



SIL Verification Report

Hybrit Pilot Plant

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Table of Contents

1	Purpose and Scope	11
1.1	General Project Information	11
2	SIF-011 SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322 H2)	12
2.1	General SIF Information	12
2.2	Safety Integrity Levels	12
2.3	SIL Verification	13
2.3.1	Sensor Part Configuration	14
2.3.2	Logic Solver Part Configuration	15
2.3.3	Final Element Part Configuration	16
3	SIF-011 Beräkning av grupp 1, 3 & 4	19
3.1	General SIF Information	19
3.2	Safety Integrity Levels	19
3.3	SIL Verification	20
3.3.1	Sensor Part Configuration	21
3.3.2	Logic Solver Part Configuration	21
3.3.3	Final Element Part Configuration	22
4	SIF-011 Beräkning av grupp 2	27
4.1	General SIF Information	27
4.2	Safety Integrity Levels	27
4.3	SIL Verification	28
4.3.1	Sensor Part Configuration	29
4.3.2	Logic Solver Part Configuration	29
4.3.3	Final Element Part Configuration	30
5	SIF-012 SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322 L2)	33
5.1	General SIF Information	33
5.2	Safety Integrity Levels	33
5.3	SIL Verification	33
5.3.1	Sensor Part Configuration	35
5.3.2	Logic Solver Part Configuration	36
5.3.3	Final Element Part Configuration	37
6	SIF-013 SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322 H3)	41
6.1	General SIF Information	41
6.2	Safety Integrity Levels	41
6.3	SIL Verification	41
6.3.1	Sensor Part Configuration	43
6.3.2	Logic Solver Part Configuration	44
6.3.3	Final Element Part Configuration	45
7	SIF-014 SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322 H4) (ej klar)	47
7.1	General SIF Information	47

7.2	Safety Integrity Levels	47
7.3	SIL Verification	47
7.3.1	Sensor Part Configuration.....	49
7.3.2	Logic Solver Part Configuration.....	50
7.3.3	Final Element Part Configuration.....	51
8	SIF-015 SIL1 Temp toppgas (320TZ323, 320TZ324 & 320TZ325 H2)	53
8.1	General SIF Information	53
8.2	Safety Integrity Levels	53
8.3	SIL Verification	54
8.3.1	Sensor Part Configuration.....	55
8.3.2	Logic Solver Part Configuration.....	56
8.3.3	Final Element Part Configuration.....	57
9	SIF-016 SIL1 Temp överhettad ånga (271TI552 L2)	62
9.1	General SIF Information	62
9.2	Safety Integrity Levels	62
9.3	SIL Verification	62
9.3.1	Sensor Part Configuration.....	64
9.3.2	Logic Solver Part Configuration.....	65
9.3.3	Final Element Part Configuration.....	66
10	SIF-017 SIL1 Tryck överhettad ånga (271PI552 L2) Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)	68
10.1	General SIF Information	68
10.2	Safety Integrity Levels	68
10.3	SIL Verification	68
10.3.1	Sensor Part Configuration	70
10.3.2	Logic Solver Part Configuration	71
10.3.3	Final Element Part Configuration	72
11	SIF-018 SIL1 Tryck överhettad ånga friblåsning (271PI320 < (343PZ305/7/8 + 0,1 barg)) Permissive Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)	74
11.1	General SIF Information	74
11.2	Safety Integrity Levels	74
11.3	SIL Verification	74
11.3.1	Sensor Part Configuration	76
11.3.2	Logic Solver Part Configuration	78
11.3.3	Final Element Part Configuration	79
12	SIF-019 SIL2 Flöde reduktionsgas till PGH (343XI302 L2)	81
12.1	General SIF Information	81
12.2	Safety Integrity Levels	81
12.3	SIL Verification	82
12.3.1	Sensor Part Configuration	83
12.3.2	Logic Solver Part Configuration	87

12.3.3	Final Element Part Configuration	87
13	SIF-020 SIL2 Difftryck O2-transferline (251XI313)	91
13.1	General SIF Information	91
13.2	Safety Integrity Levels	91
13.3	SIL Verification	91
13.3.1	Sensor Part Configuration	93
13.3.2	Logic Solver Part Configuration	95
13.3.3	Final Element Part Configuration	96
14	SIF-021 SIL1 Gastryck transferline (344PZ321 L2)	98
14.1	General SIF Information	98
14.2	Safety Integrity Levels	98
14.3	SIL Verification	98
14.3.1	Sensor Part Configuration	100
14.3.2	Logic Solver Part Configuration	101
14.3.3	Final Element Part Configuration	102
15	SIF-022 SIL2 Tryckvakt O2-inj (251XI312 L2)	104
15.1	General SIF Information	104
15.2	Safety Integrity Levels	104
15.3	SIL Verification	104
15.3.1	Sensor Part Configuration	106
15.3.2	Logic Solver Part Configuration	107
15.3.3	Final Element Part Configuration	108
16	SIF-023 SIL2 Flödesmätare O2 inj (251XI554 L2)	110
16.1	General SIF Information	110
16.2	Safety Integrity Levels	110
16.3	SIL Verification	110
16.3.1	Sensor Part Configuration	112
16.3.2	Logic Solver Part Configuration	115
16.3.3	Final Element Part Configuration	116
17	SIF-024 SIL2 Tryckvakt N2 vid O2 inj (253PZ362, 253PZ363, 253PZ364 L2)	118
17.1	General SIF Information	118
17.2	Safety Integrity Levels	118
17.3	SIL Verification	118
17.3.1	Sensor Part Configuration	120
17.3.2	Logic Solver Part Configuration	122
17.3.3	Final Element Part Configuration	123
18	SIF-025 SIL2 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318 H2)	125
18.1	General SIF Information	125
18.2	Safety Integrity Levels	125
18.3	SIL Verification	126
18.3.1	Sensor Part Configuration	127

18.3.2	Logic Solver Part Configuration	129
18.3.3	Final Element Part Configuration	129
19	SIF-026 SIL3 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318 L2)	133
19.1	General SIF Information	133
19.2	Safety Integrity Levels	133
19.3	SIL Verification	133
19.3.1	Sensor Part Configuration	135
19.3.2	Logic Solver Part Configuration	136
19.3.3	Final Element Part Configuration	137
20	SIF-027 SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305 H2)	139
20.1	General SIF Information	139
20.2	Safety Integrity Levels	139
20.3	SIL Verification	140
20.3.1	Sensor Part Configuration	141
20.3.2	Logic Solver Part Configuration	142
20.3.3	Final Element Part Configuration	143
21	SIF-028 SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305 L2)	146
21.1	General SIF Information	146
21.2	Safety Integrity Levels	146
21.3	SIL Verification	146
21.3.1	Sensor Part Configuration	148
21.3.2	Logic Solver Part Configuration	149
21.3.3	Final Element Part Configuration	150
22	SIF-029 SIL1 Temperatur strålningsdel (344TZ310, 344TZ311 & 344TZ312 H2)	152
22.1	General SIF Information	152
22.2	Safety Integrity Levels	152
22.3	SIL Verification	153
22.3.1	Sensor Part Configuration	154
22.3.2	Logic Solver Part Configuration	155
22.3.3	Final Element Part Configuration	156
23	SIF-031 SIL1 Tryck förbränningskammare (344PZ306, 344PZ307 & 344PZ308 H2)	159
23.1	General SIF Information	159
23.2	Safety Integrity Levels	159
23.3	SIL Verification	160
23.3.1	Sensor Part Configuration	161
23.3.2	Logic Solver Part Configuration	163
23.3.3	Final Element Part Configuration	163
24	SIF-032 SIL2 Flamvakt (344YS314 SG1)	167
24.1	General SIF Information	167
24.2	Safety Integrity Levels	167
24.3	SIL Verification	168

24.3.1	Sensor Part Configuration	169
24.3.2	Logic Solver Part Configuration	170
24.3.3	Final Element Part Configuration	171
25	SIF-033 SIL2 Gastemp transferline efter O2-inj (344TZ322, 344TZ323 & 344TZ324 H2).....	174
25.1	General SIF Information	174
25.2	Safety Integrity Levels	174
25.3	SIL Verification	174
25.3.1	Sensor Part Configuration	176
25.3.2	Logic Solver Part Configuration	177
25.3.3	Final Element Part Configuration	178
26	SIF-034 SIL3 Gastemp transferline efter O2-inj (344TZ322, 344TZ323 & 344TZ324 L2)	180
26.1	General SIF Information	180
26.2	Safety Integrity Levels	180
26.3	SIL Verification	180
26.3.1	Sensor Part Configuration	182
26.3.2	Logic Solver Part Configuration	183
26.3.3	Final Element Part Configuration	184
27	SIF-035 SIL2 Difftrycksmätning över syreinjektionslans (344XI340_DEV H2/L2).....	186
27.1	General SIF Information	186
27.2	Safety Integrity Levels	186
27.3	SIL Verification	187
27.3.1	Sensor Part Configuration	188
27.3.2	Logic Solver Part Configuration	192
27.3.3	Final Element Part Configuration	193
28	SIF-036 SIL1 Gastryck N2 (253PZ362, 253PZ363 & 253PZ364 L2)	195
28.1	General SIF Information	195
28.2	Safety Integrity Levels	195
28.3	SIL Verification	196
28.3.1	Sensor Part Configuration	197
28.3.2	Logic Solver Part Configuration	198
28.3.3	Final Element Part Configuration	199
29	SIF-037 SIL2 Tryckvakt efter N2 bufferttank 253TA005 (253PZ323, 253PZ324, 253PZ325 L2)	202
29.1	General SIF Information	202
29.2	Safety Integrity Levels	202
29.3	SIL Verification	203
29.3.1	Sensor Part Configuration	204
29.3.2	Logic Solver Part Configuration	205
29.3.3	Final Element Part Configuration	206
30	SIF-038 SIL2 Hög vätskenivå WESP (345LI354/5/6 H2).....	210
30.1	General SIF Information	210

30.2	Safety Integrity Levels	210
30.3	SIL Verification	210
30.3.1	Sensor Part Configuration	212
30.3.2	Logic Solver Part Configuration	213
30.3.3	Final Element Part Configuration	214
31	SIF-039 SIL2 Tryck efter WESP red. gaskrets (345PI361 H2)	217
31.1	General SIF Information	217
31.2	Safety Integrity Levels	217
31.3	SIL Verification	217
31.3.1	Sensor Part Configuration	219
31.3.2	Logic Solver Part Configuration	220
31.3.3	Final Element Part Configuration	221
32	SIF-040 SIL2 Tryck efter WESP processgaskrets (353PI361 H2)	225
32.1	General SIF Information	225
32.2	Safety Integrity Levels	225
32.3	SIL Verification	225
32.3.1	Sensor Part Configuration	227
32.3.2	Logic Solver Part Configuration	228
32.3.3	Final Element Part Configuration	229
33	SIF-041 SIL1 Temp kondensor reduktionsgas (345TI301 H2)	233
33.1	General SIF Information	233
33.2	Safety Integrity Levels	233
33.3	SIL Verification	234
33.3.1	Sensor Part Configuration	235
33.3.2	Logic Solver Part Configuration	236
33.3.3	Final Element Part Configuration	237
34	SIF-042 SIL1 Temp kondensor processgas (353TI301 H2)	241
34.1	General SIF Information	241
34.2	Safety Integrity Levels	241
34.3	SIL Verification	241
34.3.1	Sensor Part Configuration	243
34.3.2	Logic Solver Part Configuration	244
34.3.3	Final Element Part Configuration	245
35	SIF-043 SIL2 Skintemp förquench reduktionsgas (345TI307 H2)	248
35.1	General SIF Information	248
35.2	Safety Integrity Levels	248
35.3	SIL Verification	248
35.3.1	Sensor Part Configuration	250
35.3.2	Logic Solver Part Configuration	251
35.3.3	Final Element Part Configuration	252

36	SIF-044 SIL2 Skintemp quench reduktionsgas (345TI308 H2)	256
36.1	General SIF Information	256
36.2	Safety Integrity Levels	256
36.3	SIL Verification	256
36.3.1	Sensor Part Configuration	258
36.3.2	Logic Solver Part Configuration	259
36.3.3	Final Element Part Configuration	260
37	SIF-045 SIL2 Skintemp förquench processgas (353TI307 H2)	264
37.1	General SIF Information	264
37.2	Safety Integrity Levels	264
37.3	SIL Verification	264
37.3.1	Sensor Part Configuration	266
37.3.2	Logic Solver Part Configuration	267
37.3.3	Final Element Part Configuration	268
38	SIF-046 SIL2 Skintemp quench processgas (353TI308 H2)	271
38.1	General SIF Information	271
38.2	Safety Integrity Levels	271
38.3	SIL Verification	271
38.3.1	Sensor Part Configuration	273
38.3.2	Logic Solver Part Configuration	274
38.3.3	Final Element Part Configuration	275
39	SIF-047 SIL1 Flödesmätare till förquench reduktionsgaskrets (345FI305 L2)	278
39.1	General SIF Information	278
39.2	Safety Integrity Levels	278
39.3	SIL Verification	278
39.3.1	Sensor Part Configuration	280
39.3.2	Logic Solver Part Configuration	281
39.3.3	Final Element Part Configuration	282
40	SIF-048 SIL1 Flödesmätare till förquench processgaskrets (353FI305 L2)	286
40.1	General SIF Information	286
40.2	Safety Integrity Levels	286
40.3	SIL Verification	286
40.3.1	Sensor Part Configuration	288
40.3.2	Logic Solver Part Configuration	289
40.3.3	Final Element Part Configuration	290
41	SIF-075 SIL2 Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck ((253PZ323, 253PZ324, 253PZ325) - (343PZ305, 343PZ307, 343PZ308) < 0,5) => Close N2 till reduktionskrets (254SV582)	293
41.1	General SIF Information	293
41.2	Safety Integrity Levels	293
41.3	SIL Verification	294
41.3.1	Sensor Part Configuration	295

41.3.2	Logic Solver Part Configuration	297
41.3.3	Final Element Part Configuration	298
42	SIF-079 SIL2 252PZ337 (H2/L2) => close ånga, H2, O2	300
42.1	General SIF Information	300
42.2	Safety Integrity Levels	300
42.3	SIL Verification	301
42.3.1	Sensor Part Configuration	302
42.3.2	Logic Solver Part Configuration	303
42.3.3	Final Element Part Configuration	304
43	SIF-080 SIL1 271XI323 Ånga ej tillräckligt överhettad => permissive 271SV556	308
43.1	General SIF Information	308
43.2	Safety Integrity Levels	308
43.3	SIL Verification	308
43.3.1	Sensor Part Configuration	310
43.3.2	Logic Solver Part Configuration	311
43.3.3	Final Element Part Configuration	312
44	SIF-081 SIL2 Mätvärde på tillsatt flöde naturgas 252XI565*K	314
44.1	General SIF Information	314
44.2	Safety Integrity Levels	314
44.3	SIL Verification	314
44.3.1	Sensor Part Configuration	316
44.3.2	Logic Solver Part Configuration	318
44.3.3	Final Element Part Configuration	318
45	SIF-082 SIL2 252PZ337 (H2/L2) => close ånga, NG, O2	321
45.1	General SIF Information	321
45.2	Safety Integrity Levels	321
45.3	SIL Verification	322
45.3.1	Sensor Part Configuration	323
45.3.2	Logic Solver Part Configuration	324
45.3.3	Final Element Part Configuration	325
46	SILver™ Group Reuse Overview	329
46.1	Sensor Groups Reused	329
46.2	Logic Solver Groups Reused	329
46.3	Final Element Groups Reused	329
47	Abbreviations and Definitions	331
47.1	Abbreviations	331
47.2	Definitions	331
48	Disclaimer, Assumptions, Equipment Data	333
48.1	Disclaimer	333
48.2	Assumptions SILver™	333
48.3	Equipment Data	333

1 Purpose and Scope

This document, automatically generated by the exida exSILentia® software, details the results, selections, and assumptions made during the Safety Integrity Level verification for the Safety Instrumented System (SIS) and its associated Safety Instrumented Functions (SIF) for the Hybrit Pilot Plant project.

1.1 General Project Information

Project Identification: 001
Project Name: Hybrit Pilot Plant
Revision: 0
Company:
Project Description:

2 SIF-011 SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-011 SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H2) Safety Instrumented Function.

2.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H2)
 SIF Tag: SIF-011
 SIF Description: SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H2) =>

Open Topptryck nödventil eft. våtelfilter red. gas (345SV599 SIL2)
 Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)
 Close Avstängningsventil H2S till PGH (256SV522 SIL2)
 Trip Reduktionsgaskompressor (346PU101 SIL2)
 Trip Processgaskompressor (346PU101 SIL2)
 Close H2 till reduktionskrets (341SV672/341RV675 SIL2)
 Close NG till reduktionskrets (252SV562/252RV565 SIL2)
 Close NG till kylkrets (252SV614/252RV618 SIL2)
 Close N2 till kylkrets (254SV594/254RV595 SIL2)

Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)
 Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)
 Close avstängningsventil N2 till reaktortopp (253SV607 SIL2)
 Open Avstängningsventiler venturi reduktionsgas (345SV520/21/22 SIL2)
 Open Avstängningsventiler venturi processgas (353SV520/21/22 SIL2)

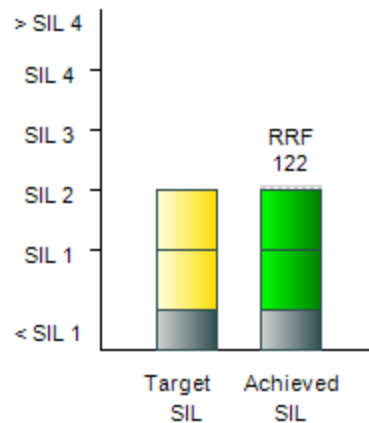
2.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 122.9



2.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-011 SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

The SIF operates in the Low demand mode.

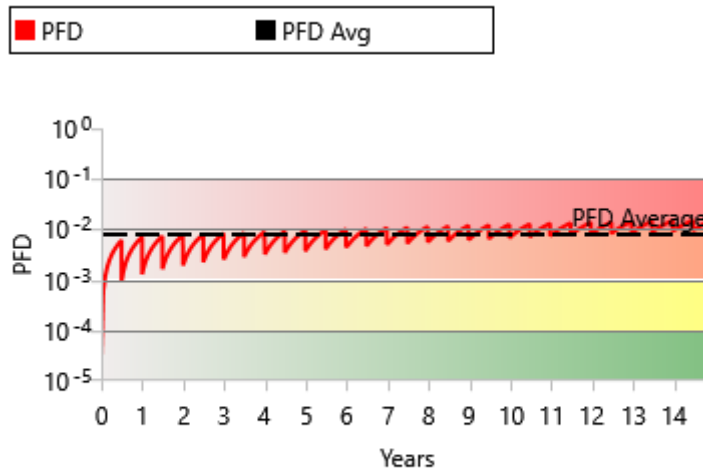
Analyst:

Analysis Date: 10/4/2019

Design References

Comments:

SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H2)

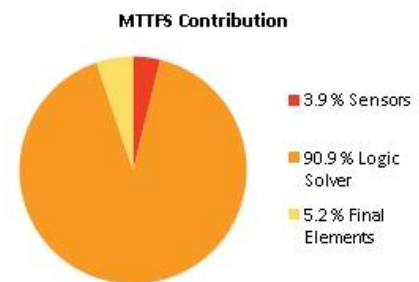
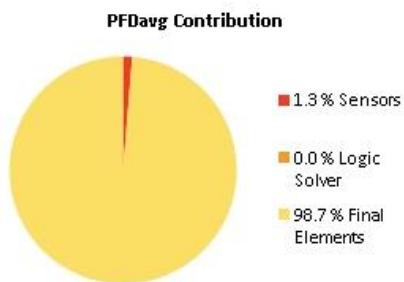


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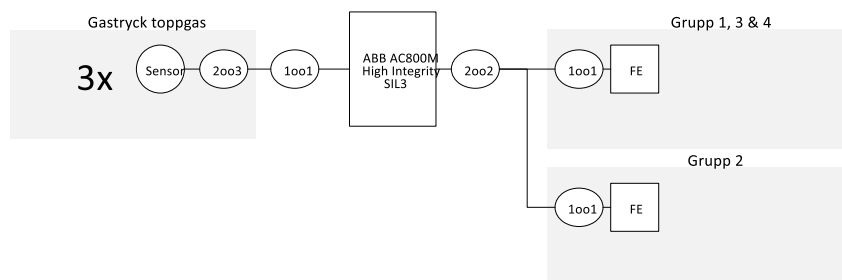
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-011 SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 1 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
8.13E-03	122.9	2	2	N/A	65.86



The SIF Conceptual Design is modeled as shown in the picture below.



2.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-011 SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H2) Safety Instrumented Function is quantified as follows.

Table 2 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.05E-04	3	N/A	1698.47	1	2

Number of Sensor group(s): 1

Voting between groups: 1001
 β -factor [%]: N/A

2.3.1.1 Sensor Group 1: Gastryck toppgas

The information and reliability data underneath describe the Gastryck toppgas sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2003 Voting type: Identical
 HFT: 1 β -factor [%]: 10

Equipment Leg
 (each): „

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
MRT [Hours]:	24				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Gastryck toppgas.

Table 3 Reliability Data Sensor Group Gastryck toppgas

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.96E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.17E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	78.5
									ROUTE 2 _H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

2.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-011 SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322 | H2) Safety Instrumented Function is quantified as follows.

Table 4 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A
Logic Solver Voting:		Architecture Type:	B	
HFT:	1	β-factor [%]:	1	
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 5 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		100.0

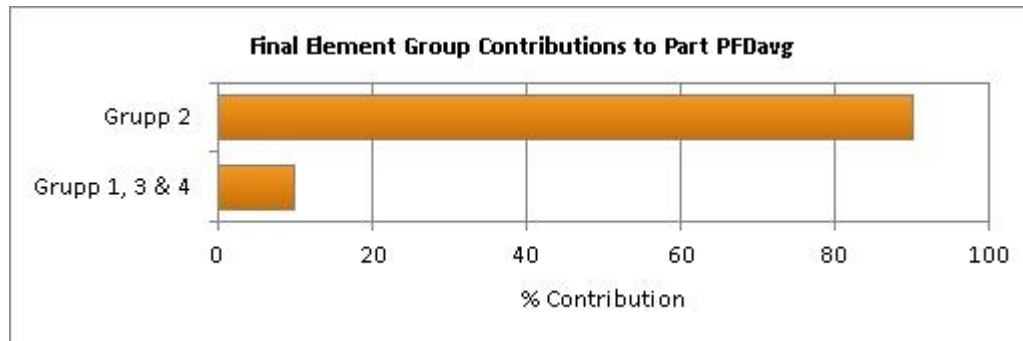
2.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-011 SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322 | H2) Safety Instrumented Function is quantified as follows.

Table 6 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
8.03E-03	2	N/A	1258.18	0	2

Number of Final Element group(s): 2
Voting between groups: 2oo2
 β -factor [%]: 10



2.3.3.1 Final Element Group 1: Grupp 1, 3 & 4

The information and reliability data underneath describe the Grupp 1, 3 & 4 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1
Voting type: Identical
HFT: 0
 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Beräkning av grupp 1, 3 & 4		-	-	A	2
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

MRT [Hours]: 24

Proof Test Interval [Months]: 6
Proof Test Coverage [%]: 95

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Grupp 1, 3 & 4.

Table 7 Reliability Data Final Element Group Grupp 1, 3 & 4

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Beräkning av grupp 1, 3 & 4		9.66E-08		1.38E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
SFF [%]							59.3
ROUTE 2 _H COMPLIANT							-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

2.3.3.2 Final Element Group 2: Grupp 2

The information and reliability data underneath describe the Grupp 2 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Beräkning av grupp 2		-	-	A	2
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

MRT [Hours]: 24

Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 95

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Grupp 2.

Table 8 Reliability Data Final Element Group Grupp 2

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Beräkning av grupp 2		3.98E-11		1.27E-06			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		7.6
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

3 SIF-011 Beräkning av grupp 1, 3 & 4

This chapter details the SIL Verification results, selections, and assumptions for the SIF-011 Beräkning av grupp 1, 3 & 4 Safety Instrumented Function.

3.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: Beräkning av grupp 1, 3 & 4
 SIF Tag: SIF-011
 SIF Description: SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H2) =>

Open Topptryck nödventil eft. våtelfilter red. gas (345SV599 SIL2)
 Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)
 Close Avstängningsventil H2S till PGH (256SV522 SIL2)
 Trip Reduktionsgaskompressor (346PU101 SIL2)
 Trip Processgaskompressor (346PU101 SIL2)
 Close H2 till reduktionskrets (341SV672/341RV675 SIL2)
 Close NG till reduktionskrets (252SV562/252RV565 SIL2)
 Close NG till kylkrets (252SV614/252RV618 SIL2)
 Close N2 till kylkrets (254SV594/254RV595 SIL2)

Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)
 Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)
 Close avstängningsventil N2 till reaktortopp (253SV607 SIL2)
 Open Avstängningsventiler venturi reduktionsgas (345SV520/21/22 SIL2)
 Open Avstängningsventiler venturi processgas (353SV520/21/22 SIL2)

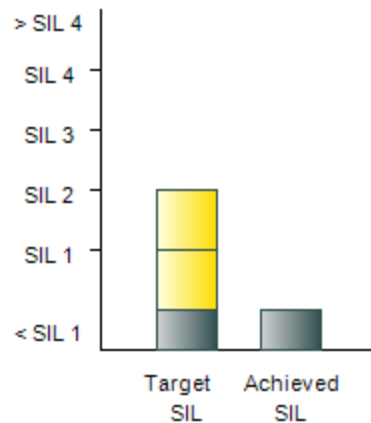
3.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 0

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 0 with RRF = 0



3.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-011 Beräkning av grupp 1, 3 & 4 Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:

Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-011 Beräkning av grupp 1, 3 & 4 Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 9 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
0.00E+00	0	0	0	N/A	∞



The SIF Conceptual Design is modeled as shown in the picture below.

3.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-011 Beräkning av grupp 1, 3 & 4 Safety Instrumented Function is quantified as follows.

Table 10 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
0.00E+00	0	N/A	∞	0	2

Number of Sensor group(s): 0
Voting between groups: 0oo0
β-factor [%]: N/A

3.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-011 Beräkning av grupp 1, 3 & 4 Safety Instrumented Function is quantified as follows.

Table 11 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
0.00E+00	0	N/A	∞	0	2

The information and reliability data underneath describe the logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name:

Equipment:

SERH VERSION PIU SIL
Cap

N/A

Logic Solver Voting:

Architecture Type:

B

HFT: 0

β -factor [%]:

MRT [Hours]:

Proof Test Interval [Months]:

Proof Test Coverage [%]:

Proof Test Execution:

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group .

Table 12 Reliability Data Logic Solver Group

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
						SFF [%]	0.0

3.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-011 Beräkning av grupp 1, 3 & 4 Safety Instrumented Function is quantified as follows.

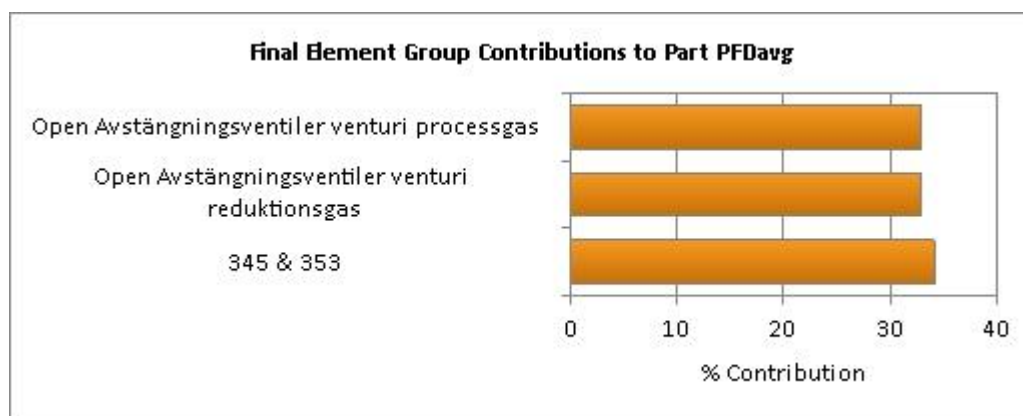
Table 13 Final Element Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.05E-04	3	N/A	1181.76	1	2

Number of Final Element group(s): 3

Voting between groups: 3oo3

β -factor [%]: 10



3.3.3.1 Final Element Group 1: 345 & 353

The information and reliability data underneath describe the 345 & 353 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	1oo2	Voting type:		Diverse		
HFT:	1	β-factor [%]:		10		
Equipment Leg (1):		SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N			-	-	A	3
Rotork			-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01		-	-	A	3
Clean Service: Yes, Close on Trip: No, Full Stroke: Yes						
Equipment Leg (2):		SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N			-	-	A	3
Sirca AP-APM			-	-	A	3
GEFA HG1			-	-	A	3
Clean Service: Yes, Close on Trip: No, Full Stroke: Yes						
MRT [Hours]:	24					
Proof Test Interval [Months]:	6	Proof Test Coverage [%]:		100		
Proof Test Execution:	Duration 1 hours					

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group 345 & 353.

Table 14 Reliability Data Final Element Group 345 & 353

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
345SV599							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				3.70E-07			2.20E-06
						SFF [%]	5.1
						ROUTE 2 _H COMPLIANT	-
345SV524							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
GEFA HG1				1.56E-07			
						SFF [%]	6.6
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

3.3.3.2 Final Element Group 2: Open Avstängningsventiler venturi reduktionsgas

The information and reliability data underneath describe the Open Avstängningsventiler venturi reduktionsgas final element group as it has been analyzed in this Safety Integrity Level verification.

exSILentia-VER-002-01-b SILver Detailed-Eng

Voting within group: 1003

Voting type:

Diverse

HFT: 2

β -factor [%]:

10

Equipment Leg (1):

SERH VERSION

2_H

PIU

AC TYPE

SIL CAP

Parker N

-

-

A

3

Sirca AP-APM

-

-

A

3

GEFA HG1

-

-

A

3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

Equipment Leg (2):

SERH VERSION

2_H

PIU

AC TYPE

SIL CAP

Parker N

-

-

A

3

Sirca AP-APM

-

-

A

3

GEFA HG1

-

-

A

3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

Equipment Leg (3):

SERH VERSION

2_H

PIU

AC TYPE

SIL CAP

Parker N

-

-

A

3

Sirca AP-APM

-

-

A

3

Flowserve Worcester Series 44/59/459/599

2013.3.03

-

-

A

3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

MRT [Hours]: 24

Proof Test Interval [Months]:

6

Proof Test Coverage [%]:

100

Proof Test Execution:

Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Open Avstängningsventiler venturi reduktionsgas.

Table 15 Reliability Data Final Element Group Open Avstängningsventiler venturi reduktionsgas

Component	Failure Rates [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
345SV520							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
GEFA HG1				1.56E-07			
					SFF [%]		6.6
					Route 2 _H Compliant		-
345SV521							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
GEFA HG1				1.56E-07			
					SFF [%]		6.6
					Route 2 _H Compliant		-
345SV522							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
Flowserve Worcester Series 44/59/459/599		1.46E-07		3.22E-07			8.80E-07
					SFF [%]		17.3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

3.3.3.3 Final Element Group 3: Open Avstängningsventiler venturi processgas

The information and reliability data underneath describe the Open Avstängningsventiler venturi processgas final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo3 Voting type: Diverse
HFT: 2 β -factor [%]: 10

Equipment Leg (1):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
GEFA HG1		-	-	A	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

Equipment Leg (2):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
GEFA HG1		-	-	A	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

Equipment Leg (3):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
Flowserve Worcester Series 44/59/459/599	2013.3.03	-	-	A	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 100
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Open Avstängningsventiler venturi processgas.

Table 16 Reliability Data Final Element Group Open Avstängningsventiler venturi processgas

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
353SV520							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
GEFA HG1				1.56E-07			
					SFF [%]		6.6
					ROUTE 2 _H COMPLIANT		-

Component	Failure Rates [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
353SV521							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
GEFA HG1				1.56E-07			
					SFF [%]		6.6
					ROUTE 2 _H COMPLIANT		-
353SV522							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
Flowserve Worcester Series 44/59/459/599		1.46E-07		3.22E-07			8.80E-07
					SFF [%]		17.3
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

4 SIF-011 Beräkning av grupp 2

This chapter details the SIL Verification results, selections, and assumptions for the SIF-011 Beräkning av grupp 2 Safety Instrumented Function.

4.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: Beräkning av grupp 2
 SIF Tag: SIF-011
 SIF Description: SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H2) =>

Open Topptryck nödventil eft. våtelfilter red. gas (345SV599 SIL2)
 Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)
 Close Avstängningsventil H2S till PGH (256SV522 SIL2)
 Trip Reduktionsgaskompressor (346PU101 SIL2)
 Trip Processgaskompressor (346PU101 SIL2)
 Close H2 till reduktionskrets (341SV672/341RV675 SIL2)
 Close NG till reduktionskrets (252SV562/252RV565 SIL2)
 Close NG till kylkrets (252SV614/252RV618 SIL2)
 Close N2 till kylkrets (254SV594/254RV595 SIL2)

Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)
 Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)
 Close avstängningsventil N2 till reaktortopp (253SV607 SIL2)
 Open Avstängningsventiler venturi reduktionsgas (345SV520/21/22 SIL2)
 Open Avstängningsventiler venturi processgas (353SV520/21/22 SIL2)

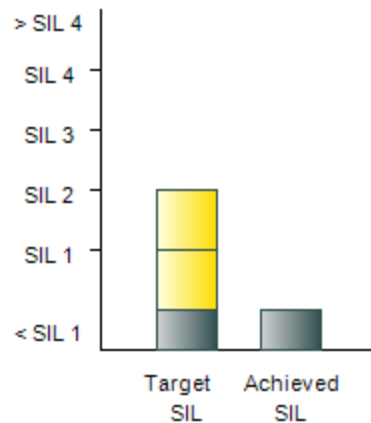
4.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 0

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 0 with RRF = 0



4.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-011 Beräkning av grupp 2 Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst:

Analysis Date: 10/4/2019

Design References

Comments:

Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-011 Beräkning av grupp 2 Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 17 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
0.00E+00	0	0	0	N/A	∞



The SIF Conceptual Design is modeled as shown in the picture below.

4.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-011 Beräkning av grupp 2 Safety Instrumented Function is quantified as follows.

Table 18 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
0.00E+00	0	N/A	∞	0	2

Number of Sensor group(s): 0
Voting between groups: 0oo0
β-factor [%]: N/A

4.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-011 Beräkning av grupp 2 Safety Instrumented Function is quantified as follows.

Table 19 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
0.00E+00	0	N/A	∞	0	2

The information and reliability data underneath describe the logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name:

Equipment:

SERH VERSION PIU SIL
Cap

N/A

Logic Solver Voting:

Architecture Type:

B

HFT:

0

β -factor [%]:

MRT [Hours]:

Proof Test Interval [Months]:

Proof Test Coverage [%]:

Proof Test Execution:

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group .

Table 20 Reliability Data Logic Solver Group

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
						SFF [%]	0.0

4.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-011 Beräkning av grupp 2 Safety Instrumented Function is quantified as follows.

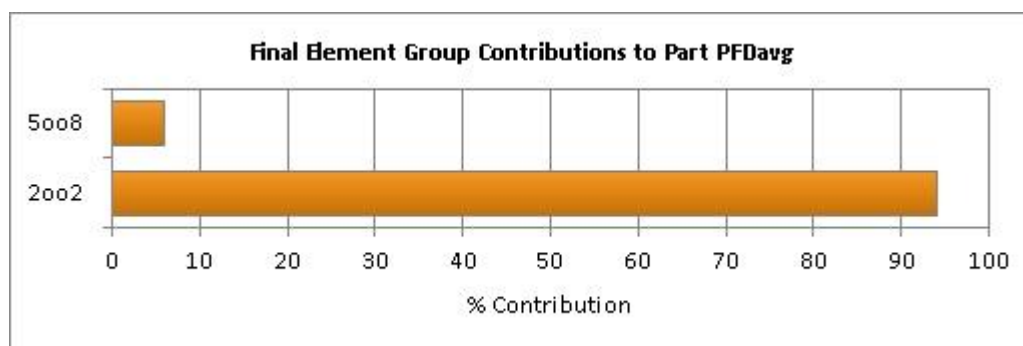
Table 21 Final Element Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
5.57E-03	2	N/A	28653.33	0	2

Number of Final Element group(s): 2

Voting between groups: 2oo2

β -factor [%]: 10



4.3.3.1 Final Element Group 1: 2oo2

The information and reliability data underneath describe the 2oo2 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	2oo2	Voting type:		Diverse		
HFT:	0	β-factor [%]:		10		
Equipment Leg (1):		SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Parker N			-	-	A	3
Rotork			-	-	A	3
Transworld Mars			-	-	A	3
Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes						
Equipment Leg (2):		SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Parker N			-	-	A	3
Rotork			-	-	A	3
Transworld Mars			-	-	A	3
Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes						
MRT [Hours]:	24					
Proof Test Interval [Months]:	6	Proof Test Coverage [%]:		100		
Proof Test Execution:	Duration 1 hours					

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group 2oo2.

Table 22 Reliability Data Final Element Group 2oo2

Component	Failure Rates [1/hr]						
	SD	SU	DD	DU	AD	AU	NE
252SV562							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Transworld Mars				3.73E-07			
					SFF [%]		5.1
					Route 2 _H Compliant		-
341SV672							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Transworld Mars				3.73E-07			
					SFF [%]		5.1
					Route 2 _H Compliant		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

4.3.3.2 Final Element Group 2: 5oo8

The information and reliability data underneath describe the 5oo8 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	5oo8	Voting type:	Identical
HFT:	3	β -factor [%]:	10

Equipment Leg

(each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]:	24		
Proof Test Interval [Months]:	6	Proof Test Coverage [%]:	100
Proof Test Execution:	Duration 1 hours		

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group 5008.

Table 23 Reliability Data Final Element Group 5008

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				5.55E-07			2.02E-06
						SFF [%]	4.4
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

5 SIF-012 SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|L2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-012 SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|L2) Safety Instrumented Function.

5.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322 L2)
SIF Tag	SIF-012
SIF Description:	SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322 L2) => Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2) Open N2 till reduktionskrets (254SV582 SIL2) Close Avstängningsventil 1 O2 inj. (251SV555 SIL2) Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

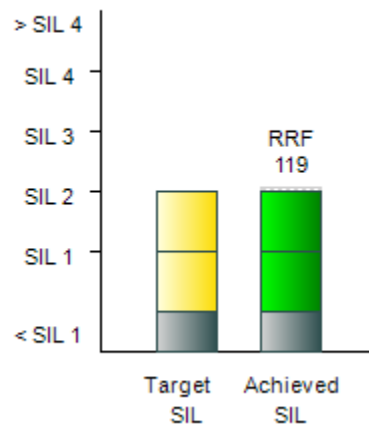
5.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 119.2



5.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-012 SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|L2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 10 years

Startup time: 24 hours

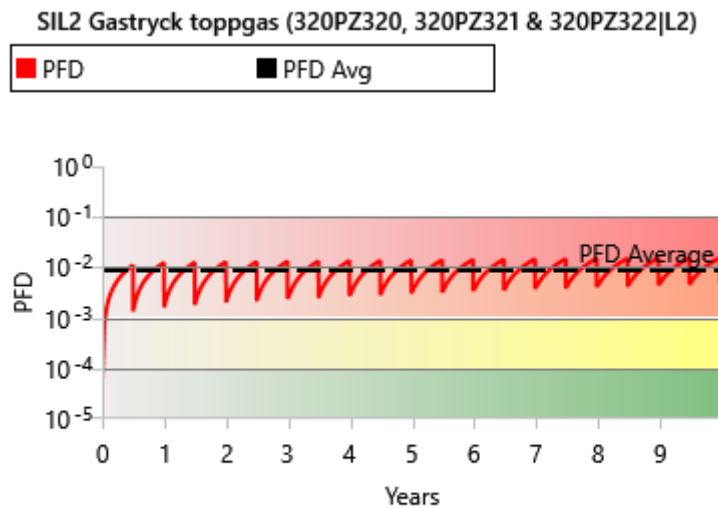
The SIF operates in the Low demand mode.

Analyst:

Analysis Date: 10/4/2019

Design References

Comments:

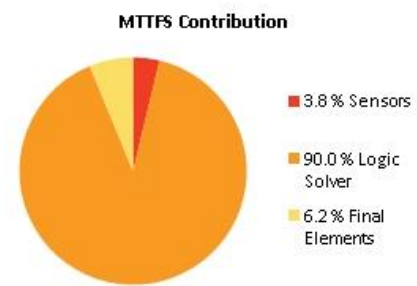
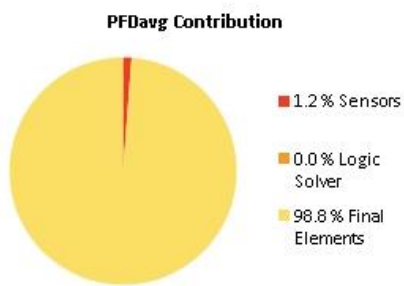


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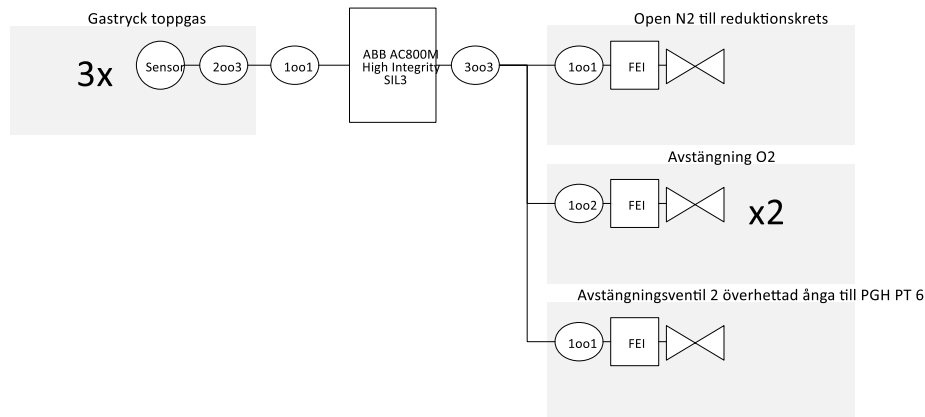
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-012 SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|L2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 24 Functional Safety Performance

PFDavg	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFDavg	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
8.38E-03	119.2	2	2	N/A	64.07



The SIF Conceptual Design is modeled as shown in the picture below.



5.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-012 SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|L2) Safety Instrumented Function is quantified as follows.

Table 25 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.00E-04	3	N/A	1703.33	1	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

5.3.1.1 Sensor Group 1: Gastryck toppgas

The information and reliability data underneath describe the Gastryck toppgas sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
„ Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3
MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Gastryck toppgas.

Table 26 Reliability Data Sensor Group Gastryck toppgas

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.96E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.17E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		78.5
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

5.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-012 SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322 | L2) Safety Instrumented Function is quantified as follows.

Table 27 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.73E-08	3	N/A	71.17	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment: ABB AC800M High Integrity SIL3 SERH VERSION 2013.1.02 PIU - SIL Cap N/A

Logic Solver Voting: Architecture Type: B

HFT: 1 β-factor [%]: 1

MRT [Hours]: 24
 Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
 Proof Test Execution: Duration 1 hours

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 28 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

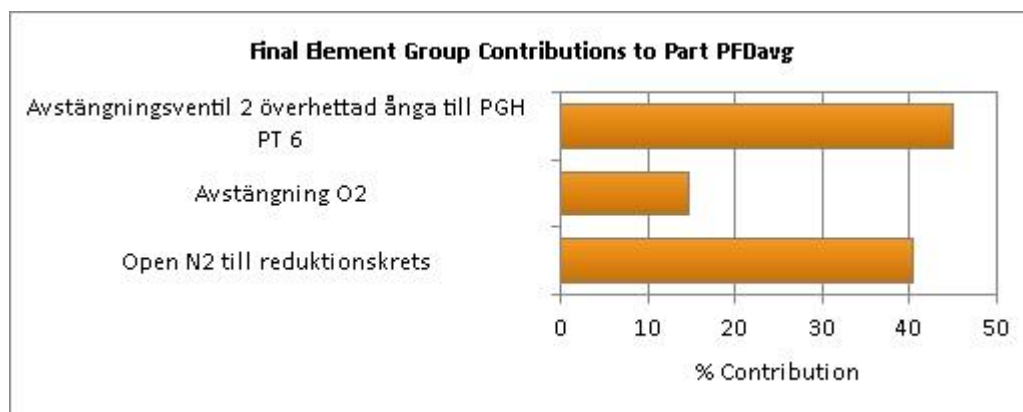
5.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-012 SIL2 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322 | L2) Safety Instrumented Function is quantified as follows.

Table 29 Final Element Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
8.29E-03	2	N/A	1030.27	0	2

Number of Final Element group(s): 3
 Voting between groups: 3oo3
 β -factor [%]: 10



5.3.3.1 Final Element Group 1: Open N2 till reduktionskrets

The information and reliability data underneath describe the Open N2 till reduktionskrets final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Transworld Mars		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 98
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Open N2 till reduktionskrets.

Table 30 Reliability Data Final Element Group Open N2 till reduktionskrets

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Transworld Mars				3.73E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		13.2
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

5.3.3.2 Final Element Group 2: Avstängning O2

The information and reliability data underneath describe the Avstängning O2 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg
(each): ,

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker Lucifer N-Series	2008.3.03	-	-	A	-
Rotork		-	-	A	3

Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]:	24		
Proof Test Interval [Months]:	6	Proof Test Coverage [%]:	98
Proof Test Execution:	Duration 1 hours		

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2.

Table 31 Reliability Data Final Element Group Avstängning O2

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker Lucifer N-Series		1.53E-07		9.93E-07			1.45E-06
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
SFF [%]							8.1
ROUTE 2H COMPLIANT							-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

5.3.3.3 Final Element Group 3: Avstängningsventil 2 överhettad ånga till PGH PT 6

The information and reliability data underneath describe the Avstängningsventil 2 överhettad ånga till PGH PT 6 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	1oo1	Voting type:	Identical
HFT:	0	β-factor [%]:	N/A

Equipment Leg (each):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]:	24		
Proof Test Interval [Months]:	6	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängningsventil 2 överhettad ånga till PGH PT 6.

Table 32 Reliability Data Final Element Group Avstängningsventil 2 överhettad ånga till PGH PT 6

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				5.55E-07			2.02E-06
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		45.3
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

6 SIF-013 SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H3)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-013 SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H3) Safety Instrumented Function.

6.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322 H3)
SIF Tag	SIF-013
SIF Description:	SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322 H3) => Open Topptryck nödventil eft. spraykylare red. gas (345SV524 SIL1)

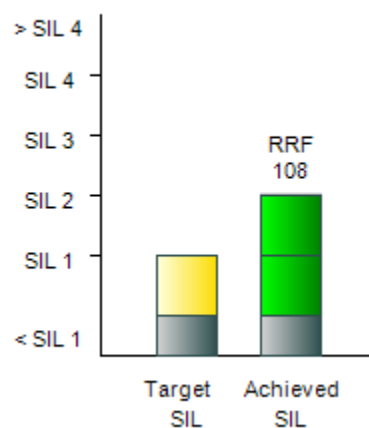
6.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 108.6



6.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-013 SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H3) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time:	15 years
Startup time:	24 hours

The SIF operates in the Low demand mode.

Analyst:

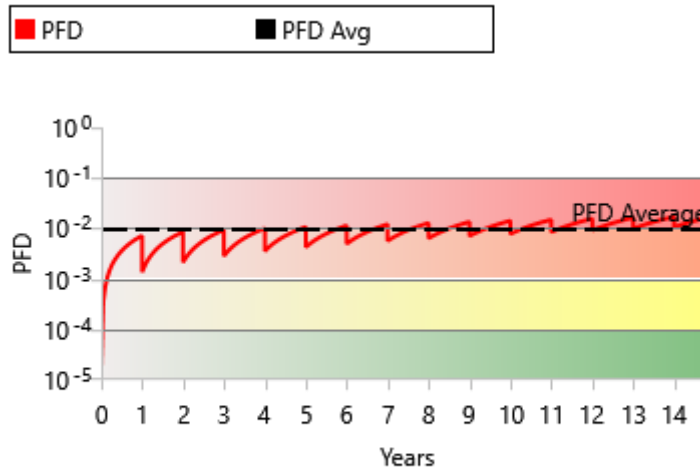
Analysis Date:

10/4/2019

Design References

Comments:

SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H3)



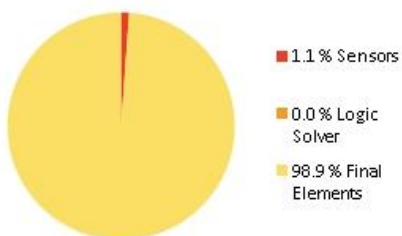
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Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-013 SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H3) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

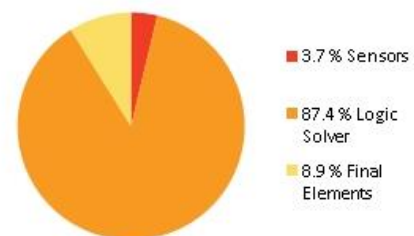
Table 33 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
9.21E-03	108.6	2	2	N/A	63.31

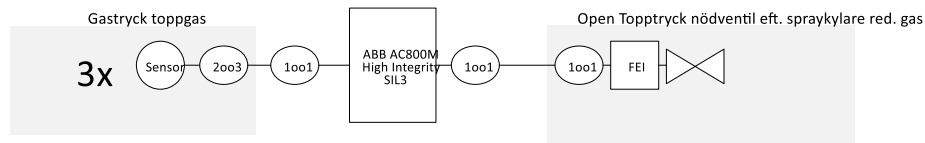
PFD_{avg} Contribution



MTTFS Contribution



The SIF Conceptual Design is modeled as shown in the picture below.



6.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-013 SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H3) Safety Instrumented Function is quantified as follows.

Table 34 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.05E-04	3	N/A	1698.47	1	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

6.3.1.1 Sensor Group 1: Gastryck toppgas

The information and reliability data underneath describe the Gastryck toppgas sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2
Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off					
ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
MRT [Hours]: 24					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 99		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Gastryck toppgas.

Table 35 Reliability Data Sensor Group Gastrick toppgas

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.96E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.17E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		78.5
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

6.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-013 SIL1 Gastrick toppgas (320PZ320, 320PZ321 & 320PZ322|H3) Safety Instrumented Function is quantified as follows.

Table 36 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β-factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 37 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

6.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-013 SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322 | H3) Safety Instrumented Function is quantified as follows.

Table 38 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
9.10E-03	2	N/A	711	0	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

6.3.3.1 Final Element Group 1: Open Topptryck nödventil eft. spraykylare red. gas

The information and reliability data underneath describe the Open Topptryck nödventil eft. spraykylare red. gas final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
GEFA HG1		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Open Topptryck nödventil eft. spraykylare red. gas.

Table 39 Reliability Data Final Element Group Open Toptryck nödventil eft. spraykylare red. gas

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
GEFA HG1				1.56E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		16.7
					ROUTE 2 _h COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

7 SIF-014 SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H4) (ej klar)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-014 SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H4) (ej klar) Safety Instrumented Function.

7.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322 H4) (ej klar)
SIF Tag	SIF-014
SIF Description:	SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322 H4) => Open Topptryck nödventil eft. spraykylare processgas. (353SV524 SIL1)

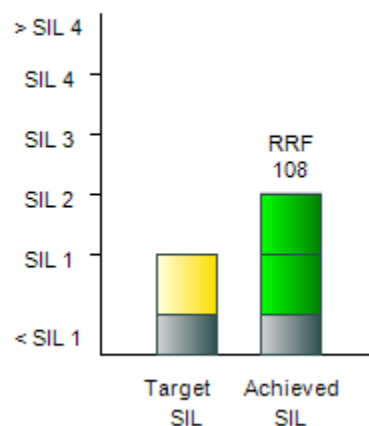
7.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 108.6



7.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-014 SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H4) (ej klar) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time:	15 years
Startup time:	24 hours

The SIF operates in the Low demand mode.

Analyst:

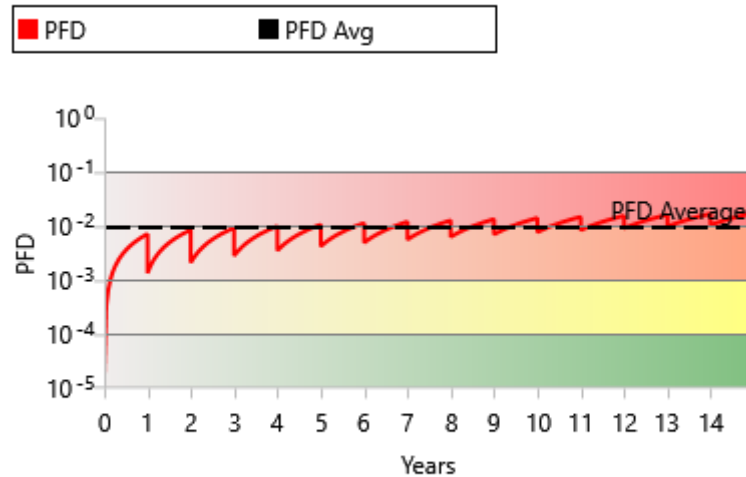
Analysis Date:

10/4/2019

Design References

Comments:

SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H4) (ej klar)

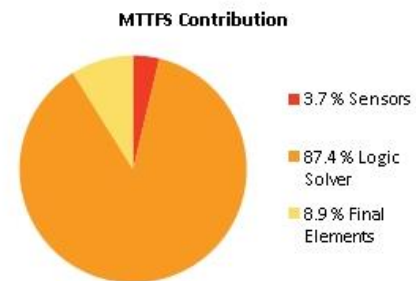
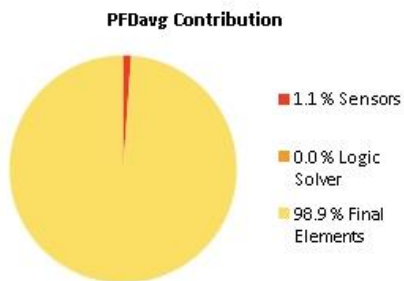


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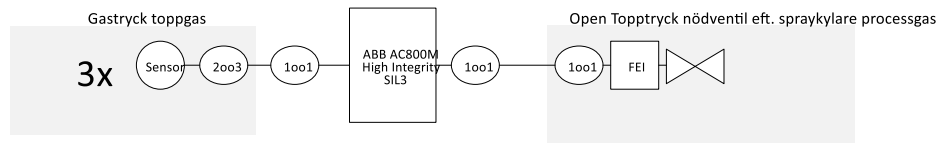
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-014 SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H4) (ej klar) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 40 Functional Safety Performance

PFDavg	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFDavg	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
9.21E-03	108.6	2	2	N/A	63.31



The SIF Conceptual Design is modeled as shown in the picture below.



7.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-014 SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H4) (ej klar) Safety Instrumented Function is quantified as follows.

Table 41 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.05E-04	3	N/A	1698.47	1	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

7.3.1.1 Sensor Group 1: Gastryck toppgas

The information and reliability data underneath describe the Gastryck toppgas sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2
Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off					
ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
MRT [Hours]: 24					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 99		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Gastryck toppgas.

Table 42 Reliability Data Sensor Group Gastryck toppgas

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.96E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.17E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		78.5
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

7.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-014 SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322|H4) (ej klar) Safety Instrumented Function is quantified as follows.

Table 43 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β-factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 44 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

7.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-014 SIL1 Gastryck toppgas (320PZ320, 320PZ321 & 320PZ322 | H4) (ej klar) Safety Instrumented Function is quantified as follows.

Table 45 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
9.10E-03	2	N/A	711	0	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

7.3.3.1 Final Element Group 1: Open Topptryck nödventil eft. spraykylare processgas

The information and reliability data underneath describe the Open Topptryck nödventil eft. spraykylare processgas final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
GEFA HG1		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Open Topptryck nödventil eft. spraykylare processgas.

Table 46 Reliability Data Final Element Group Open Toptryck nödventil eft. spraykylare processgas

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
GEFA HG1				1.56E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		16.7
					ROUTE 2 _h COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

8 SIF-015 SIL1 Temp toppgas (320TZ323, 320TZ324 & 320TZ325|H2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-015 SIL1 Temp toppgas (320TZ323, 320TZ324 & 320TZ325|H2) Safety Instrumented Function.

8.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL1 Temp toppgas (320TZ323, 320TZ324 & 320TZ325 H2)
SIF Tag	SIF-015
SIF Description:	SIL1 Temp toppgas (320TZ323, 320TZ324 & 320TZ325 H2) => Close H2 till reduktionskrets (341SV672/341RV675 SIL2) Close NG till reduktionskrets (252SV562/252RV565 SIL2) Close NG till kylkrets (252SV614/252RV618 SIL2) Close Avstängningsventil 1 O2 inj. (251SV555 SIL2) Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

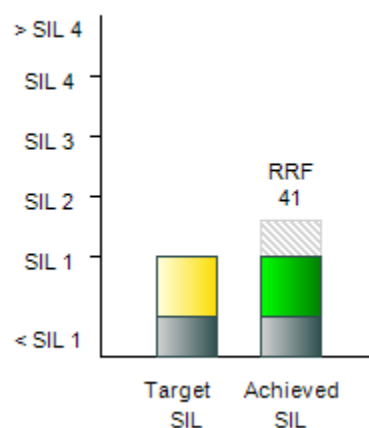
8.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 1 with RRF = 41.5



8.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-015 SIL1 Temp toppgas (320TZ323, 320TZ324 & 320TZ325|H2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

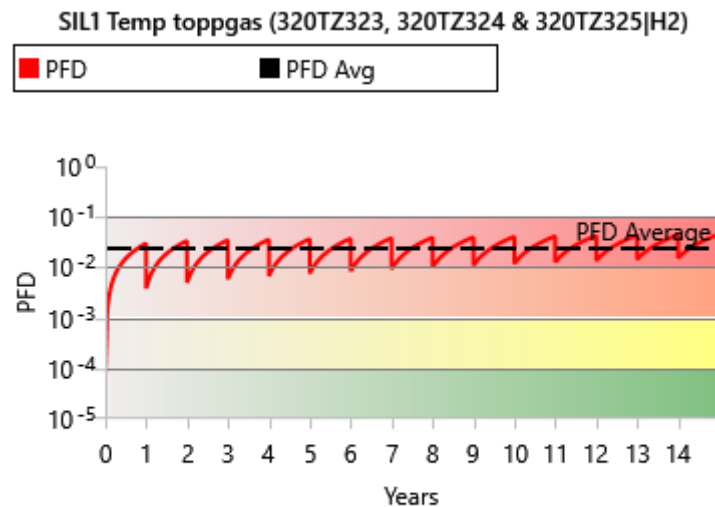
The SIF operates in the Low demand mode.

Analyst:

Analysis Date: 10/4/2019

Design References

Comments:

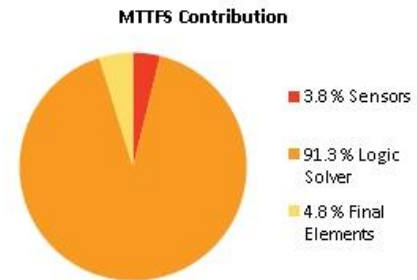
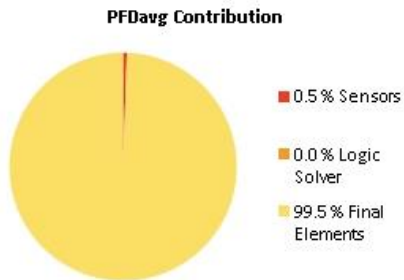


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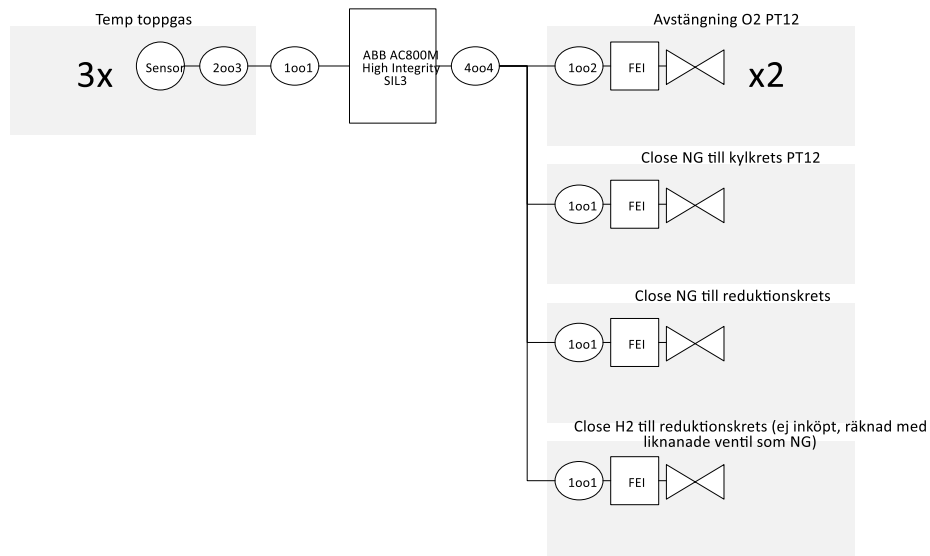
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-015 SIL1 Temp toppgas (320TZ323, 320TZ324 & 320TZ325|H2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 47 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
2.41E-02	41.5	1	2	N/A	65.01



The SIF Conceptual Design is modeled as shown in the picture below.



8.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-015 SIL1 Temp toppgas (320TZ323, 320TZ324 & 320TZ325 | H2) Safety Instrumented Function is quantified as follows.

Table 48 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.31E-04	3	N/A	1693.48	1	2

Number of Sensor group(s): 1

Voting between groups: 1oo1
 β -factor [%]: N/A

8.3.1.1 Sensor Group 1: Temp toppgas

The information and reliability data underneath describe the Temp toppgas sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
 HFT: 1 β -factor [%]: 10

Equipment Leg (each): „

PR electronics PR5435 / PR5437, Single, 4..20mA

SERH VERSION

2_H

PIU

AC TYPE

SIL CAP

2019.4.01

-

✓

B

-

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]

2013.1.02

-

-

B

3

MRT [Hours]: 24

Proof Test Interval [Months]:

12

Proof Test Coverage [%]:

90

Proof Test Execution:

Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Temp toppgas.

Table 49 Reliability Data Sensor Group Temp toppgas

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
PR electronics PR5435 / PR5437, Single, 4..20mA	4.90E-08 <i>0.00E+00</i>	1.70E-08 <i>0.00E+00</i>	3.58E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.75E-07</i>	0.00E+00 <i>0.00E+00</i>	2.70E-08 <i>7.60E-08</i>	1.90E-08 <i>1.90E-08</i>	1.10E-08 <i>1.10E-08</i>	2.13E-07 <i>2.13E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]		84.9	
							ROUTE 2 _H COMPLIANT		-	

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

8.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-015 SIL1 Temp toppgas (320TZ323, 320TZ324 & 320TZ325 | H2) Safety Instrumented Function is quantified as follows.

Table 50 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.86E-08	3	N/A	71.17	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A
Logic Solver Voting:		Architecture Type:	B	
HFT:	1	β-factor [%]:	1	
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 51 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

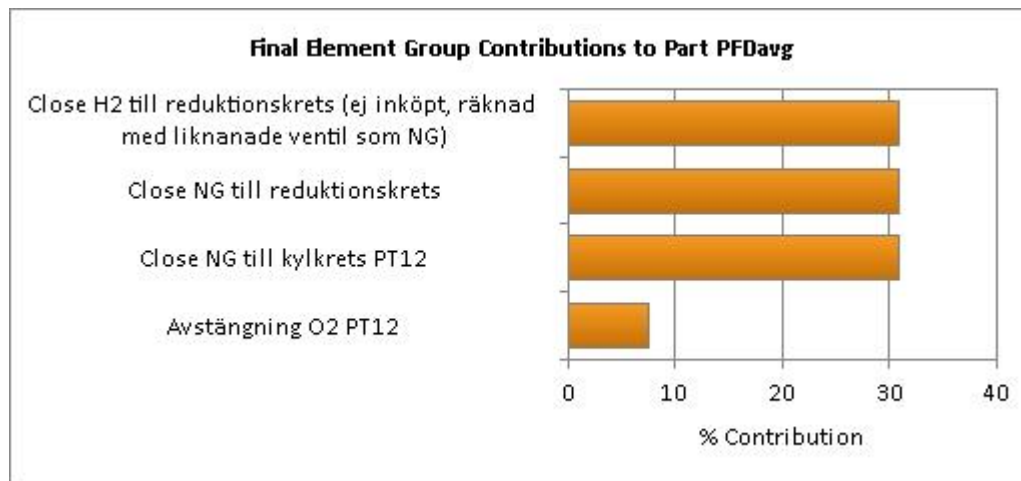
8.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-015 SIL1 Temp toppgas (320TZ323, 320TZ324 & 320TZ325 | H2) Safety Instrumented Function is quantified as follows.

Table 52 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.39E-02	2	N/A	1347.75	0	2

Number of Final Element group(s): 4
Voting between groups: 4oo4
 β -factor [%]:



8.3.3.1 Final Element Group 1: Avstängning O2 PT12

The information and reliability data underneath describe the Avstängning O2 PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg (each):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2 PT12.

Table 53 Reliability Data Final Element Group Avstängning O2 PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve,				1.90E-06			6.72E-07

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
metal seat, < 8"							
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	5.9
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

8.3.3.2 Final Element Group 2: Close NG till kylkrets PT12

The information and reliability data underneath describe the Close NG till kylkrets PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Transworld Mars		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 97
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close NG till kylkrets PT12.

Table 54 Reliability Data Final Element Group Close NG till kylkrets PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Transworld Mars				3.73E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	13.2
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

8.3.3.3 Final Element Group 3: Close NG till reduktionskrets

The information and reliability data underneath describe the Close NG till reduktionskrets final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Transworld Mars		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 97
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close NG till reduktionskrets.

Table 55 Reliability Data Final Element Group Close NG till reduktionskrets

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Transworld Mars				3.73E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
SFF [%]							13.2
ROUTE 2 _H COMPLIANT							-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

8.3.3.4 Final Element Group 4: Close H2 till reduktionskrets (ej inköpt, räknad med liknanade ventil som NG)

The information and reliability data underneath describe the Close H2 till reduktionskrets (ej inköpt, räknad med liknanade ventil som NG) final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Transworld Mars		-	-	A	3

ABB AC800M High Integrity SIL3 [DigitalOut] 2013.1.02 - - B 3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 97

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close H2 till reduktionskrets (ej inköpt, räknad med liknanade ventil som NG).

Table 56 Reliability Data Final Element Group Close H2 till reduktionskrets (ej inköpt, räknad med liknanade ventil som NG)

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Transworld Mars				3.73E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	13.2
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

9 SIF-016 SIL1 Temp överhettad ånga (271TI552|L2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-016 SIL1 Temp överhettad ånga (271TI552|L2) Safety Instrumented Function.

9.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL1 Temp överhettad ånga (271TI552|L2)
 SIF Tag: SIF-016
 SIF Description: SIL1 Temp överhettad ånga (271TI552|L2) =>

Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)

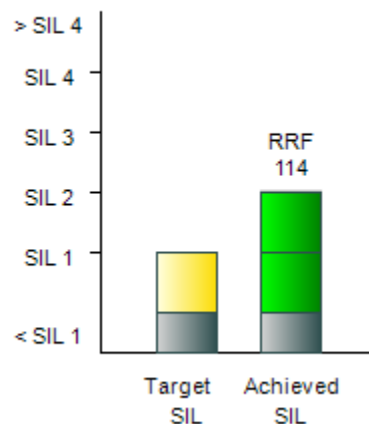
9.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 114.5



9.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-016 SIL1 Temp överhettad ånga (271TI552|L2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
 Startup time: 24 hours

The SIF operates in the Low demand mode.

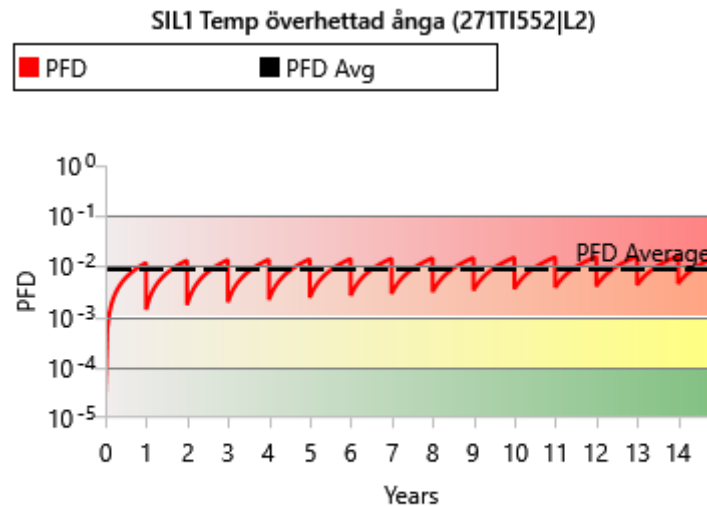
Analyst:

Analysis Date:

10/4/2019

Design References

Comments:

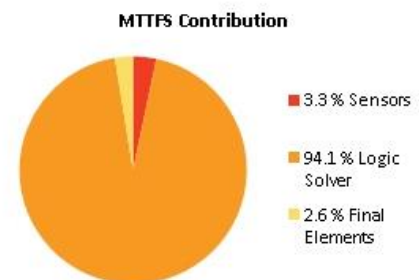
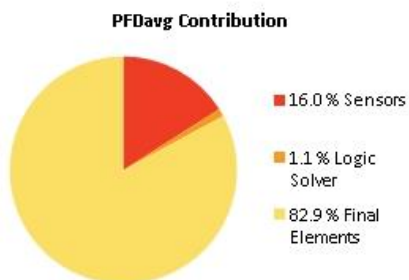


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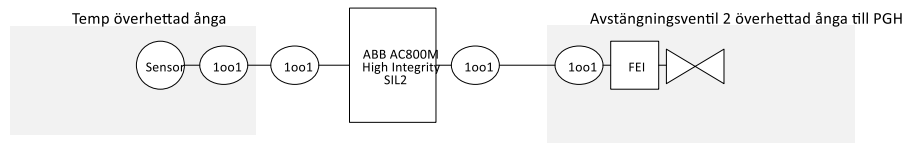
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-016 SIL1 Temp överhettad ånga (271TI552|L2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table S7 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
8.73E-03	114.5	2	2	N/A	18.29



The SIF Conceptual Design is modeled as shown in the picture below.



9.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-016 SIL1 Temp överhettad ånga (271TI552|L2) Safety Instrumented Function is quantified as follows.

Table 58 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.40E-03	2	N/A	554.98	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

9.3.1.1 Sensor Group 1: Temp överhettad ånga

The information and reliability data underneath describe the Temp överhettad ånga sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser iTemp TMT 182	2006.2.02	-	-	B	-
Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off					
ABB AC800M High Integrity SIL2 [AnalogIn]	2012.1.02	-	-	B	2
MRT [Hours]: 24					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 90		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Temp överhettad ånga.

Table 59 Reliability Data Sensor Group Temp överhettad ånga

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser iTemp TMT 182	1.66E-08 <i>0.00E+00</i>	9.16E-08 <i>0.00E+00</i>	9.18E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>1.83E-07</i>	0.00E+00 <i>0.00E+00</i>	1.07E-07 <i>1.24E-07</i>	0.00E+00 <i>0.00E+00</i>	1.22E-09 <i>1.22E-09</i>	9.78E-08 <i>9.78E-08</i>
ABB AC800M High Integrity SIL2 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		62.6
								ROUTE 2H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

9.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-016 SIL1 Temp överhettad ånga (271TI552|L2) Safety Instrumented Function is quantified as follows.

Table 60 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
9.63E-05	2	N/A	19.43	0	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL2 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL2

Equipment:	ABB AC800M High Integrity SIL2	SERH VERSION	PIU	SIL Cap
		2012.1.02	-	N/A
Logic Solver Voting:	Architecture Type:	B		
HFT:	β-factor [%]:			
MRT [Hours]:				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:		
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL2.

Table 61 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL2

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/TP855)	4.51E-07	7.54E-07	1.46E-06	1.11E-10	3.63E-09	1.19E-08	6.79E-07
Power Supply: SD823 + SS823 + MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
					SFF [%]		100.0

9.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-016 SIL1 Temp överhettad ånga (271TI552|L2) Safety Instrumented Function is quantified as follows.

Table 62 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
7.25E-03	2	N/A	712.45	0	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

9.3.3.1 Final Element Group 1: Avstängningsventil 2 överhettad ånga till PGH

The information and reliability data underneath describe the Avstängningsventil 2 överhettad ånga till PGH final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängningsventil 2 överhettad ånga till PGH.

Table 63 Reliability Data Final Element Group Avstängningsventil 2 överhettad ånga till PGH

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				5.55E-07			2.02E-06
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		11.5
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

10 SIF-017 SIL1 Tryck överhettad ånga (271PI552|L2) Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-017 SIL1 Tryck överhettad ånga (271PI552|L2) Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2) Safety Instrumented Function.

10.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL1 Tryck överhettad ånga (271PI552 L2) Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)
SIF Tag	SIF-017
SIF Description:	SIL1 Tryck överhettad ånga (271PI552 L2) => Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)

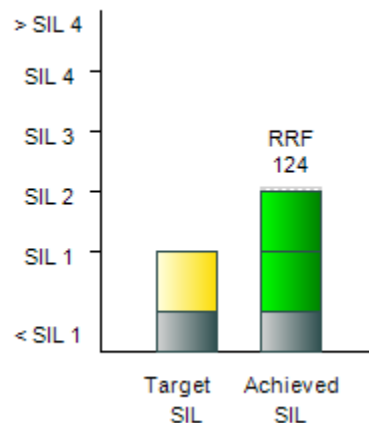
10.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 124.6



10.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-017 SIL1 Tryck överhettad ånga (271PI552|L2) Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

exSILentia-VER-002-01-b SILver Detailed-Eng

Mission Time: 15 years
Startup time: 24 hours

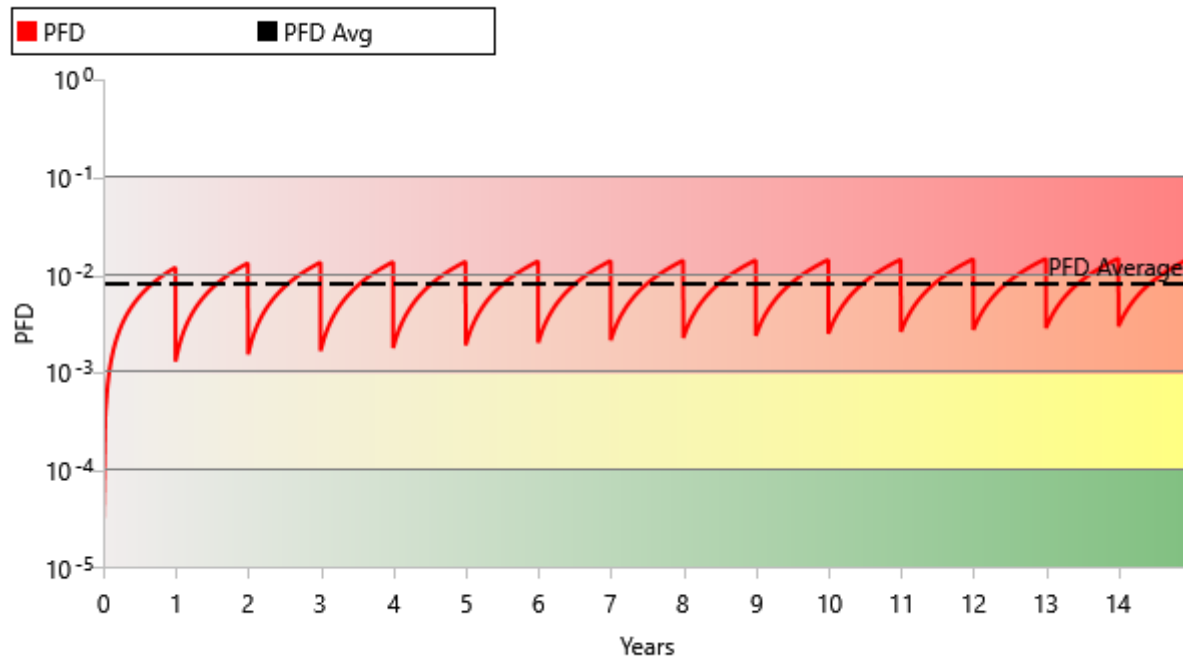
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

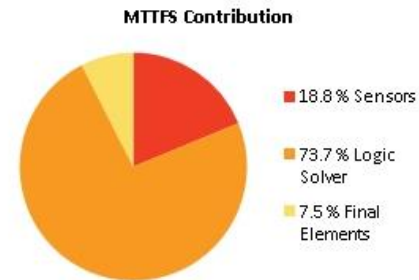
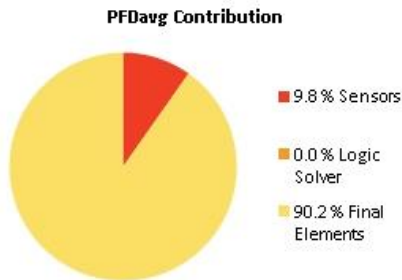
SIL1 Tryck överhettad ånga (271PI552|L2) Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)



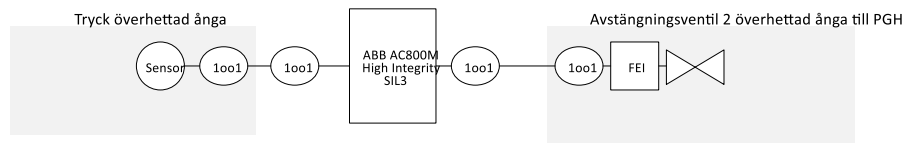
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-017 SIL1 Tryck överhettad ånga (271PI552|L2) Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 64 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
8.02E-03	124.6	2	2	N/A	53.42



The SIF Conceptual Design is modeled as shown in the picture below.



10.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-017 SIL1 Tryck överhettad ånga (271PI552|L2) Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2) Safety Instrumented Function is quantified as follows.

Table 65 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
7.86E-04	2	N/A	284.42	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

10.3.1.1 Sensor Group 1: Tryck överhettad ånga

The information and reliability data underneath describe the Tryck överhettad ånga sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

SERH VERSION

2H

PIU

AC TYPE

SIL CAP

Endress+Hauser Cerabar S PMP71, PMP72
HT, PMP75 2006.2.02 - - B 2

Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Tryck överhettad ånga.

Table 66 Reliability Data Sensor Group Tryck överhettad ånga

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.34E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]		75.5	
							ROUTE 2H COMPLIANT		-	

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

10.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-017 SIL1 Tryck överhettad ånga (271PI552|L2) Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2) Safety Instrumented Function is quantified as follows.

Table 67 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High
Integrity SIL3

Equipment:

	SERH VERSION	PIU	SIL Cap
ABB AC800M High Integrity SIL3	2013.1.02	-	N/A

Logic Solver Voting: Architecture Type: B
HFT: 1 β -factor [%]: 1
MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 68 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

10.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-017 SIL1 Tryck överhettad ånga (271PI552|L2) Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2) Safety Instrumented Function is quantified as follows.

Table 69 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
7.25E-03	2	N/A	712.45	0	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
 β -factor [%]: N/A

10.3.3.1 Final Element Group 1: Avstängningsventil 2 överhettad ånga till PGH

The information and reliability data underneath describe the Avstängningsventil 2 överhettad ånga till PGH final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

SERH VERSION 2H PIU AC TYPE SIL CAP

Parker N	-	-	A	3
Rotork	-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängningsventil 2 överhettad ånga till PGH.

Table 70 Reliability Data Final Element Group Avstängningsventil 2 överhettad ånga till PGH

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				5.55E-07			2.02E-06
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	11.5
						ROUTE 2h COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

11 SIF-018 SIL1 Tryck överhettad ånga friblåsning ($271PI320 < (343PZ305/7/8 + 0,1 \text{ barg})$) Permissive Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-018 SIL1 Tryck överhettad ånga friblåsning ($271PI320 < (343PZ305/7/8 + 0,1 \text{ barg})$) Permissive Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2) Safety Instrumented Function.

11.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL1 Tryck överhettad ånga friblåsning ($271PI320 < (343PZ305/7/8 + 0,1 \text{ barg})$) Permissive Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)
SIF Tag	SIF-018
SIF Description:	SIL1 Tryck överhettad ånga friblåsning ($271PI320 < (343PZ305/7/8 + 0,1 \text{ barg})$) => Permissive Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)

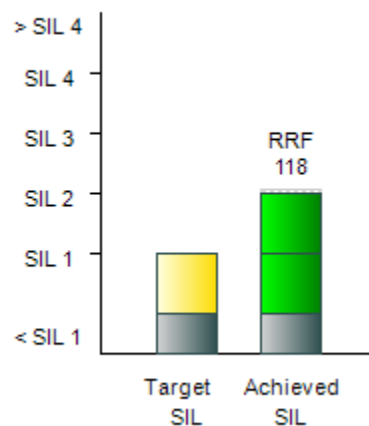
11.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 118.5



11.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-018 SIL1 Tryck överhettad ånga friblåsning ($271PI320 < (343PZ305/7/8 + 0,1 \text{ barg})$) Permissive Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

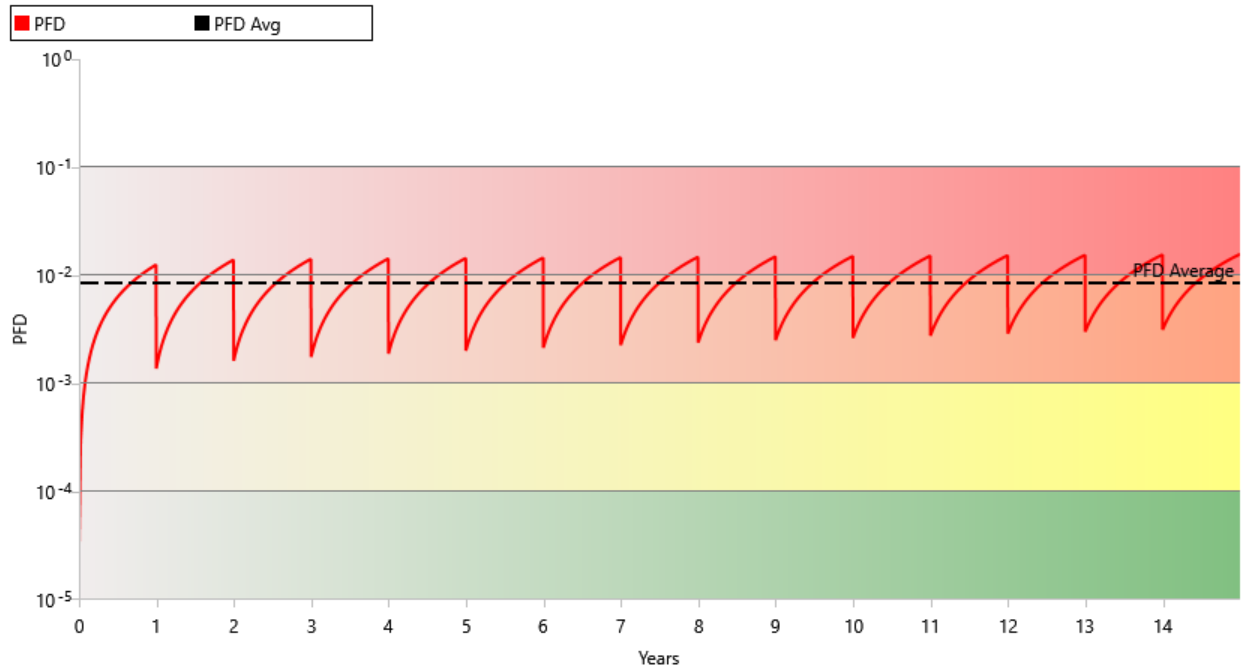
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

SIL1 Tryck överhettad ånga friblåsning (271PI320 < (343PZ305/7/8 + 0,1 barg)) Permissive Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)



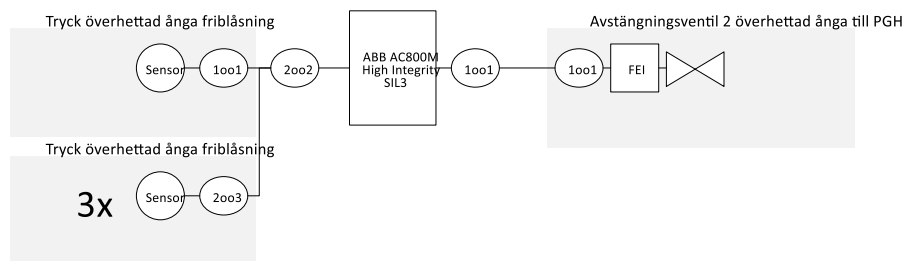
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-018 SIL1 Tryck överhettad ånga friblåsning (271PI320 < (343PZ305/7/8 + 0,1 barg)) Permissive Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 71 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
8.44E-03	118.5	2	2	N/A	65.36



The SIF Conceptual Design is modeled as shown in the picture below.



11.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-018 SIL1 Tryck överhettad ånga friblåsning (271PI320 < (343PZ305/7/8 + 0,1 barg)) Permissive Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2) Safety Instrumented Function is quantified as follows.

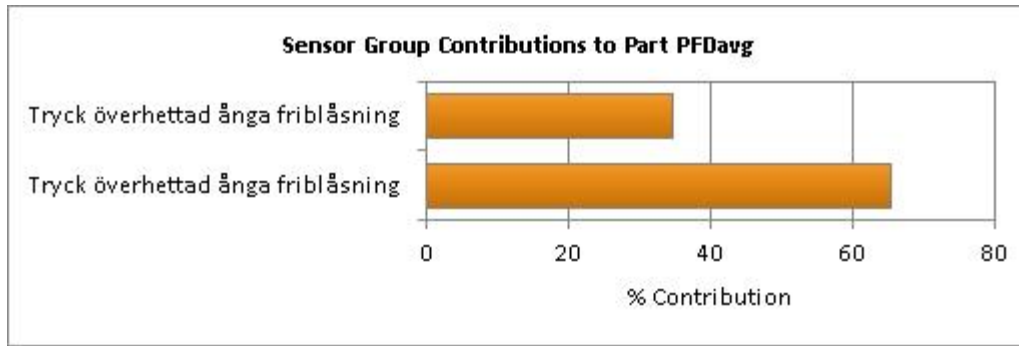
Table 72 Sensor Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.20E-03	2	N/A	10362.29	0	2

Number of Sensor group(s): 2

Voting between groups: 2oo2

β -factor [%]:



11.3.1.1 Sensor Group 1: Tryck överhettad ånga friblåsning

The information and reliability data underneath describe the Tryck överhettad ånga friblåsning sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β -factor [%]: N/A

Equipment Leg (each):

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
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MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Tryck överhettad ånga friblåsning.

Table 73 Reliability Data Sensor Group Tryck överhettad ånga friblåsning

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.34E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	75.5
									ROUTE 2H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

11.3.1.2 Sensor Group 2: Tryck överhettad ånga friblåsning

The information and reliability data underneath describe the Tryck överhettad ånga friblåsning sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg
(each): „

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
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MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Tryck överhettad ånga friblåsning.

Table 74 Reliability Data Sensor Group Tryck överhettad ånga friblåsning

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>6.85E-08</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>4.44E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]		18.5	
							ROUTE 2H COMPLIANT		-	

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

11.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-018 SIL1 Tryck överhettad ånga friblåsning (271PI320 < (343PZ305/7/8 + 0,1 barg)) Permissive Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2) Safety Instrumented Function is quantified as follows.

Table 75 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A
Logic Solver Voting:		Architecture Type:	B	
HFT:	1	β-factor [%]:	1	
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 76 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

11.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-018 SIL1 Tryck överhettad ånga friblåsning (271PI320 < (343PZ305/7/8 + 0,1 barg)) Permissive Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2) Safety Instrumented Function is quantified as follows.

Table 77 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
7.25E-03	2	N/A	712.45	0	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
 β -factor [%]: N/A

11.3.3.1 Final Element Group 1: Avstängningsventil 2 överhettad ånga till PGH

The information and reliability data underneath describe the Avstängningsventil 2 överhettad ånga till PGH final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	ACTYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängningsventil 2 överhettad ånga till PGH.

Table 78 Reliability Data Final Element Group Avstängningsventil 2 överhettad ånga till PGH

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				5.55E-07			2.02E-06
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	11.5
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

12 SIF-019 SIL2 Flöde reduktionsgas till PGH (343XI302|L2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-019 SIL2 Flöde reduktionsgas till PGH (343XI302|L2) Safety Instrumented Function.

12.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL2 Flöde reduktionsgas till PGH (343XI302|L2)
SIF Tag SIF-019
SIF Description: SIL2 Flöde reduktionsgas till PGH (343XI302|L2) =>

Close Avstängningsventil 1 H2 till brännare PGH (341SV665 SIL2)
Close Avstängningsventil 2 H2 till brännare PGH (341SV667 SIL2)

Close Avstängningsventil 1 NG till brännare PGH (252SV544 SIL2)
Close Avstängningsventil 2 NG till brännare PGH (252SV546 SIL2)

Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)
Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

Endast 1 bränsle åt gången.

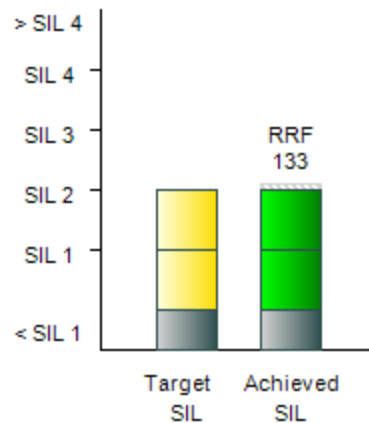
12.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 133.7



12.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-019 SIL2 Flöde reduktionsgas till PGH (343XI302|L2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

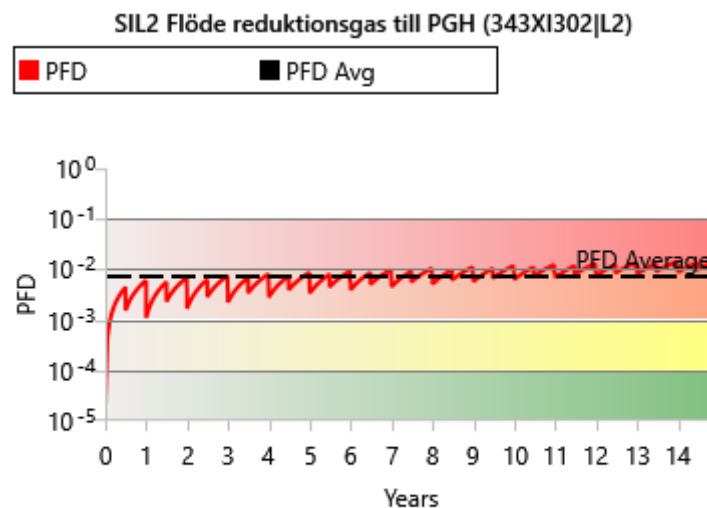
The SIF operates in the Low demand mode.

Analyst:

Analysis Date: 10/4/2019

Design References

Comments:

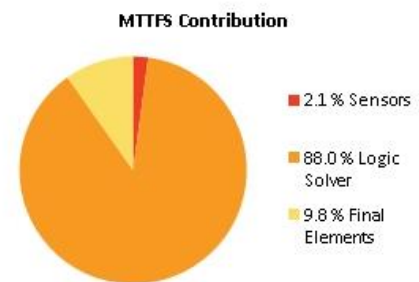
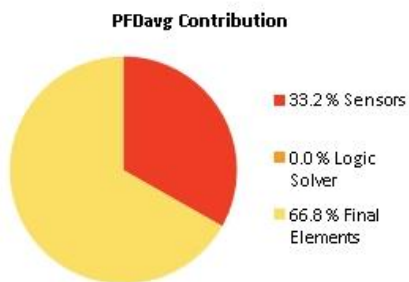


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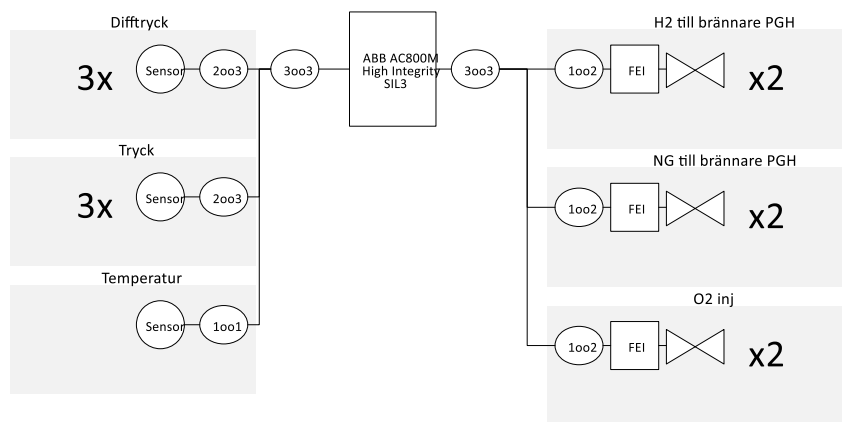
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-019 SIL2 Flöde reduktionsgas till PGH (343XI302|L2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 79 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
7.47E-03	133.7	2	2	N/A	62.67



The SIF Conceptual Design is modeled as shown in the picture below.



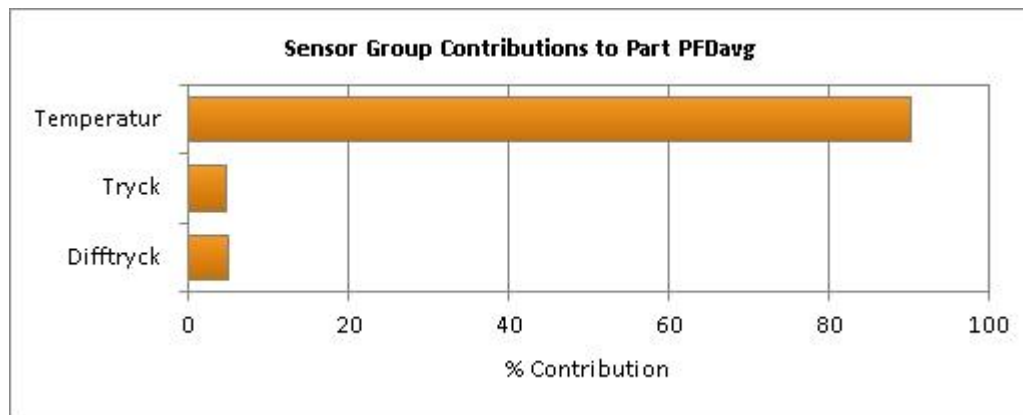
12.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-019 SIL2 Flöde reduktionsgas till PGH (343XI302|L2) Safety Instrumented Function is quantified as follows.

Table 80 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.49E-03	2	N/A	2921.8	0	2

Number of Sensor group(s): 3
Voting between groups: 3oo3
 β -factor [%]:



12.3.1.1 Sensor Group 1: Difftryck

The information and reliability data underneath describe the Difftryck sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg

(each):	„	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Deltabar S PMD75, FMD77, FMD78		2006.2.02	-	-	B	2

Trip Direction: Low, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
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MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Difftryck.

Table 81 Reliability Data Sensor Group Difftryck

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Deltabar S PMD75, FMD77, FMD78	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.89E-08 <i>1.37E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	4.30E-07 <i>4.30E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		75.0
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

12.3.1.2 Sensor Group 2: Tryck

The information and reliability data underneath describe the Tryck sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical

HFT: 1 β -factor [%]: 10

Equipment Leg

(each): „ SERH VERSION 2_H PIU AC TYPE SIL CAP

Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75 2006.2.02 - - B 2

Trip Direction: Low, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Tryck.

Table 82 Reliability Data Sensor Group Tryck

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.34E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
							SFF [%]			75.5
							ROUTE 2 _H COMPLIANT			-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

12.3.1.3 Sensor Group 3: Temperatur

The information and reliability data underneath describe the Temperatur sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β -factor [%]: N/A

Equipment Leg (each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser iTemp TMT 182	2006.2.02	-	-	B	-

Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
---	-----------	---	---	---	---

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Temperatur.

Table 83 Reliability Data Sensor Group Temperatur

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser iTemp TMT 182	1.66E-08 <i>0.00E+00</i>	9.16E-08 <i>0.00E+00</i>	9.18E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>1.08E-07</i>	0.00E+00 <i>0.00E+00</i>	1.07E-07 <i>1.99E-07</i>	0.00E+00 <i>0.00E+00</i>	1.22E-09 <i>1.22E-09</i>	9.78E-08 <i>9.78E-08</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]			40.0
							ROUTE 2 _H COMPLIANT			-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

12.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-019 SIL2 Flöde reduktionsgas till PGH (343XI302 | L2) Safety Instrumented Function is quantified as follows.

Table 84 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.86E-08	3	N/A	71.17	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	SERH VERSION	PIU	SIL Cap
ABB AC800M High Integrity SIL3	2013.1.02	-	N/A
Logic Solver Voting:	Architecture Type:	B	
HFT:	β-factor [%]:	1	
MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 85 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

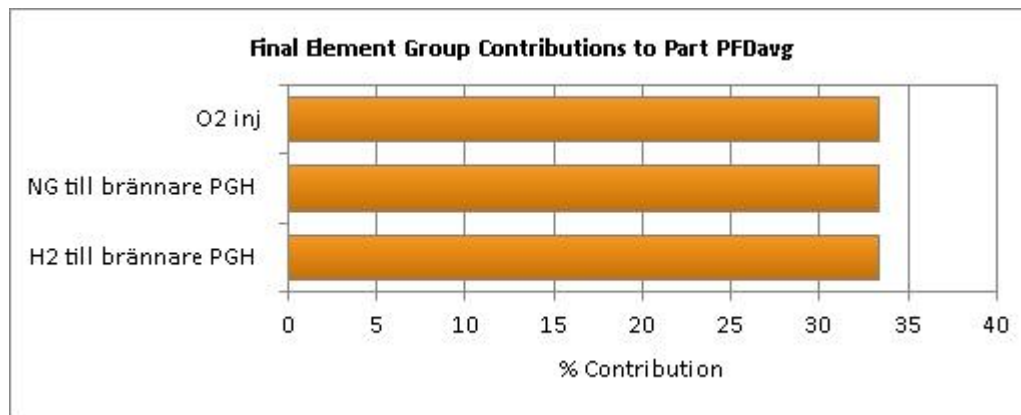
12.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-019 SIL2 Flöde reduktionsgas till PGH (343XI302 | L2) Safety Instrumented Function is quantified as follows.

Table 86 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
5.00E-03	3	N/A	638.97	1	2

Number of Final Element group(s): 3
Voting between groups: 3oo3
 β -factor [%]: 10



12.3.3.1 Final Element Group 1: H2 till brännare PGH

The information and reliability data underneath describe the H2 till brännare PGH final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	ACTYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 95
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group H2 till brännare PGH.

Table 87 Reliability Data Final Element Group H2 till brännare PGH

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		5.9
					ROUTE 2H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

12.3.3.2 Final Element Group 2: NG till brännare PGH

The information and reliability data underneath describe the NG till brännare PGH final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 95
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group NG till brännare PGH.

Table 88 Reliability Data Final Element Group NG till brännare PGH

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		28.4

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

12.3.3.3 Final Element Group 3: O2 inj

The information and reliability data underneath describe the O2 inj final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2

Voting type: Identical

HFT: 1

β-factor [%]: 10

Equipment Leg
(each): ,

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24

Proof Test Interval [Months]: 6

Proof Test Coverage [%]: 95

Proof Test Execution:

Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group O2 inj.

Table 89 Reliability Data Final Element Group O2 inj

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		5.9
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

13 SIF-020 SIL2 Difftryck O2-transferline (251XI313)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-020 SIL2 Difftryck O2-transferline (251XI313) Safety Instrumented Function.

13.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Difftryck O2-transferline (251XI313)
SIF Tag	SIF-020
SIF Description:	SIL2 Gastryck transferline (251XI313) => Close Avstängningsventil 1 O2 inj. (251SV555 SIL2) Close Avstängningsventil 2 O2 inj. (251SV556 SIL2) Open Avluftningsventil O2 (F.O) (251SV557 SIL2)

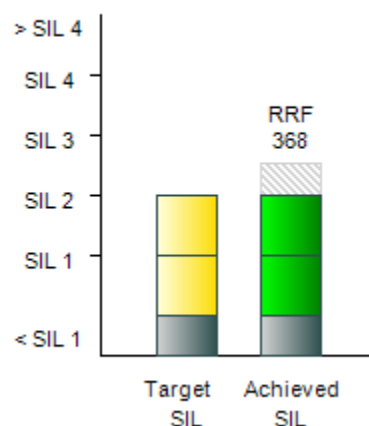
13.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 368.5



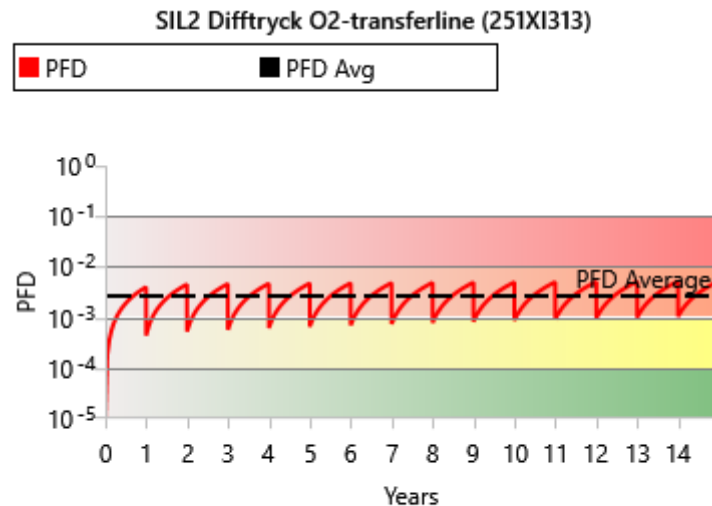
13.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-020 SIL2 Difftryck O2-transferline (251XI313) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:

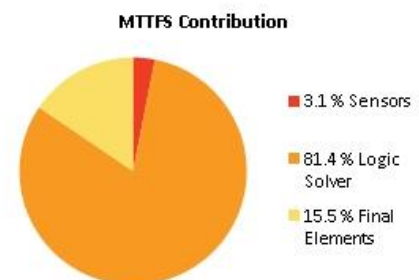
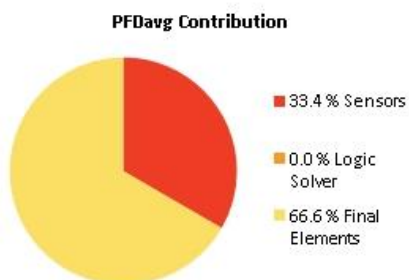


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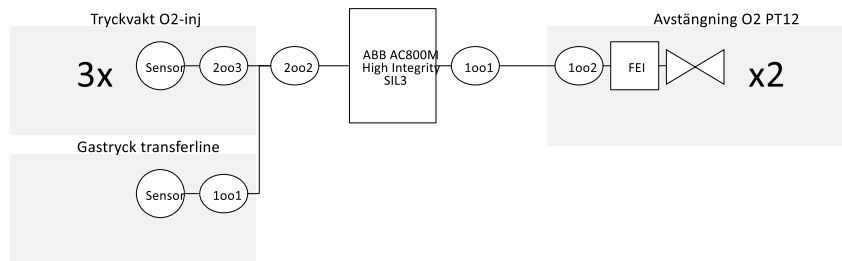
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-020 SIL2 Difftryck O2-transferline (251XI313) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 90 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
2.71E-03	368.5	2	2	N/A	58.62



The SIF Conceptual Design is modeled as shown in the picture below.



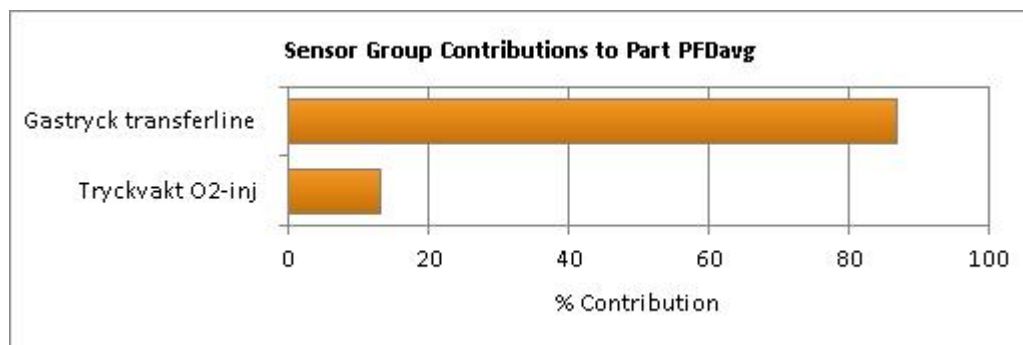
13.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-020 SIL2 Difftryck O2-transferline (251XI313) Safety Instrumented Function is quantified as follows.

Table 91 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
9.06E-04	2	N/A	1895.34	0	2

Number of Sensor group(s): 2
Voting between groups: 2oo2
 β -factor [%]:



13.3.1.1 Sensor Group 1: Tryckvakt O2-inj

The information and reliability data underneath describe the Tryckvakt O2-inj sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical

HFT: 1 β -factor 10
[%]:

Equipment Leg

(each): „ SERH VERSION 2_H PIU AC TYPE SIL CAP

Endress+Hauser Cerabar S PMP71, PMP72 2006.2.02 - - B 2
HT, PMP75

Trip Direction: Low, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Tryckvakt O2-inj.

Table 92 Reliability Data Sensor Group Tryckvakt O2-inj

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.34E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	75.5
									ROUTE 2 _H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

13.3.1.2 Sensor Group 2: Gastryck transferline

The information and reliability data underneath describe the Gastryck transferline sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001 Voting type: Identical

HFT: 0 β -factor N/A
[%]:

Equipment Leg

(each): SERH VERSION 2_H PIU AC TYPE SIL CAP

Endress+Hauser Cerabar S PMP71, PMP72 2006.2.02 - - B 2
HT, PMP75

Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
 Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Gastryck transferline.

Table 93 Reliability Data Sensor Group Gastryck transferline

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.34E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		75.5
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

13.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-020 SIL2 Difftryck O2-transferline (251XI313) Safety Instrumented Function is quantified as follows.

Table 94 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.84E-08	3	N/A	72.03	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A

Logic Solver Voting: Architecture Type: B

HFT: 1 β-factor [%]: 1

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 95 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

13.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-020 SIL2 Difftryck O2-transferline (251XI313) Safety Instrumented Function is quantified as follows.

Table 96 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.81E-03	3	N/A	377.66	1	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

13.3.3.1 Final Element Group 1: Avstängning O2 PT12

The information and reliability data underneath describe the Avstängning O2 PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each):	SERH VERSION	2H	PIU	ACTYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution:

Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2 PT12.

Table 97 Reliability Data Final Element Group Avstängning O2 PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		5.9
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

14 SIF-021 SIL1 Gastryck transferline (344PZ321|L2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-021 SIL1 Gastryck transferline (344PZ321|L2) Safety Instrumented Function.

14.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL1 Gastryck transferline (344PZ321|L2)
 SIF Tag: SIF-021
 SIF Description: SIL1 Gastryck transferline (344PZ321|L2) =>

Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)
 Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

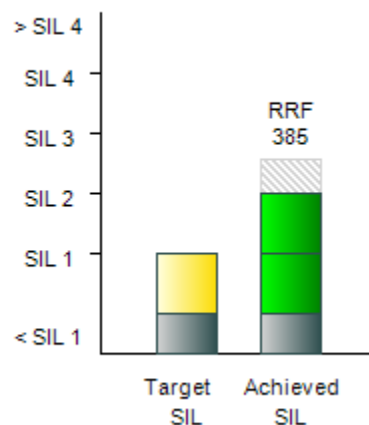
14.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 385.5



14.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-021 SIL1 Gastryck transferline (344PZ321|L2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

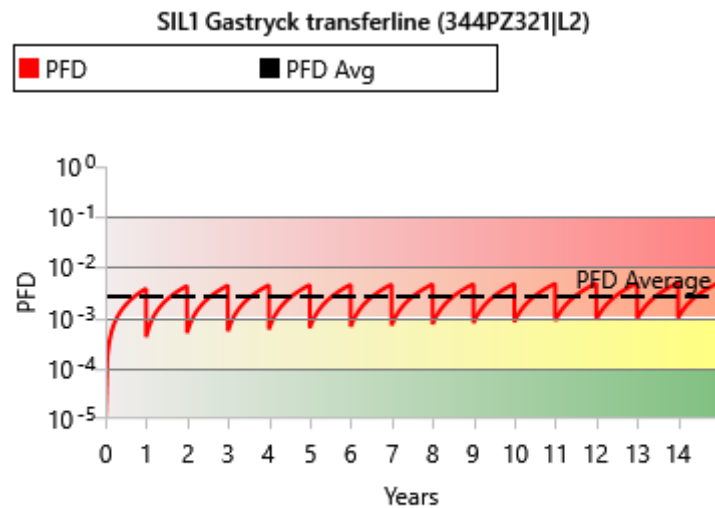
The SIF operates in the Low demand mode.

Analyst:

Analysis Date: 10/4/2019

Design References

Comments:

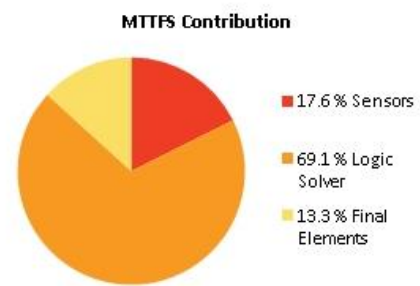
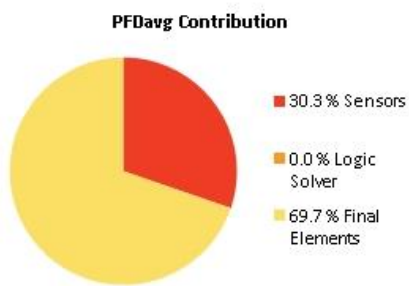


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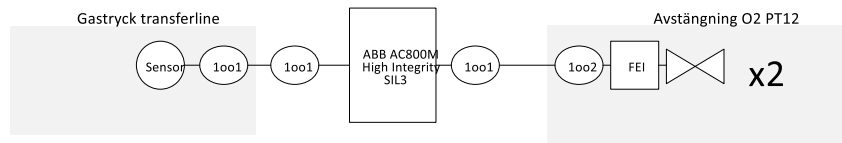
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-021 SIL1 Gastryck transferline (344PZ321|L2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 98 Functional Safety Performance

PFDavg	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFDavg	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
2.59E-03	385.5	2	2	N/A	50.09



The SIF Conceptual Design is modeled as shown in the picture below.



14.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-021 SIL1 Gastryck transferline (344PZ321|L2) Safety Instrumented Function is quantified as follows.

Table 99 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
7.86E-04	2	N/A	284.42	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

14.3.1.1 Sensor Group 1: Gastryck transferline

The information and reliability data underneath describe the Gastryck transferline sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2
Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off					
ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
MRT [Hours]: 24					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 99		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Gastryck transferline.

Table 100 Reliability Data Sensor Group Gastryck transferline

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.34E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		75.5
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

14.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-021 SIL1 Gastryck transferline (344PZ321|L2) Safety Instrumented Function is quantified as follows.

Table 101 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A
Logic Solver Voting:	Architecture Type:	B		
HFT:	β-factor [%]:	1		
MRT [Hours]:				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 102 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

14.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-021 SIL1 Gastryck transferline (344PZ321|L2) Safety Instrumented Function is quantified as follows.

Table 103 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.81E-03	3	N/A	377.66	1	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

14.3.3.1 Final Element Group 1: Avstängning O2 PT12

The information and reliability data underneath describe the Avstängning O2 PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2 PT12.

Table 104 Reliability Data Final Element Group Avstängning O2 PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		5.9
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

15 SIF-022 SIL2 Tryckvakt O2-inj (251XI312 | L2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-022 SIL2 Tryckvakt O2-inj (251XI312 | L2) Safety Instrumented Function.

15.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL2 Tryckvakt O2-inj (251XI312 | L2)
 SIF Tag: SIF-022
 SIF Description: SIL2 Tryckvakt O2-inj (251XI312 | L2) =>

Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)
 Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

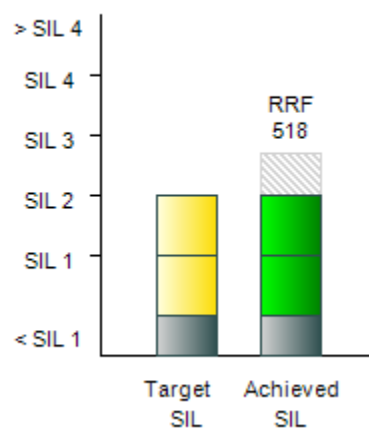
15.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 518.2



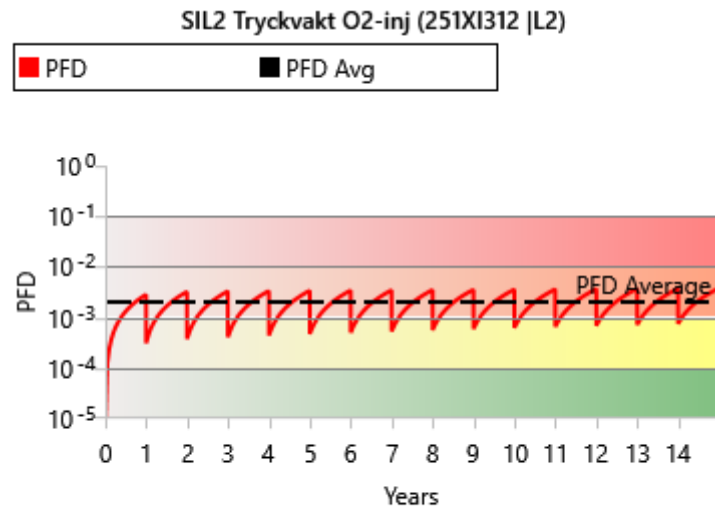
15.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-022 SIL2 Tryckvakt O2-inj (251XI312 | L2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:

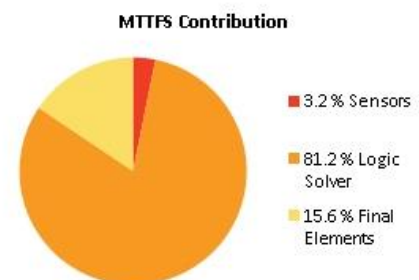
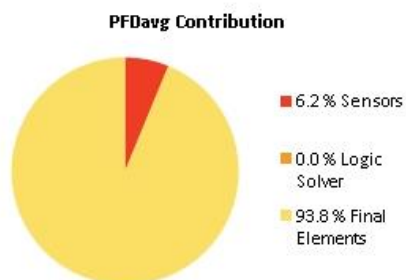


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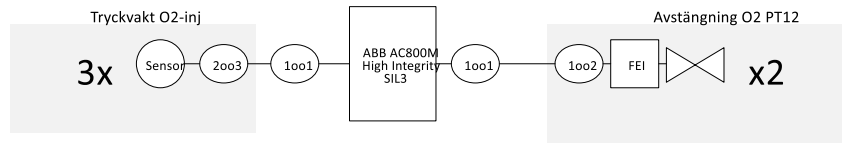
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-022 SIL2 Tryckvakt O2-inj (251XI312 |L2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 105 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
1.93E-03	518.2	2	3	N/A	58.87



The SIF Conceptual Design is modeled as shown in the picture below.



15.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-022 SIL2 Tryckvakt O2-inj (251XI312 |L2) Safety Instrumented Function is quantified as follows.

Table 106 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.20E-04	3	N/A	1857.58	1	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

15.3.1.1 Sensor Group 1: Tryckvakt O2-inj

The information and reliability data underneath describe the Tryckvakt O2-inj sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical

HFT: 1 β-factor [%]: 10

Equipment Leg (each): „

Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75

SERH VERSION

2H

PIU

AC TYPE

SIL CAP

2006.2.02

-

-

B

2

Trip Direction: Low, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]

2013.1.02

-

-

B

3

MRT [Hours]: 24

Proof Test Interval [Months]:

12

Proof Test Coverage [%]:

99

Proof Test Execution:

Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Tryckvakt O2-inj.

Table 107 Reliability Data Sensor Group Tryckvakt O2-inj

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.34E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		75.5
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

15.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-022 SIL2 Tryckvakt O2-inj (251XI312 | L2) Safety Instrumented Function is quantified as follows.

Table 108 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A
Logic Solver Voting:	Architecture Type:	B		
HFT:	β-factor [%]:	1		
MRT [Hours]:				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 109 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

15.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-022 SIL2 Tryckvakt O2-inj (251XI312 | L2) Safety Instrumented Function is quantified as follows.

Table 110 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.81E-03	3	N/A	377.66	1	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

15.3.3.1 Final Element Group 1: Avstängning O2 PT12

The information and reliability data underneath describe the Avstängning O2 PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2 PT12.

Table 111 Reliability Data Final Element Group Avstängning O2 PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		5.9
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

16 SIF-023 SIL2 Flödesmätare O2 inj (251XI554|L2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-023 SIL2 Flödesmätare O2 inj (251XI554|L2) Safety Instrumented Function.

16.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Flödesmätare O2 inj (251XI554 L2)
SIF Tag	SIF-023
SIF Description:	SIL2 Flödesmätare O2 inj (251XI554 L2) => Close Avstängningsventil 1 O2 inj. (251SV555 SIL2) Close Avstängningsventil 2 O2 inj. (251SV556 SIL2) Open Avluftsventil O2 (F.O) (251SV557 SIL2)

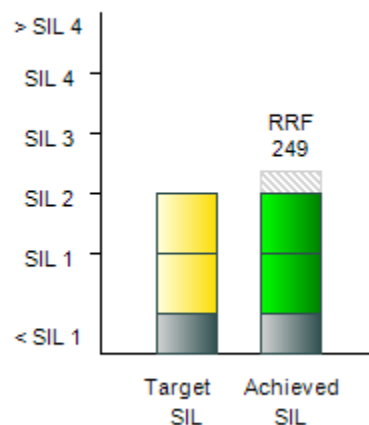
16.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 249.5



16.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-023 SIL2 Flödesmätare O2 inj (251XI554|L2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

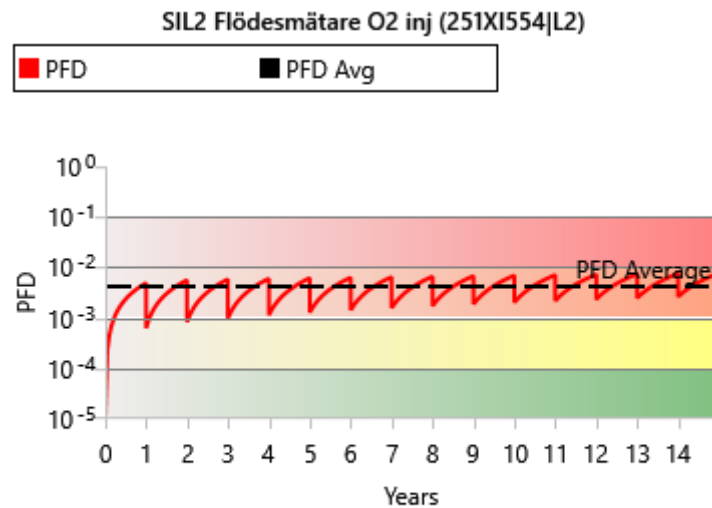
The SIF operates in the Low demand mode.

Analyst:

Analysis Date: 10/4/2019

Design References

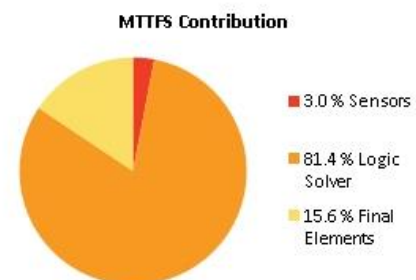
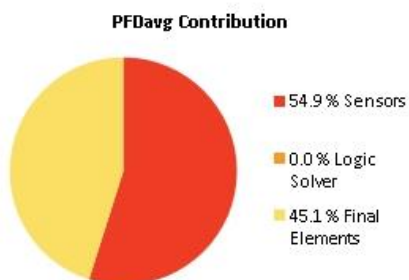
Comments:



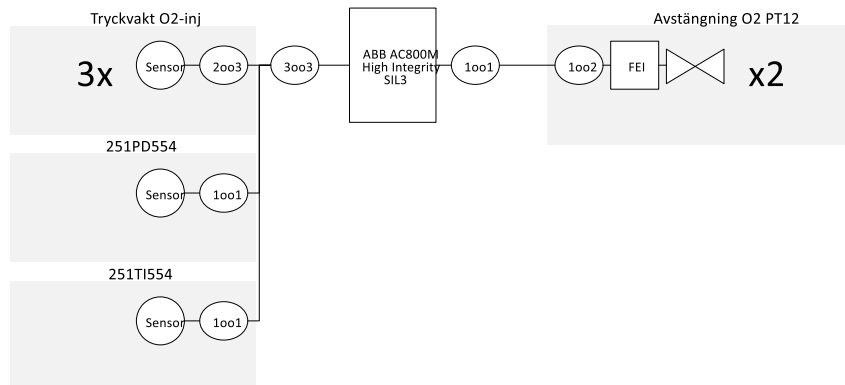
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-023 SIL2 Flödesmätare O2 inj (251XI554|L2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 112 Functional Safety Performance

PFDavg	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFDavg	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
4.01E-03	249.5	2	2	N/A	58.98



The SIF Conceptual Design is modeled as shown in the picture below.



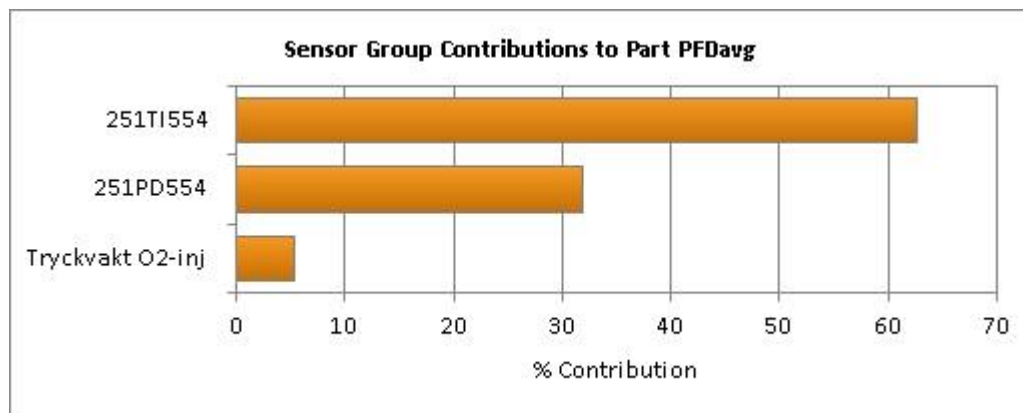
16.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-023 SIL2 Flödesmätare O2 inj (251XI554|L2) Safety Instrumented Function is quantified as follows.

Table 113 Sensor Part Functional Safety Performance

PFD _{avg}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.20E-03	2	N/A	1971.41	0	2

Number of Sensor group(s): 3
Voting between groups: 3oo3
β-factor [%]: 10



16.3.1.1 Sensor Group 1: Tryckvakt O2-inj

The information and reliability data underneath describe the Tryckvakt O2-inj sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical

HFT: 1 β -factor [%]: 10

Equipment Leg (each): „

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: Low, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Tryckvakt O2-inj.

Table 114 Reliability Data Sensor Group Tryckvakt O2-inj

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.34E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	75.5
									ROUTE 2 _H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

16.3.1.2 Sensor Group 2: 251PD554

The information and reliability data underneath describe the 251PD554 sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β -factor [%]: N/A

Equipment Leg (each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
--	--------------	----------------	-----	---------	---------

Endress+Hauser Deltabar S PMD75, FMD77, FMD78 2006.2.02 - - B 2

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group 251PD554.

Table 115 Reliability Data Sensor Group 251PD554

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Deltabar S PMD75, FMD77, FMD78	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.96E-07</i>	0.00E+00 <i>0.00E+00</i>	6.89E-08 <i>1.21E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	4.30E-07 <i>4.30E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]		78.0	
							ROUTE 2H COMPLIANT		-	

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

16.3.1.3 Sensor Group 3: 251TI554

The information and reliability data underneath describe the 251TI554 sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β-factor [%]: N/A

Equipment Leg (each):

Endress+Hauser iTemp TMT 182 2006.2.02 - - B -

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group 251TI554.

Table 116 Reliability Data Sensor Group 251TI554

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser iTemp TMT 182	1.66E-08 <i>0.00E+00</i>	9.16E-08 <i>0.00E+00</i>	9.18E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>1.83E-07</i>	0.00E+00 <i>0.00E+00</i>	1.07E-07 <i>1.24E-07</i>	0.00E+00 <i>0.00E+00</i>	1.22E-09 <i>1.22E-09</i>	9.78E-08 <i>9.78E-08</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		62.6
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

16.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-023 SIL2 Flödesmätare O2 inj (251XI554|L2) Safety Instrumented Function is quantified as follows.

Table 117 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β-factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 118 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

16.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-023 SIL2 Flödesmätare O2 inj (251XI554|L2) Safety Instrumented Function is quantified as follows.

Table 119 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.81E-03	3	N/A	377.66	1	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

16.3.3.1 Final Element Group 1: Avstängning O2 PT12

The information and reliability data underneath describe the Avstängning O2 PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2 PT12.

Table 120 Reliability Data Final Element Group Avstängning O2 PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		5.9
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

17 SIF-024 SIL2 Tryckvakt N2 vid O2 inj (253PZ362, 253PZ363, 253PZ364|L2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-024 SIL2 Tryckvakt N2 vid O2 inj (253PZ362, 253PZ363, 253PZ364|L2) Safety Instrumented Function.

17.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Tryckvakt N2 vid O2 inj (253PZ362, 253PZ363, 253PZ364 L2)
SIF Tag	SIF-024
SIF Description:	Difftryck N2/transfer line (253XI364) (253PZ362, 253PZ363, 253PZ364) - 344PZ321 < 1 bar(g) L2) => Close Avstängningsventil 1 O2 inj. (251SV555 SIL2) Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

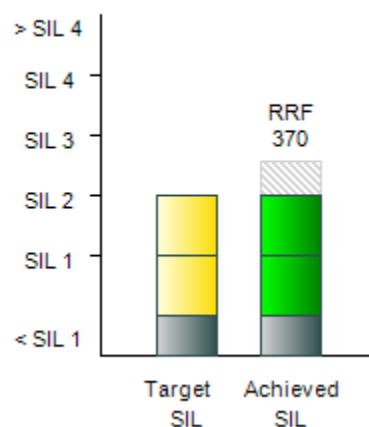
17.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 370.1



17.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-024 SIL2 Tryckvakt N2 vid O2 inj (253PZ362, 253PZ363, 253PZ364|L2) Safety Instrumented Function. In order to perform

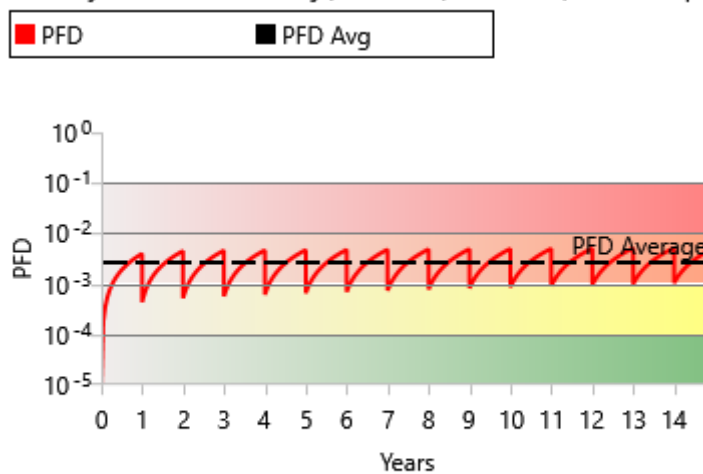
the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:

SIL2 Tryckvakt N2 vid O2 inj (253PZ362, 253PZ363, 253PZ364|L2)



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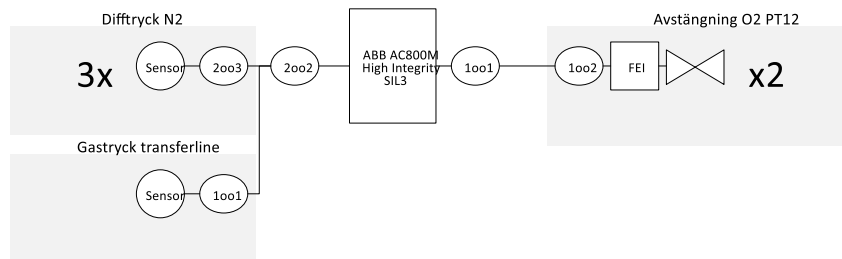
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-024 SIL2 Tryckvakt N2 vid O2 inj (253PZ362, 253PZ363, 253PZ364|L2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 121 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
2.70E-03	370.1	2	2	N/A	58.9



The SIF Conceptual Design is modeled as shown in the picture below.



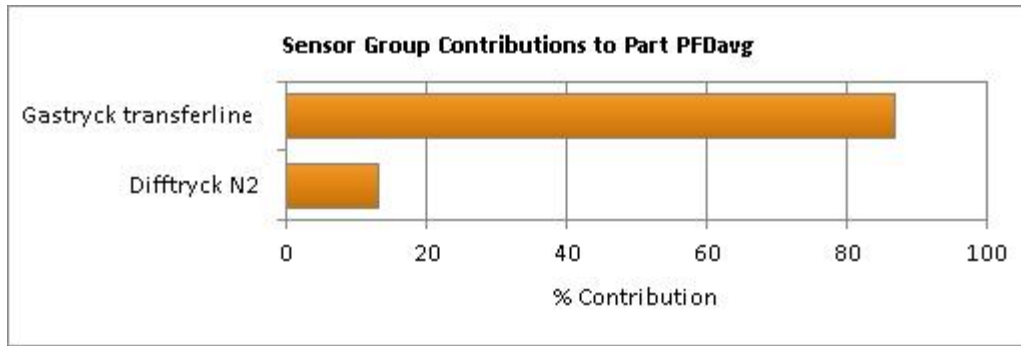
17.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-024 SIL2 Tryckvakt N2 vid O2 inj (253PZ362, 253PZ363, 253PZ364|L2) Safety Instrumented Function is quantified as follows.

Table 122 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
8.94E-04	2	N/A	1888.91	0	2

Number of Sensor group(s): 2
Voting between groups: 2oo2
β-factor [%]: 10



17.3.1.1 Sensor Group 1: Difftryck N2

The information and reliability data underneath describe the Difftryck N2 sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical

HFT: 1 β -factor: 10
[%]:

Equipment Leg (each): „

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: Low, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Difftryck N2.

Table 123 Reliability Data Sensor Group Difftryck N2

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.34E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		75.5
								ROUTE 2H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

17.3.1.2 Sensor Group 2: Gastryck transferline

The information and reliability data underneath describe the Gastryck transferline sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
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MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Gastryck transferline.

Table 124 Reliability Data Sensor Group Gastryck transferline

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.34E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	75.5
									ROUTE 2 _H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

17.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-024 SIL2 Tryckvakt N2 vid O2 inj (253PZ362, 253PZ363, 253PZ364|L2) Safety Instrumented Function is quantified as follows.

Table 125 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A
Logic Solver Voting:		Architecture Type:	B	
HFT:	1	β-factor [%]:	1	
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 126 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

17.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-024 SIL2 Tryckvakt N2 vid O2 inj (253PZ362, 253PZ363, 253PZ364 | L2) Safety Instrumented Function is quantified as follows.

Table 127 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.81E-03	3	N/A	377.66	1	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
 β -factor [%]: N/A

17.3.3.1 Final Element Group 1: Avstängning O2 PT12

The information and reliability data underneath describe the Avstängning O2 PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2 PT12.

Table 128 Reliability Data Final Element Group Avstängning O2 PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	5.9
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

18 SIF-025 SIL2 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|H2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-025 SIL2 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|H2) Safety Instrumented Function.

18.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL2 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|H2)
SIF Tag: SIF-025
SIF Description: SIL2 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|H2) =>

Close Avstängningsventil 1 H2 till brännare PGH (341SV665 SIL2)
Close Avstängningsventil 2 H2 till brännare PGH (341SV667 SIL2)

Close Avstängningsventil 1 NG till brännare PGH (252SV544 SIL2)
Close Avstängningsventil 2 NG till brännare PGH (252SV546 SIL2)

Trip Reduktionsgaskompressor (346PU101 SIL2)
Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)
Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

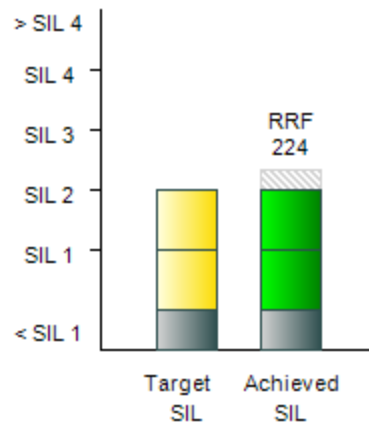
18.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 224.5



18.3 SIL Verification

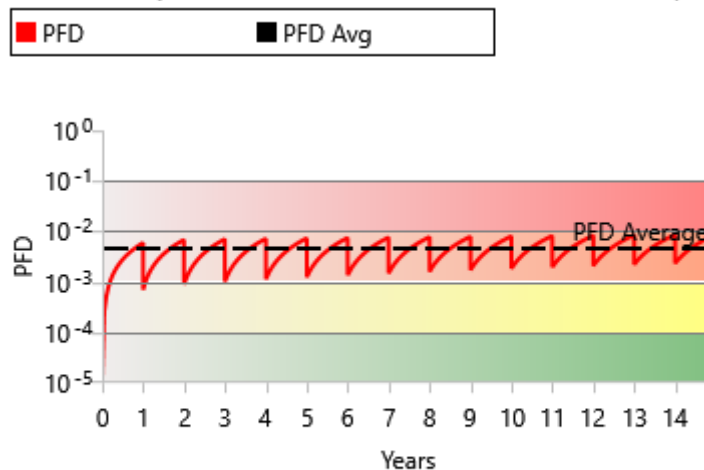
This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-025 SIL2 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|H2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:

SIL2 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|H2)

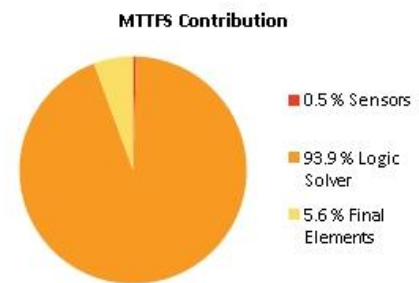
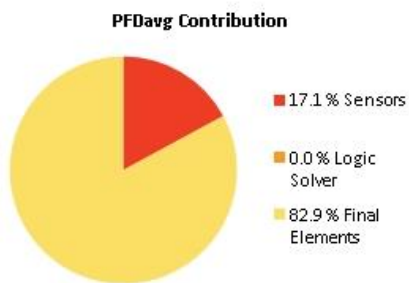


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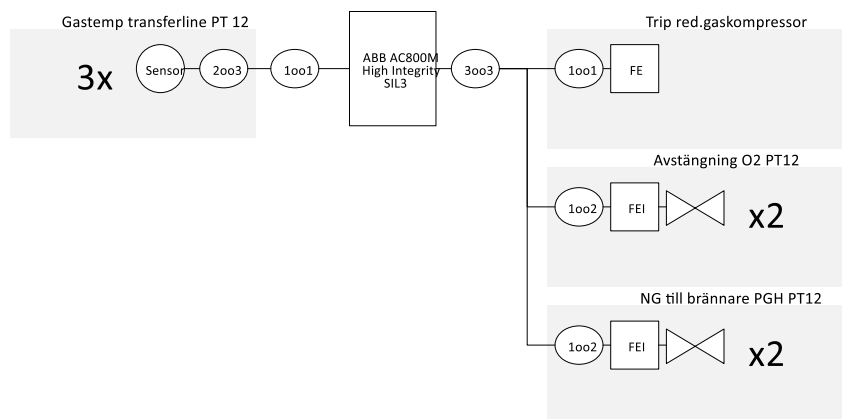
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-025 SIL2 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|H2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 129 Functional Safety Performance

PFD _{avg}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{avg}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
4.45E-03	224.5	2	2	N/A	65.31



The SIF Conceptual Design is modeled as shown in the picture below.



18.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-025 SIL2 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|H2) Safety Instrumented Function is quantified as follows.

Table 130 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
7.62E-04	3	N/A	14107.8	1	2

Number of Sensor group(s): 1
 Voting between groups: 1oo1
 β-factor [%]: N/A

18.3.1.1 Sensor Group 1: Gastemp transferline PT 12

The information and reliability data underneath describe the Gastemp transferline PT 12 sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
 HFT: 1 β-factor [%]: 10

Equipment Leg (each): „
 PR electronics PR5435 / PR5437, Single, 4..20mA
 SERH VERSION: 2019.4.01 2H: - PIU: ✓ AC TYPE: B SIL CAP: -
 Trip Direction: Low, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off
 ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3
 MRT [Hours]: 24
 Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90
 Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Gastemp transferline PT 12.

Table 131 Reliability Data Sensor Group Gastemp transferline PT 12

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
PR electronics PR5435 / PR5437, Single, 4..20mA	4.90E-08 <i>0.00E+00</i>	1.70E-08 <i>0.00E+00</i>	3.58E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>4.90E-08</i>	0.00E+00 <i>0.00E+00</i>	2.70E-08 <i>4.02E-07</i>	1.90E-08 <i>1.90E-08</i>	1.10E-08 <i>1.10E-08</i>	2.13E-07 <i>2.13E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		20.2
								ROUTE 2H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

18.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-025 SIL2 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|H2) Safety Instrumented Function is quantified as follows.

Table 132 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.58E-08	3	N/A	69.52	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β-factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 133 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

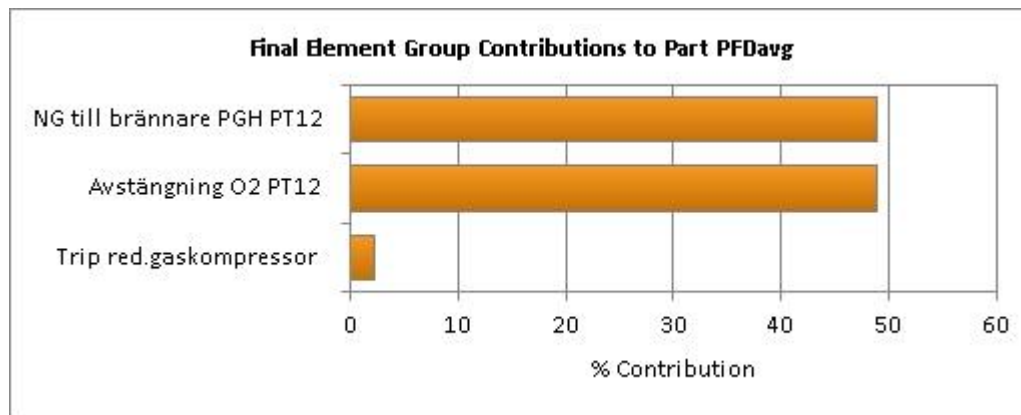
18.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-025 SIL2 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|H2) Safety Instrumented Function is quantified as follows.

Table 134 Final Element Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
3.69E-03	2	N/A	1168.89	0	2

Number of Final Element group(s): 3
Voting between groups: 3oo3
 β -factor [%]:



18.3.3.1 Final Element Group 1: Trip red.gaskompressor

The information and reliability data underneath describe the Trip red.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	ACTYPE	SIL CAP
ASC880 R2 STO		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip red.gaskompressor.

Table 135 Reliability Data Final Element Group Trip red.gaskompressor

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
ASC880 R2 STO		1.08E-08		1.35E-08			
ABB AC800M High	8.53E-08	1.91E-08					4.07E-08

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Integrity SIL3 [DigitalOut]							
					SFF [%]		89.5
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

18.3.3.2 Final Element Group 2: Avstängning O2 PT12

The information and reliability data underneath describe the Avstängning O2 PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical

HFT: 1 β-factor [%]: 10

Equipment Leg

(each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2 PT12.

Table 136 Reliability Data Final Element Group Avstängning O2 PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		5.9
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

18.3.3.3 Final Element Group 3: NG till brännare PGH PT12

The information and reliability data underneath describe the NG till brännare PGH PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical

HFT: 1 β -factor [%]: 10

Equipment Leg
(each): ,

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group NG till brännare PGH PT12.

Table 137 Reliability Data Final Element Group NG till brännare PGH PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		5.9
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

19 SIF-026 SIL3 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|L2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-026 SIL3 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|L2) Safety Instrumented Function.

19.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL3 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|L2)
 SIF Tag: SIF-026
 SIF Description: SIL3 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|L2) =>

Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)

Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

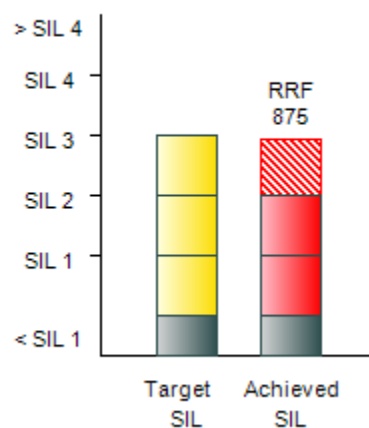
19.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 3 with RRF ≥ 1000

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 875.6



19.3 SIL Verification

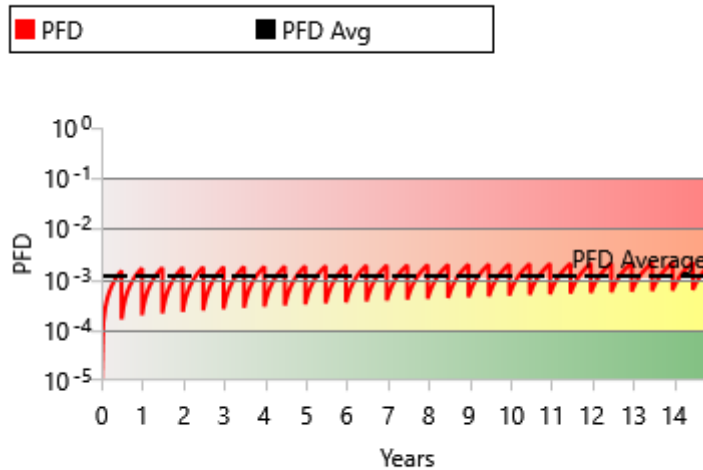
This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-026 SIL3 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|L2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:

SIL3 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|L2)



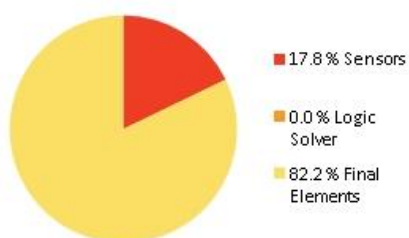
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Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-026 SIL3 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|L2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

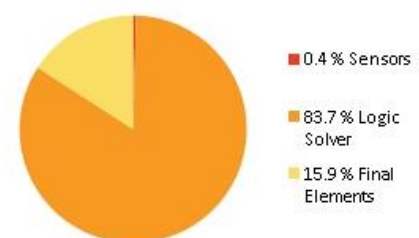
Table 138 Functional Safety Performance

PFDavg	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFDavg	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
1.14E-03	875.6	2	3	N/A	59.58

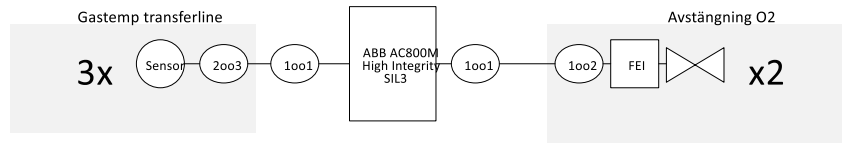
PFDavg Contribution



MTTFS Contribution



The SIF Conceptual Design is modeled as shown in the picture below.



19.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-026 SIL3 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|L2) Safety Instrumented Function is quantified as follows.

Table 139 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.04E-04	3	N/A	14231.71	1	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

19.3.1.1 Sensor Group 1: Gastemp transferline

The information and reliability data underneath describe the Gastemp transferline sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical

HFT: 1 β-factor [%]: 10

Equipment Leg
(each): „

PR electronics PR5435 / PR5437, Single,
4..20mA

SERH VERSION

2_H

PIU

AC TYPE

SIL CAP

2019.4.01

-

✓

B

-

Trip Direction: Low, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]

2013.1.02

-

-

B

3

MRT [Hours]: 24

Proof Test Interval [Months]:

6

Proof Test Coverage [%]:

99

Proof Test Execution:

Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Gastemp transferline.

Table 140 Reliability Data Sensor Group Gastemp transferline

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
PR electronics PR5435 / PR5437, Single, 4..20mA	4.90E-08 <i>0.00E+00</i>	1.70E-08 <i>0.00E+00</i>	3.58E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>4.90E-08</i>	0.00E+00 <i>0.00E+00</i>	2.70E-08 <i>4.02E-07</i>	1.90E-08 <i>1.90E-08</i>	1.10E-08 <i>1.10E-08</i>	2.13E-07 <i>2.13E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		20.2
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

19.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-026 SIL3 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|L2) Safety Instrumented Function is quantified as follows.

Table 141 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.86E-08	3	N/A	71.17	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β-factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 142 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		100.0

19.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-026 SIL3 Gastemp transferline (344TZ316, 344TZ317 & 344TZ318|L2) Safety Instrumented Function is quantified as follows.

Table 143 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
9.39E-04	3	N/A	375.25	1	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

19.3.3.1 Final Element Group 1: Avstängning O2

The information and reliability data underneath describe the Avstängning O2 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 99
Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2.

Table 144 Reliability Data Final Element Group Avstängning O2

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		28.4
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

20 SIF-027 SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305|H2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-027 SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305|H2) Safety Instrumented Function.

20.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305|H2)
SIF Tag: SIF-027
SIF Description: SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305|H2) =>

Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)

Close Avstängningsventil 1 H2 till brännare PGH (341SV665 SIL2)

Close Avstängningsventil 2 H2 till brännare PGH (341SV667 SIL2)

Close Avstängningsventil 1 NG till brännare PGH (252SV544 SIL2)

Close Avstängningsventil 2 NG till brännare PGH (252SV546 SIL2)

Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)

Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

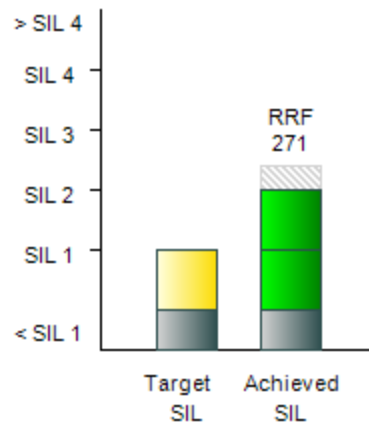
20.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 271.5



20.3 SIL Verification

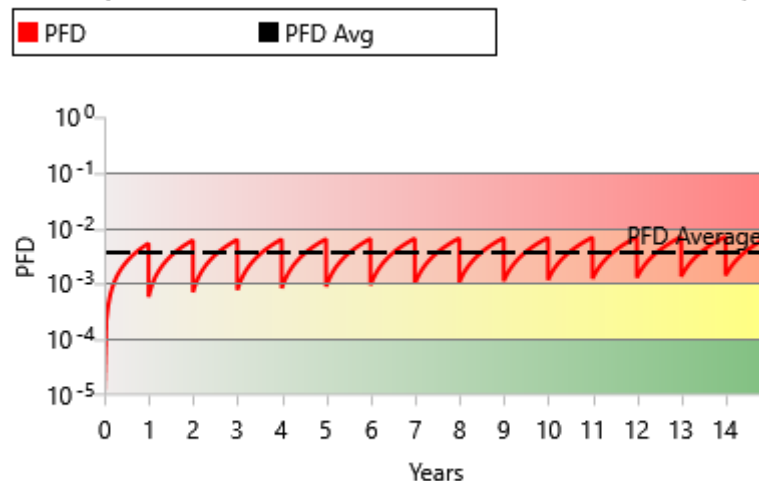
This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-027 SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305|H2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:

SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305|H2)

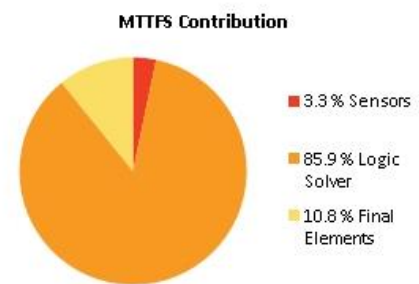
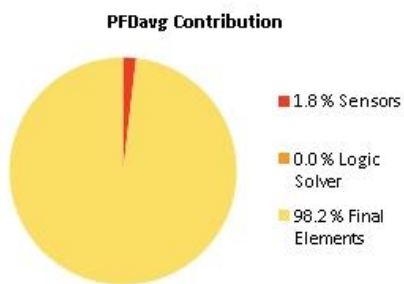


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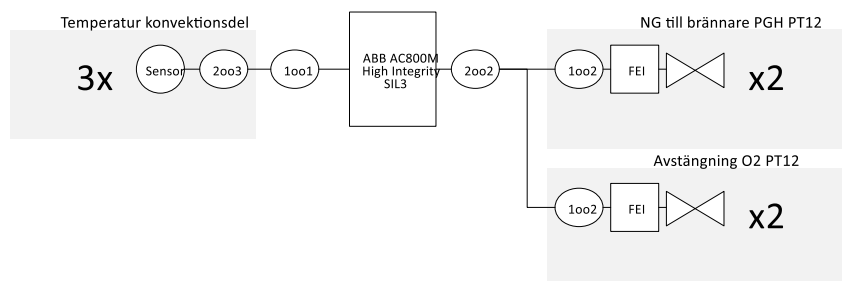
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-027 SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305|H2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 145 Functional Safety Performance

PFDavg	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFDavg	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
3.68E-03	271.5	2	3	N/A	61.17



The SIF Conceptual Design is modeled as shown in the picture below.



20.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-027 SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305|H2) Safety Instrumented Function is quantified as follows.

Table 146 Sensor Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.77E-05	3	N/A	1878.24	1	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
 β -factor [%]: N/A

20.3.1.1 Sensor Group 1: Temperatur konvektionsdel

The information and reliability data underneath describe the Temperatur konvektionsdel sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg
(each): „

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
PR electronics PR5435 / PR5437, Single, 4..20mA	2019.4.01	-	-	B	3

Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
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MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Temperatur konvektionsdel.

Table 147 Reliability Data Sensor Group Temperatur konvektionsdel

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
PR electronics PR5435 / PR5437, Single, 4..20mA	4.90E-08 <i>0.00E+00</i>	1.70E-08 <i>0.00E+00</i>	3.58E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.75E-07</i>	0.00E+00 <i>0.00E+00</i>	2.70E-08 <i>7.60E-08</i>	1.90E-08 <i>1.90E-08</i>	1.10E-08 <i>1.10E-08</i>	2.13E-07 <i>2.13E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	84.9
									ROUTE 2H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

20.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-027 SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305 | H2) Safety Instrumented Function is quantified as follows.

Table 148 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.86E-08	3	N/A	71.17	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β-factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 149 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

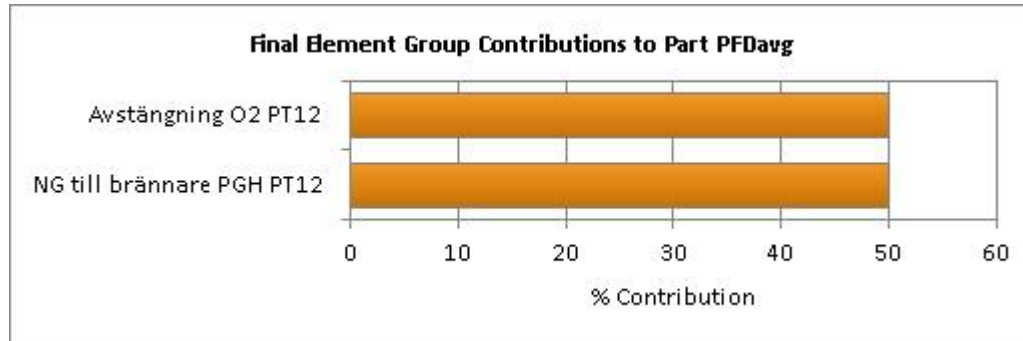
20.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-027 SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305|H2) Safety Instrumented Function is quantified as follows.

Table 150 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
3.62E-03	3	N/A	566.49	1	2

Number of Final Element group(s): 2
Voting between groups: 2oo2
 β -factor [%]:



20.3.3.1 Final Element Group 1: NG till brännare PGH PT12

The information and reliability data underneath describe the NG till brännare PGH PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg
(each): ,

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group NG till brännare PGH PT12.

Table 151 Reliability Data Final Element Group NG till brännare PGH PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
					SFF [%]		5.9
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

20.3.3.2 Final Element Group 2: Avstängning O2 PT12

The information and reliability data underneath describe the Avstängning O2 PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical

HFT: 1 β-factor [%]: 10

Equipment Leg

(each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2 PT12.

Table 152 Reliability Data Final Element Group Avstängning O2 PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		5.9
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

21 SIF-028 SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305|L2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-028 SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305|L2) Safety Instrumented Function.

21.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305 L2)
SIF Tag	SIF-028
SIF Description:	SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305 L2) => Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)

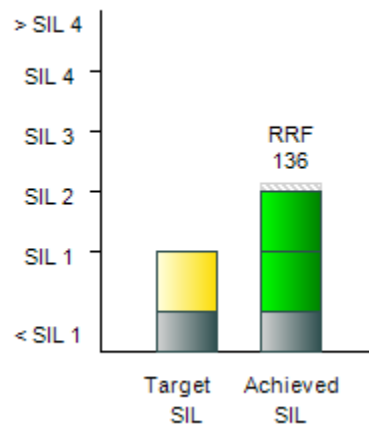
21.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 136.7



21.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-028 SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305|L2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time:	15 years
Startup time:	24 hours

The SIF operates in the Low demand mode.

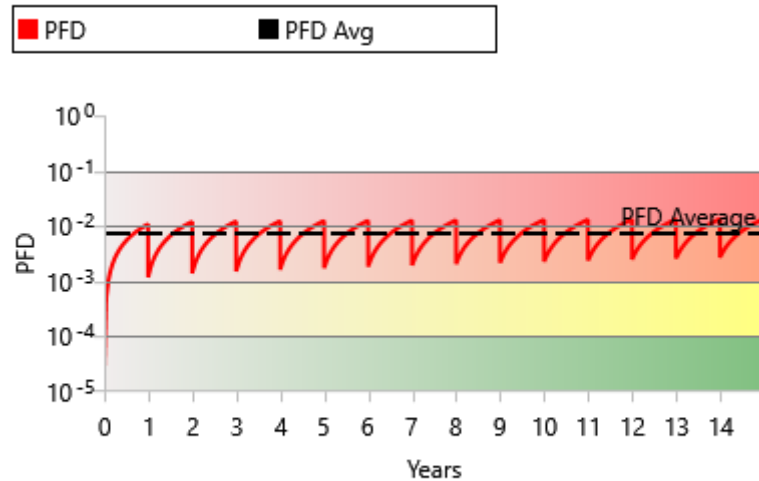
Analyst:

Analysis Date: 10/4/2019

Design References

Comments:

SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305|L2)



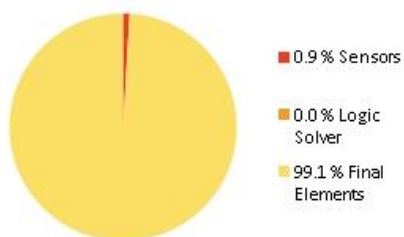
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Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-028 SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305|L2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

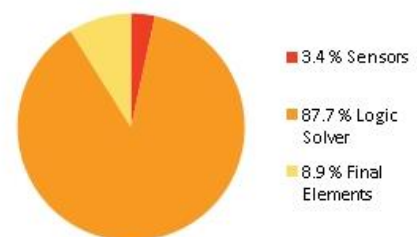
Table 153 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
7.31E-03	136.7	2	2	3	63.55

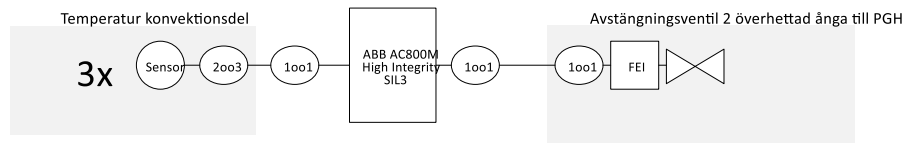
PFD_{avg} Contribution



MTTFS Contribution



The SIF Conceptual Design is modeled as shown in the picture below.



21.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-028 SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305|L2) Safety Instrumented Function is quantified as follows.

Table 154 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.77E-05	3	3	1878.24	1	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

21.3.1.1 Sensor Group 1: Temperatur konvektionsdel

The information and reliability data underneath describe the Temperatur konvektionsdel sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each):	„	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
PR electronics PR5435 / PR5437, Single, 4..20mA		2019.4.01	-	-	B	3
Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off						
ABB AC800M High Integrity SIL3 [AnalogIn]		2013.1.02	-	-	B	3
MRT [Hours]:	24					
Proof Test Interval [Months]:	12		Proof Test Coverage [%]:	99		
Proof Test Execution:	Duration 1 hours					

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Temperatur konvektionsdel.

Table 155 Reliability Data Sensor Group Temperatur konvektionsdel

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
PR electronics PR5435 / PR5437, Single, 4..20mA	4.90E-08 <i>0.00E+00</i>	1.70E-08 <i>0.00E+00</i>	3.58E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.75E-07</i>	0.00E+00 <i>0.00E+00</i>	2.70E-08 <i>7.60E-08</i>	1.90E-08 <i>1.90E-08</i>	1.10E-08 <i>1.10E-08</i>	2.13E-07 <i>2.13E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		84.9
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

21.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-028 SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305 | L2) Safety Instrumented Function is quantified as follows.

Table 156 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	3	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	3
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β-factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 157 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

21.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-028 SIL1 Temperatur konvektionsdel (344TZ303, 344TZ304 & 344TZ305 | L2) Safety Instrumented Function is quantified as follows.

Table 158 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
7.25E-03	2	3	712.45	0	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

21.3.3.1 Final Element Group 1: Avstängningsventil 2 överhettad ånga till PGH

The information and reliability data underneath describe the Avstängningsventil 2 överhettad ånga till PGH final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängningsventil 2 överhettad ånga till PGH.

Table 159 Reliability Data Final Element Group Avstängningsventil 2 överhettad ånga till PGH

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				5.55E-07			2.02E-06
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		11.5
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

22 SIF-029 SIL1 Temperatur strålningssdel (344TZ310, 344TZ311 & 344TZ312|H2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-029 SIL1 Temperatur strålningssdel (344TZ310, 344TZ311 & 344TZ312|H2) Safety Instrumented Function.

22.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL1 Temperatur strålningssdel (344TZ310, 344TZ311 & 344TZ312|H2)
SIF Tag: SIF-029
SIF Description: SIL1 Temperatur strålningssdel (344TZ310, 344TZ311 & 344TZ312|H2) =>

Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)
Close Avstängningsventil 1 H2 till brännare PGH (341SV665 SIL2)
Close Avstängningsventil 2 H2 till brännare PGH (341SV667 SIL2)

Close Avstängningsventil 1 NG till brännare PGH (252SV544 SIL2)
Close Avstängningsventil 2 NG till brännare PGH (252SV546 SIL2)

Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)
Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

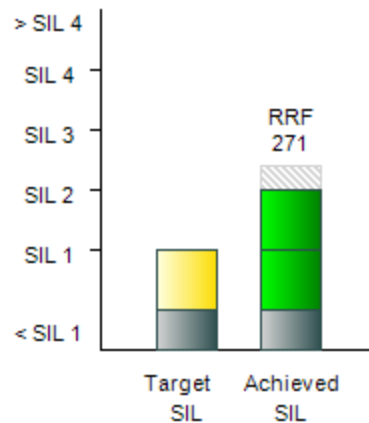
22.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 271.5



22.3 SIL Verification

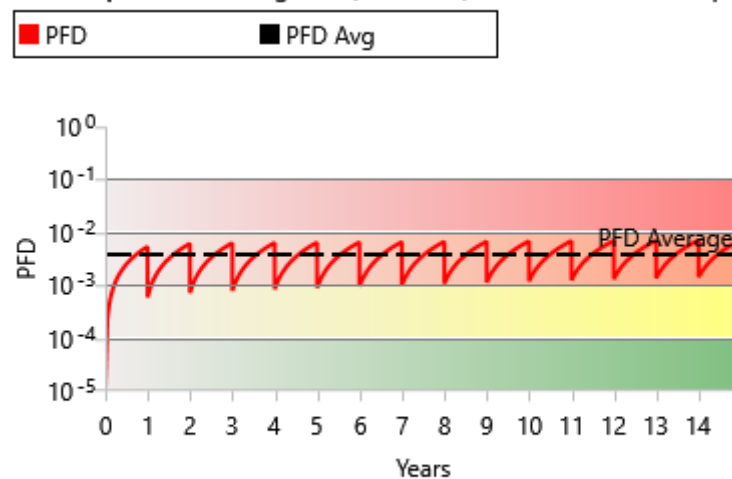
This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-029 SIL1 Temperatur strålningsdel (344TZ310, 344TZ311 & 344TZ312|H2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:

SIL1 Temperatur strålningsdel (344TZ310, 344TZ311 & 344TZ312|H2)

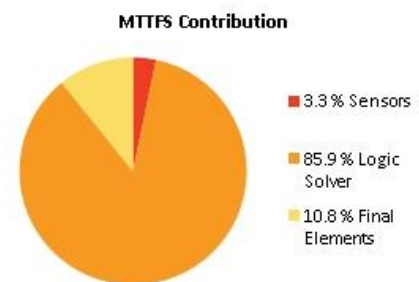
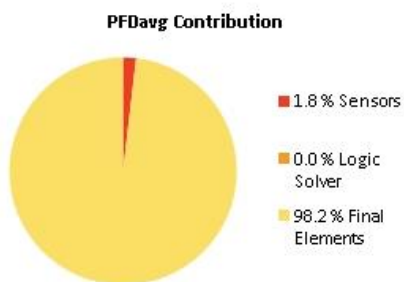


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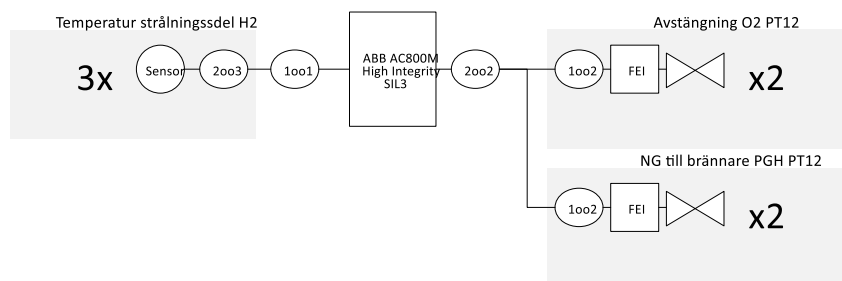
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-029 SIL1 Temperatur strålningsdel (344TZ310, 344TZ311 & 344TZ312|H2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 160 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
3.68E-03	271.5	2	3	N/A	61.17



The SIF Conceptual Design is modeled as shown in the picture below.



22.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-029 SIL1 Temperatur strålningsdel (344TZ310, 344TZ311 & 344TZ312|H2) Safety Instrumented Function is quantified as follows.

Table 161 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.77E-05	3	N/A	1878.24	1	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
 β -factor [%]: N/A

22.3.1.1 Sensor Group 1: Temperatur strålningsdel H2

The information and reliability data underneath describe the Temperatur strålningsdel H2 sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg

(each):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
PR electronics PR5435 / PR5437, Single, 4..20mA	2019.4.01	-	-	B	3

Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
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MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Temperatur strålningsdel H2.

Table 162 Reliability Data Sensor Group Temperatur strålningsdel H2

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
PR electronics PR5435 / PR5437, Single, 4..20mA	4.90E-08 <i>0.00E+00</i>	1.70E-08 <i>0.00E+00</i>	3.58E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.75E-07</i>	0.00E+00 <i>0.00E+00</i>	2.70E-08 <i>7.60E-08</i>	1.90E-08 <i>1.90E-08</i>	1.10E-08 <i>1.10E-08</i>	2.13E-07 <i>2.13E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	84.9
									ROUTE 2H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

22.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-029 SIL1 Temperatur strålningsdel (344TZ310, 344TZ311 & 344TZ312 | H2) Safety Instrumented Function is quantified as follows.

Table 163 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.86E-08	3	N/A	71.17	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β-factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 164 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

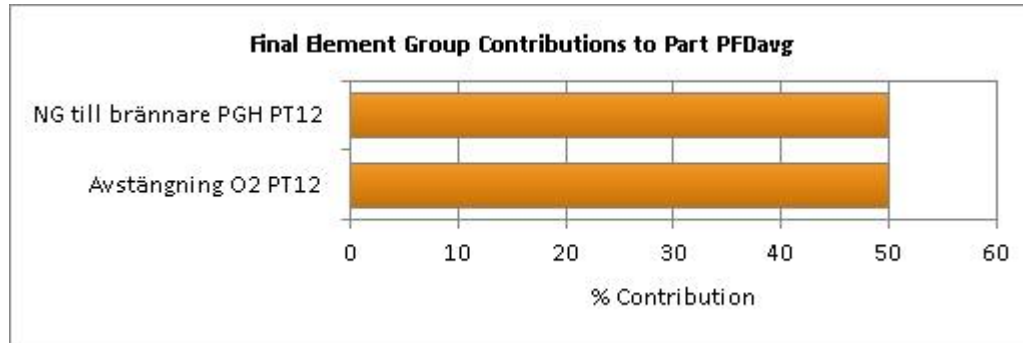
22.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-029 SIL1 Temperatur strålingssdel (344TZ310, 344TZ311 & 344TZ312|H2) Safety Instrumented Function is quantified as follows.

Table 165 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
3.62E-03	3	N/A	566.49	1	2

Number of Final Element group(s): 2
Voting between groups: 2oo2
 β -factor [%]:



22.3.3.1 Final Element Group 1: Avstängning O2 PT12

The information and reliability data underneath describe the Avstängning O2 PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg (each):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2 PT12.

Table 166 Reliability Data Final Element Group Avstängning O2 PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
					SFF [%]		5.9
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

22.3.3.2 Final Element Group 2: NG till brännare PGH PT12

The information and reliability data underneath describe the NG till brännare PGH PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical

HFT: 1 β -factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group NG till brännare PGH PT12.

Table 167 Reliability Data Final Element Group NG till brännare PGH PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		5.9
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

23 SIF-031 SIL1 Tryck förbränningskammare (344PZ306, 344PZ307 & 344PZ308|H2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-031 SIL1 Tryck förbränningskammare (344PZ306, 344PZ307 & 344PZ308|H2) Safety Instrumented Function.

23.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL1 Tryck förbränningskammare (344PZ306, 344PZ307 & 344PZ308|H2)
SIF Tag: SIF-031
SIF Description: SIL1 Tryck förbränningskammare (344PZ306, 344PZ307 & 344PZ308|H2) =>

Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)

Close Avstängningsventil 1 H2 till brännare PGH (341SV665 SIL2)

Close Avstängningsventil 2 H2 till brännare PGH (341SV667 SIL2)

Close Avstängningsventil 1 NG till brännare PGH (252SV544 SIL2)

Close Avstängningsventil 2 NG till brännare PGH (252SV546 SIL2)

Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)

Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

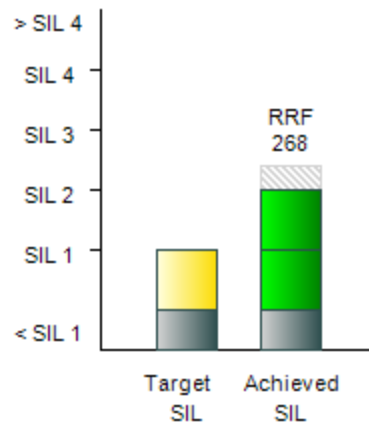
23.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 268.5



23.3 SIL Verification

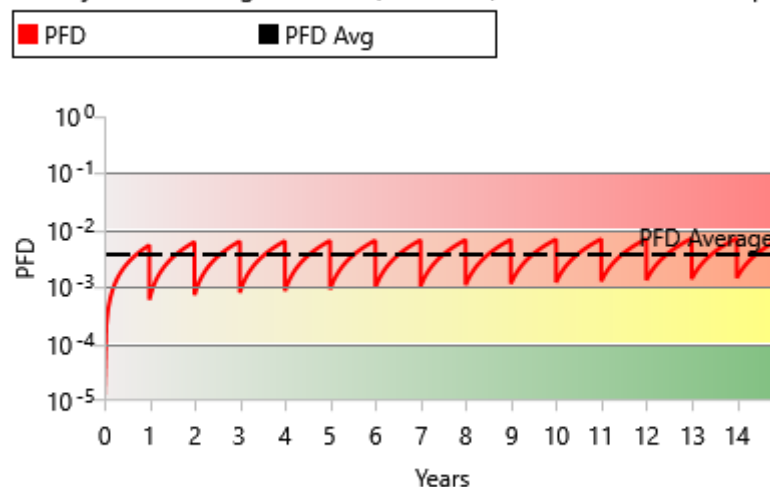
This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-031 SIL1 Tryck förbränningskammare (344PZ306, 344PZ307 & 344PZ308|H2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:

SIL1 Tryck förbränningskammare (344PZ306, 344PZ307 & 344PZ308|H2)

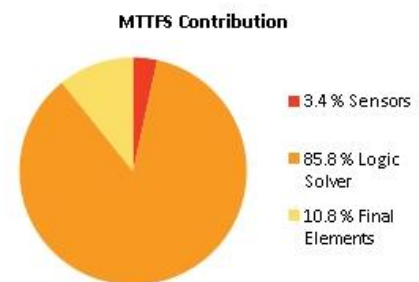
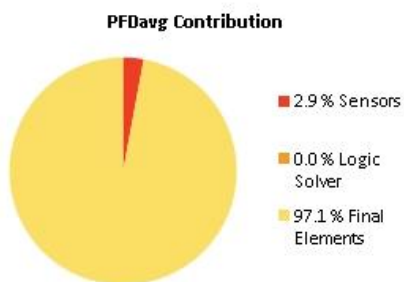


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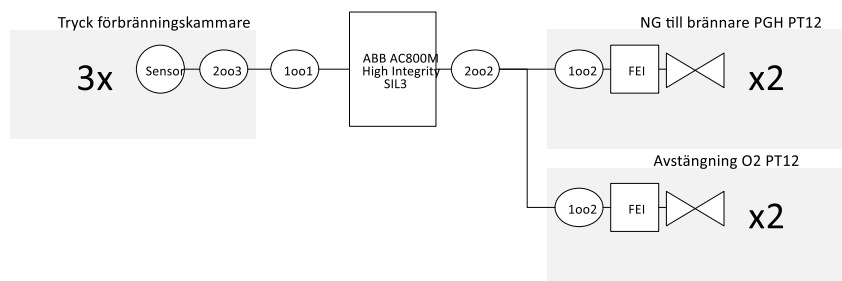
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-031 SIL1 Tryck förbränningskammare (344PZ306, 344PZ307 & 344PZ308|H2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 168 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
3.72E-03	268.5	2	3	N/A	61.06



The SIF Conceptual Design is modeled as shown in the picture below.



23.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-031 SIL1 Tryck förbränningskammare (344PZ306, 344PZ307 & 344PZ308|H2) Safety Instrumented Function is quantified as follows.

Table 169 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.08E-04	3	N/A	1776.27	1	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
 β -factor [%]: N/A

23.3.1.1 Sensor Group 1: Tryck förbränningskammare

The information and reliability data underneath describe the Tryck förbränningskammare sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg

(each):	„	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Deltabar S PMD75, FMD77, FMD78		2006.2.02	-	-	B	2

Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
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MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Tryck förbränningskammare.

Table 170 Reliability Data Sensor Group Tryck förbränningskammare

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Deltabar S PMD75, FMD77, FMD78	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.96E-07</i>	0.00E+00 <i>0.00E+00</i>	6.89E-08 <i>1.21E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	4.30E-07 <i>4.30E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	78.0
									ROUTE 2 _H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

23.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-031 SIL1 Tryck förbränningskammare (344PZ306, 344PZ307 & 344PZ308|H2) Safety Instrumented Function is quantified as follows.

Table 171 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.86E-08	3	N/A	71.17	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	SERH VERSION	PIU	SIL Cap
ABB AC800M High Integrity SIL3	2013.1.02	-	N/A
Logic Solver Voting:	Architecture Type:	B	
HFT:	β-factor [%]:	1	
MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 172 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

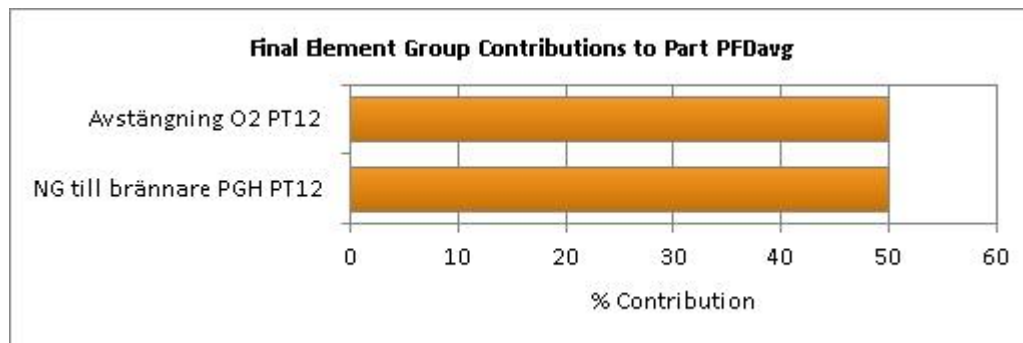
23.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-031 SIL1 Tryck förbränningskammare (344PZ306, 344PZ307 & 344PZ308|H2) Safety Instrumented Function is quantified as follows.

Table 173 Final Element Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
3.62E-03	3	N/A	566.49	1	2

Number of Final Element group(s): 2
Voting between groups: 2oo2
 β -factor [%]:



23.3.3.1 Final Element Group 1: NG till brännare PGH PT12

The information and reliability data underneath describe the NG till brännare PGH PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg (each):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group NG till brännare PGH PT12.

Table 174 Reliability Data Final Element Group NG till brännare PGH PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		5.9
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

23.3.3.2 Final Element Group 2: Avstängning O2 PT12

The information and reliability data underneath describe the Avstängning O2 PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical

HFT: 1 β -factor [%]: 10

Equipment Leg

(each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2 PT12.

Table 175 Reliability Data Final Element Group Avstängning O2 PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		5.9
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

24 SIF-032 SIL2 Flamvakt (344YS314|SG1)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-032 SIL2 Flamvakt (344YS314|SG1) Safety Instrumented Function.

24.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL2 Flamvakt (344YS314|SG1)
SIF Tag: SIF-032
SIF Description: SIL2 Flamvakt (344YS314|SG1) =>

Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)
Close Avstängningsventil 1 H2 till brännare PGH (341SV665 SIL2)
Close Avstängningsventil 2 H2 till brännare PGH (341SV667 SIL2)

Close Avstängningsventil 1 NG till brännare PGH (252SV544 SIL2)
Close Avstängningsventil 2 H2 till brännare PGH (341SV667 SIL2)

Close Avstängningsventil 1 NG till brännare PGH (252SV544 SIL2)
Close Avstängningsventil 2 NG till brännare PGH (252SV546 SIL2)

Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)
Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

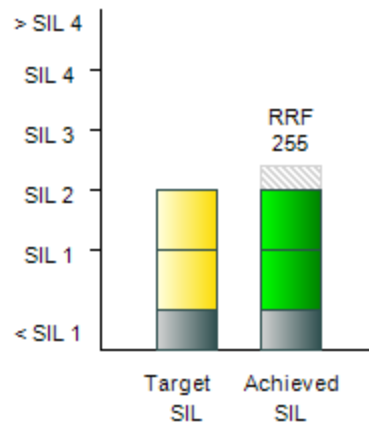
24.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 255.2



24.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-032 SIL2 Flamvakt (344YS314|SG1) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

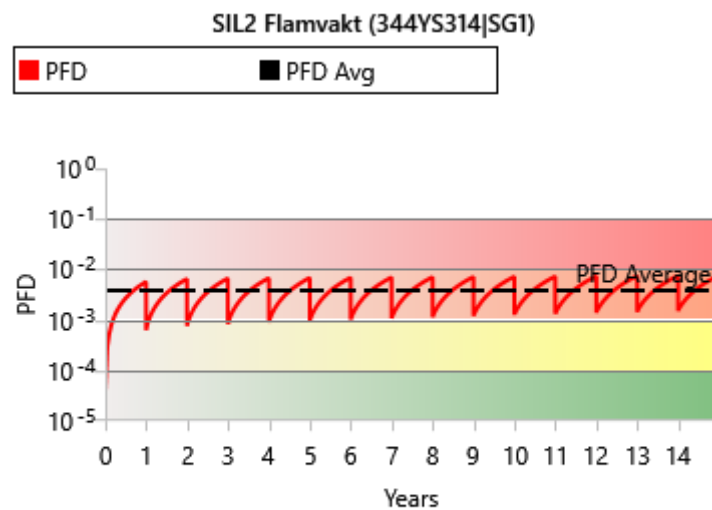
The SIF operates in the Low demand mode.

Analyst:

Analysis Date: 10/4/2019

Design References

Comments:



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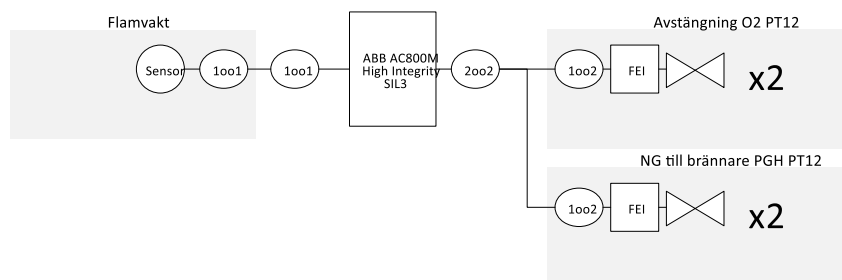
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-032 SIL2 Flamvakt (344YS314|SG1) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 176 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
3.92E-03	255.2	2	2	N/A	63.56



The SIF Conceptual Design is modeled as shown in the picture below.



24.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-032 SIL2 Flamvakt (344YS314|SG1) Safety Instrumented Function is quantified as follows.

Table 177 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
3.03E-04	2	N/A	∞	0	2

Number of Sensor group(s): 1

Voting between groups: 1001
 β -factor [%]: N/A

24.3.1.1 Sensor Group 1: Flamvakt

The information and reliability data underneath describe the Flamvakt sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

DURAG D-LX 100UA-C1

Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Flamvakt.

Table 178 Reliability Data Sensor Group Flamvakt

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
DURAG D-LX 100UA-C1	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.83E-06 <i>1.83E-06</i>	4.32E-08 <i>4.32E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>
								SFF [%]		97.7
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

24.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-032 SIL2 Flamvakt (344YS314|SG1) Safety Instrumented Function is quantified as follows.

Table 179 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.04E-08	3	N/A	71.59	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β -factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 180 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	100.0

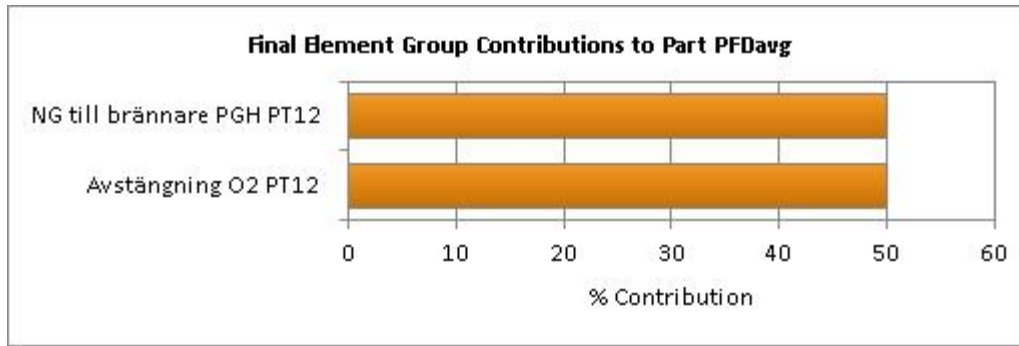
24.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-032 SIL2 Flamvakt (344YS314|SG1) Safety Instrumented Function is quantified as follows.

Table 181 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
3.62E-03	3	N/A	566.49	1	2

Number of Final Element group(s): 2
Voting between groups: 2oo2
 β -factor [%]:



24.3.3.1 Final Element Group 1: Avstängning O2 PT12

The information and reliability data underneath describe the Avstängning O2 PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2 PT12.

Table 182 Reliability Data Final Element Group Avstängning O2 PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	5.9
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

24.3.3.2 Final Element Group 2: NG till brännare PGH PT12

The information and reliability data underneath describe the NG till brännare PGH PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical

HFT: 1 β -factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group NG till brännare PGH PT12.

Table 183 Reliability Data Final Element Group NG till brännare PGH PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
SFF [%]							5.9
ROUTE 2 _H COMPLIANT							-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

25 SIF-033 SIL2 Gastemp transferline efter O2-inj (344TZ322, 344TZ323 & 344TZ324|H2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-033 SIL2 Gastemp transferline efter O2-inj (344TZ322, 344TZ323 & 344TZ324|H2) Safety Instrumented Function.

25.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Gastemp transferline efter O2-inj (344TZ322, 344TZ323 & 344TZ324 H2)
SIF Tag	SIF-033
SIF Description:	SIL2 Gastemp transferline efter O2-inj (344TZ322, 344TZ323 & 344TZ324 H2) => Close Avstängningsventil 1 O2 inj. (251SV555 SIL2) Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

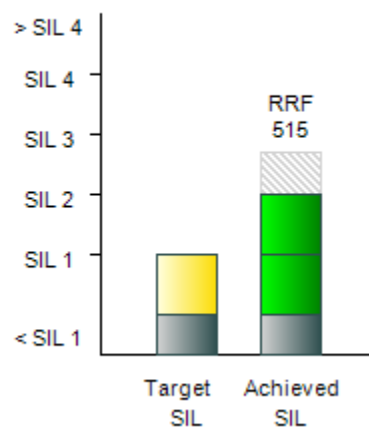
25.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 515.2



25.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-033 SIL2 Gastemp transferline efter O2-inj (344TZ322, 344TZ323 & 344TZ324|H2) Safety Instrumented Function. In order to

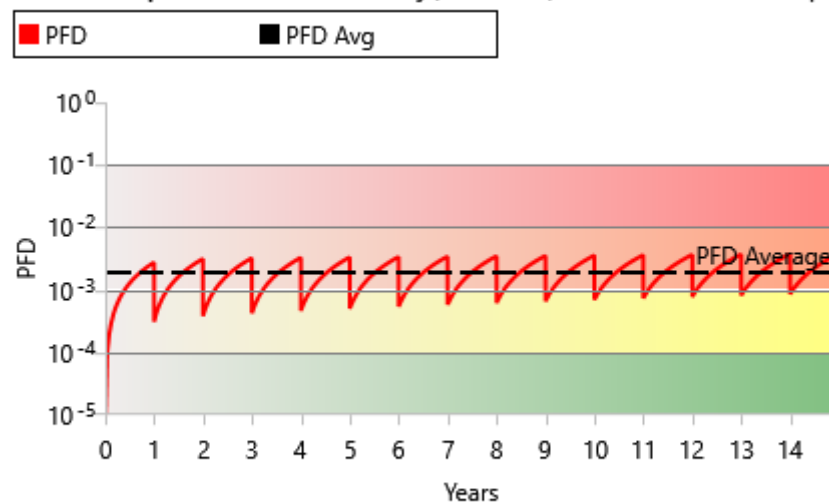
perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:

SIL2 Gastemp transferline after O2-inj (344TZ322, 344TZ323 & 344TZ324|H2)



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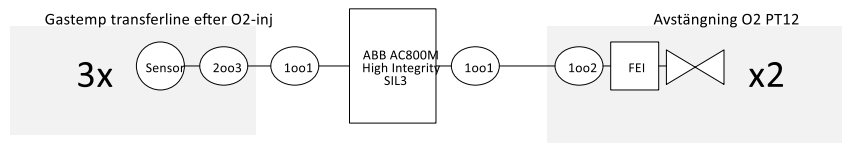
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-033 SIL2 Gastemp transferline after O2-inj (344TZ322, 344TZ323 & 344TZ324|H2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 184 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
1.94E-03	515.2	2	3	N/A	58.78



The SIF Conceptual Design is modeled as shown in the picture below.



25.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-033 SIL2 Gastemp transferline after O2-inj (344TZ322, 344TZ323 & 344TZ324|H2) Safety Instrumented Function is quantified as follows.

Table 185 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.32E-04	3	N/A	1775.15	1	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

25.3.1.1 Sensor Group 1: Gastemp transferline after O2-inj

The information and reliability data underneath describe the Gastemp transferline after O2-inj sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
PR electronics PR5435 / PR5437, Single,	2019.4.01	-	-	B	3

4..20mA

Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Gastemp transferline after O2-inj.

Table 186 Reliability Data Sensor Group Gastemp transferline after O2-inj

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
PR electronics PR5435 / PR5437, Single, 4..20mA	4.90E-08 <i>0.00E+00</i>	1.70E-08 <i>0.00E+00</i>	3.58E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.75E-07</i>	0.00E+00 <i>0.00E+00</i>	2.70E-08 <i>7.60E-08</i>	1.90E-08 <i>1.90E-08</i>	1.10E-08 <i>1.10E-08</i>	2.13E-07 <i>2.13E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]		84.9	
							ROUTE 2H COMPLIANT		-	

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

25.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-033 SIL2 Gastemp transferline after O2-inj (344TZ322, 344TZ323 & 344TZ324|H2) Safety Instrumented Function is quantified as follows.

Table 187 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment: ABB AC800M High Integrity SIL3

SERH VERSION 2013.1.02 PIU - SIL Cap N/A

Logic Solver Voting: Architecture Type: B

HFT:	1	β -factor [%]:	1
MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 188 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

25.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-033 SIL2 Gastemp transferline after O2-inj (344TZ322, 344TZ323 & 344TZ324 | H2) Safety Instrumented Function is quantified as follows.

Table 189 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.81E-03	3	N/A	377.66	1	2

Number of Final Element group(s):	1
Voting between groups:	1001
β -factor [%]:	N/A

25.3.3.1 Final Element Group 1: Avstängning O2 PT12

The information and reliability data underneath describe the Avstängning O2 PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	1002	Voting type:	Identical
HFT:	1	β -factor [%]:	10

Equipment Leg (each):	SERH VERSION	2H	PIU	ACTYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3

Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Leak Test; Duration 1 hours		

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2 PT12.

Table 190 Reliability Data Final Element Group Avstängning O2 PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	5.9
						ROUTE 2h COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

26 SIF-034 SIL3 Gastemp transferline efter O2-inj (344TZ322, 344TZ323 & 344TZ324|L2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-034 SIL3 Gastemp transferline efter O2-inj (344TZ322, 344TZ323 & 344TZ324|L2) Safety Instrumented Function.

26.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL3 Gastemp transferline efter O2-inj (344TZ322, 344TZ323 & 344TZ324|L2)

SIF Tag SIF-034

SIF Description: Gastemp transferline efter O2-inj (344TZ322, 344TZ323 & 344TZ324|L2) =>

Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)

Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

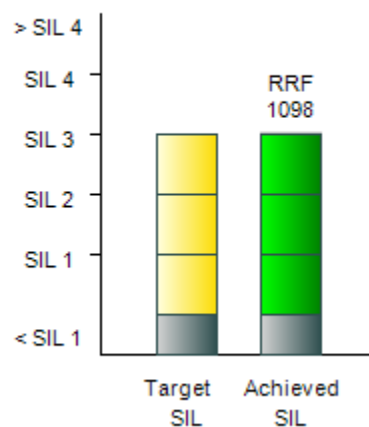
26.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 3 with RRF ≥ 1000

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 3 with RRF = 1098.6



26.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-034 SIL3 Gastemp transferline efter O2-inj (344TZ322, 344TZ323 & 344TZ324|L2) Safety Instrumented Function. In order to

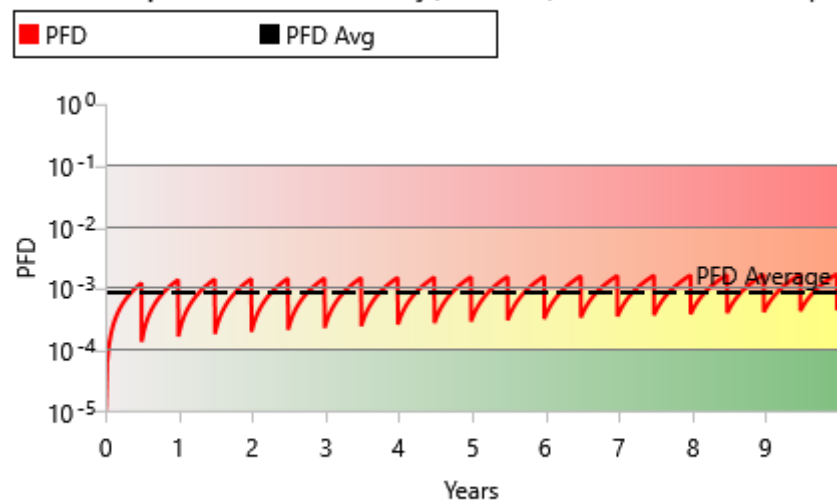
perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 10 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:

SIL3 Gastemp transferline efter O2-inj (344TZ322, 344TZ323 & 344TZ324|L2)



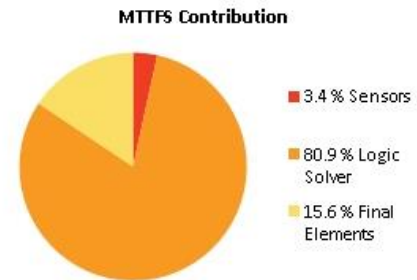
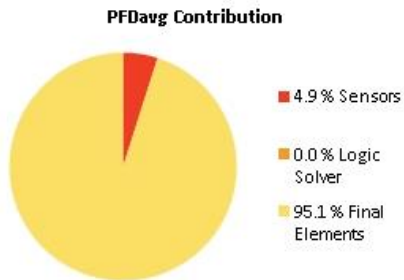
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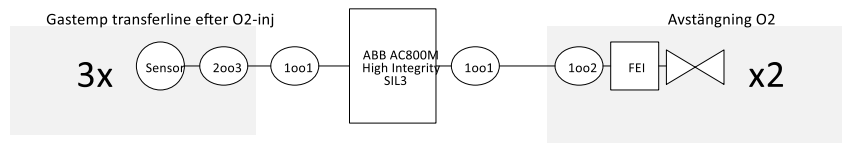
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-034 SIL3 Gastemp transferline efter O2-inj (344TZ322, 344TZ323 & 344TZ324|L2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 191 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
9.10E-04	1098.6	3	3	N/A	58.64



The SIF Conceptual Design is modeled as shown in the picture below.



26.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-034 SIL3 Gastemp transferline efter O2-inj (344TZ322, 344TZ323 & 344TZ324|L2) Safety Instrumented Function is quantified as follows.

Table 192 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.48E-05	3	N/A	1705.14	1	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

26.3.1.1 Sensor Group 1: Gastemp transferline efter O2-inj

The information and reliability data underneath describe the Gastemp transferline efter O2-inj sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each):	„	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
PR electronics PR5435 / PR5437, Single,		2019.4.01	-	-	B	3

4..20mA

Trip Direction: Low, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 90

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Gastemp transferline after O2-inj.

Table 193 Reliability Data Sensor Group Gastemp transferline after O2-inj

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
PR electronics PR5435 / PR5437, Single, 4..20mA	4.90E-08 <i>0.00E+00</i>	1.70E-08 <i>0.00E+00</i>	3.58E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>4.07E-07</i>	0.00E+00 <i>0.00E+00</i>	2.70E-08 <i>4.40E-08</i>	1.90E-08 <i>1.90E-08</i>	1.10E-08 <i>1.10E-08</i>	2.13E-07 <i>2.13E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	91.3
									ROUTE 2H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

26.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-034 SIL3 Gastemp transferline after O2-inj (344TZ322, 344TZ323 & 344TZ324|L2) Safety Instrumented Function is quantified as follows.

Table 194 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.92E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:		SERH VERSION	PIU	SIL Cap
	ABB AC800M High Integrity SIL3	2013.1.02	-	N/A

Logic Solver Voting: Architecture Type: B

HFT:	1	β -factor [%]:	1
MRT [Hours]:	24		
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 195 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

26.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-034 SIL3 Gastemp transferline after O2-inj (344TZ322, 344TZ323 & 344TZ324|L2) Safety Instrumented Function is quantified as follows.

Table 196 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
8.65E-04	3	N/A	375.04	1	2

Number of Final Element group(s):	1
Voting between groups:	1oo1
β -factor [%]:	N/A

26.3.3.1 Final Element Group 1: Avstängning O2

The information and reliability data underneath describe the Avstängning O2 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	1oo2	Voting type:	Identical
HFT:	1	β -factor [%]:	10

Equipment Leg (each):	SERH VERSION	2H	PIU	ACTYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3

Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]:	24		
Proof Test Interval [Months]:	6	Proof Test Coverage [%]:	99
Proof Test Execution:	Leak Test; Duration 1 hours		

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2.

Table 197 Reliability Data Final Element Group Avstängning O2

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	5.9
						ROUTE 2H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

27 SIF-035 SIL2 Difftrycksmätning över syreinjektionslans (344XI340_DEV|H2/L2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-035 SIL2 Difftrycksmätning över syreinjektionslans (344XI340_DEV|H2/L2) Safety Instrumented Function.

27.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL2 Difftrycksmätning över syreinjektionslans (344XI340_DEV|H2/L2)
 SIF Tag: SIF-035
 SIF Description: Eventuellt delar vi upp denna i två.

SIL2 Difftrycksmätning över syreinjektionslans (344XI340_DEV|H2/L2) =>

Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)

Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

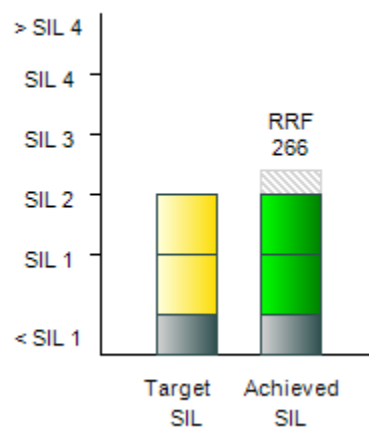
27.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 266.9



27.3 SIL Verification

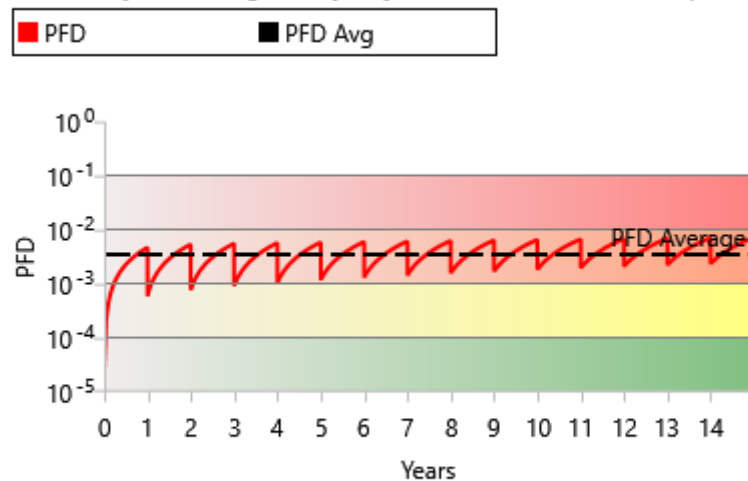
This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-035 SIL2 Difftrycksmätning över syreinjektionslans (344XI340_DEV|H2/L2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:

SIL2 Difftrycksmätning över syreinjektionslans (344XI340_DEV|H2/L2)



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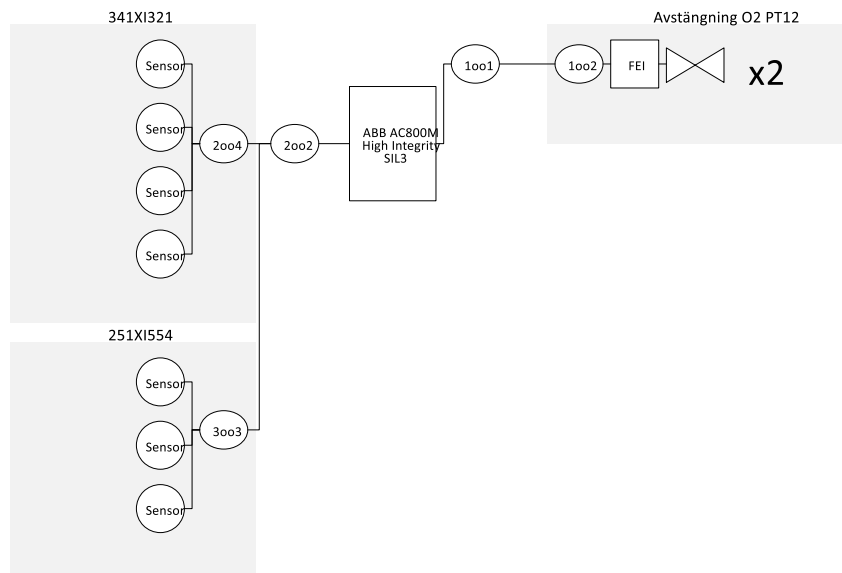
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-035 SIL2 Difftrycksmätning över syreinjektionslans (344XI340_DEV|H2/L2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 198 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
3.75E-03	266.9	2	2	N/A	60.43



The SIF Conceptual Design is modeled as shown in the picture below.



27.3.1 Sensor Part Configuration

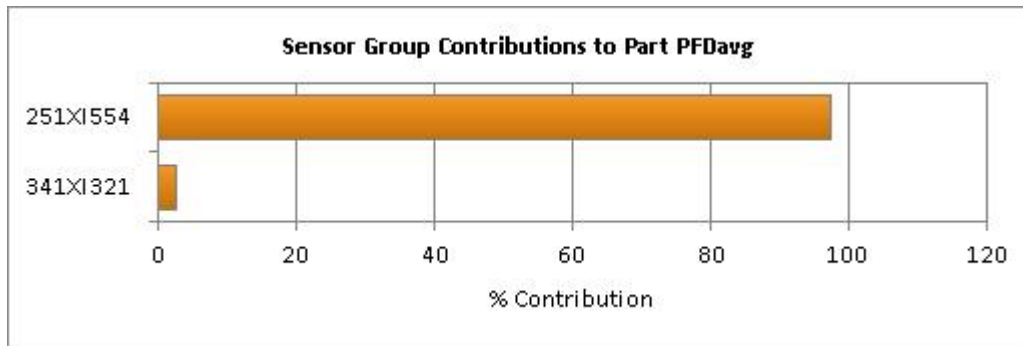
The functional safety and spurious trip behavior of the sensor part of the SIF-035 SIL2 Difftrycksmätning över syreinjektionslans (344XI340_DEV | H2/L2) Safety Instrumented Function is quantified as follows.

Table 199 Sensor Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.94E-03	2	N/A	58854.58	0	2

Number of Sensor group(s): 2

Voting between groups: 2oo2
 β -factor [%]: 10



27.3.1.1 Sensor Group 1: 341XI321

The information and reliability data underneath describe the 341XI321 sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo4 Voting type: Diverse
 HFT: 2 β -factor [%]: 10

Equipment Leg (1):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Deltabar S PMD75, FMD77, FMD78	2006.2.02	-	-	B	2

Trip Direction: None, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
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Equipment Leg (2):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser iTemp TMT 182	2006.2.02	-	-	B	-

Trip Direction: None, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
---	-----------	---	---	---	---

Equipment Leg (3):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: None, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
---	-----------	---	---	---	---

Equipment Leg (4):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: None, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
---	-----------	---	---	---	---

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 95.9

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group 341XI321.

Table 200 Reliability Data Sensor Group 341XI321

Component	Failure Rates [1/HR]									
	Fail Low	Fail High	Fail Det.	SD	SU	DD	DU	AD	AU	NE
251PD554										
Endress+Hauser Deltabar S PMD75, FMD77, FMD78	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>5.21E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.96E-07</i>	6.89E-08 <i>6.89E-08</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	4.30E-07 <i>4.30E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]		87.5	
							Route 2 _H Compliant		-	
251TI554										
Endress+Hauser iTemp TMT 182	1.66E-08 <i>0.00E+00</i>	9.16E-08 <i>0.00E+00</i>	9.18E-08 <i>0.00E+00</i>	0.00E+00 <i>1.66E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>1.83E-07</i>	1.07E-07 <i>1.07E-07</i>	0.00E+00 <i>0.00E+00</i>	1.22E-09 <i>1.22E-09</i>	9.78E-08 <i>9.78E-08</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]		67.7	
							Route 2 _H Compliant		-	
251PZ312										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>5.21E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.96E-07</i>	6.53E-08 <i>6.53E-08</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]		88.0	
							Route 2 _H Compliant		-	
251PZ313										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>5.21E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.96E-07</i>	6.53E-08 <i>6.53E-08</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]		88.0	
							Route 2 _H Compliant		-	

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

27.3.1.2 Sensor Group 2: 251XI554

The information and reliability data underneath describe the 251XI554 sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 3oo3 Voting type: Diverse

HFT: 0 β -factor [%]: 10

Equipment Leg (1):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser iTemp TMT 182	2006.2.02	-	-	B	-

Trip Direction: None, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
Equipment Leg (2):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP

Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2
---	-----------	---	---	---	---

Trip Direction: None, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
Equipment Leg (3):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP

Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2
---	-----------	---	---	---	---

Trip Direction: None, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
---	-----------	---	---	---	---

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 94.9

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group 251XI554.

Table 201 Reliability Data Sensor Group 251XI554

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
251TI554										
Endress+Hauser iTemp TMT 182	1.66E-08 <i>0.00E+00</i>	9.16E-08 <i>0.00E+00</i>	9.18E-08 <i>0.00E+00</i>	0.00E+00 <i>1.66E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>1.83E-07</i>	1.07E-07 <i>1.07E-07</i>	0.00E+00 <i>0.00E+00</i>	1.22E-09 <i>1.22E-09</i>	9.78E-08 <i>9.78E-08</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]		67.7	
							ROUTE 2 _H COMPLIANT		-	
251PZ320										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>5.21E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.96E-07</i>	6.53E-08 <i>6.53E-08</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]		88.0	
							ROUTE 2 _H COMPLIANT		-	
344PZ321										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>5.21E-08</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.96E-07</i>	6.53E-08 <i>6.53E-08</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]		88.0	
							ROUTE 2 _H COMPLIANT		-	

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

27.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-035 SIL2 Difftrycksmätning över syreinjektionslans (344XI340_DEV|H2/L2) Safety Instrumented Function is quantified as follows.

Table 202 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.84E-08	3	N/A	72.03	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β-factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 203 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

27.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-035 SIL2 Difftrycksmätning över syreinjektionslans (344XI340_DEV|H2/L2) Safety Instrumented Function is quantified as follows.

Table 204 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.81E-03	3	N/A	377.66	1	2

Number of Final Element group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

27.3.3.1 Final Element Group 1: Avstängning O2 PT12

The information and reliability data underneath describe the Avstängning O2 PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β-factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2 PT12.

Table 205 Reliability Data Final Element Group Avstängning O2 PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		5.9
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

28 SIF-036 SIL1 Gastryck N2 (253PZ362, 253PZ363 & 253PZ364|L2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-036 SIL1 Gastryck N2 (253PZ362, 253PZ363 & 253PZ364|L2) Safety Instrumented Function.

28.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL1 Gastryck N2 (253PZ362, 253PZ363 & 253PZ364|L2)
 SIF Tag: SIF-036
 SIF Description: SIL1 Gastryck N2 (253PZ362, 253PZ363 & 253PZ364|L2) =>

Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)

Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

Close 253SV652 (SIL2) (F.O)

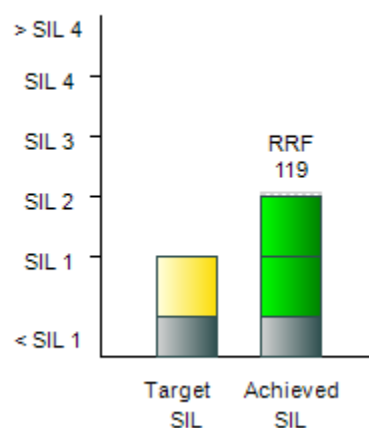
28.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 119



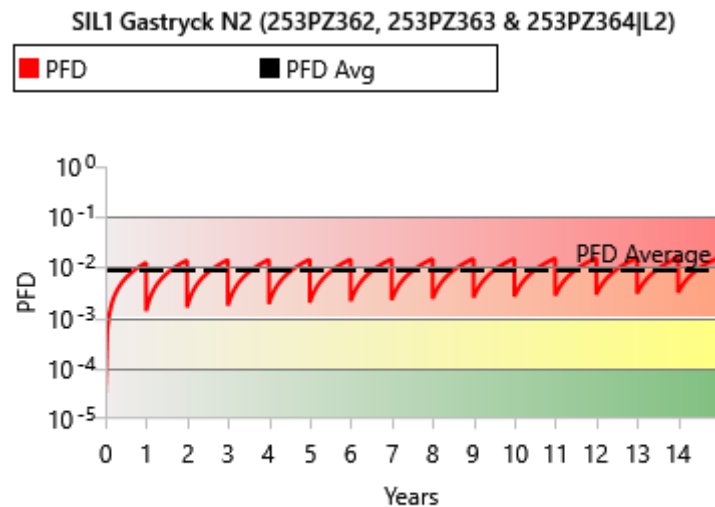
28.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-036 SIL1 Gastryck N2 (253PZ362, 253PZ363 & 253PZ364|L2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:



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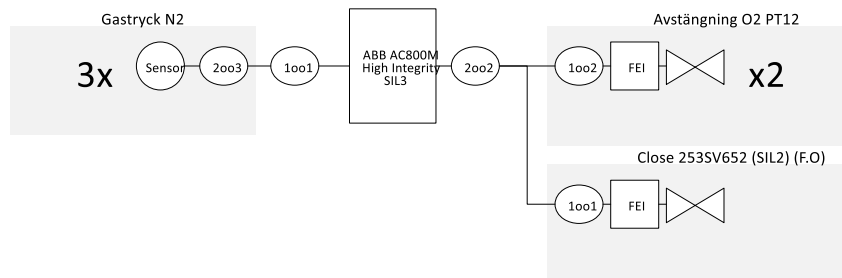
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-036 SIL1 Gastryck N2 (253PZ362, 253PZ363 & 253PZ364|L2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 206 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
8.40E-03	119	2	2	N/A	64.86



The SIF Conceptual Design is modeled as shown in the picture below.



28.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-036 SIL1 Gastryck N2 (253PZ362, 253PZ363 & 253PZ364|L2) Safety Instrumented Function is quantified as follows.

Table 207 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.17E-04	3	N/A	10354.69	1	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

28.3.1.1 Sensor Group 1: Gastryck N2

The information and reliability data underneath describe the Gastryck N2 sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical
HFT: 1 β-factor: 10

[%]:

Equipment Leg

(each):

Endress+Hauser Cerabar S PMP71, PMP72
HT, PMP75

SERH VERSION

2006.2.02

2H

-

PIU

-

AC TYPE

B

SIL CAP

2

Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm
Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]

2013.1.02

-

-

B

3

MRT [Hours]: 24

Proof Test Interval [Months]:

12

Proof Test Coverage [%]:

99

Proof Test Execution:

Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Gastryck N2.

Table 208 Reliability Data Sensor Group Gastryck N2

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>6.85E-08</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>4.44E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		18.5
								ROUTE 2H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

28.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-036 SIL1 Gastryck N2 (253PZ362, 253PZ363 & 253PZ364|L2) Safety Instrumented Function is quantified as follows.

Table 209 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
3.70E-08	3	N/A	70.75	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High
Integrity SIL3

Equipment:

SERH VERSION

PIU

SIL

exSILentia-VER-002-01-b SILver Detailed-Eng

SIL Verification (SILver™) Details

June 25, 2020

Page 198 of 333

	ABB AC800M High Integrity SIL3	2013.1.02	-	Cap N/A
Logic Solver Voting:		Architecture Type:	B	
HFT:	1	β -factor [%]:	1	
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 210 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

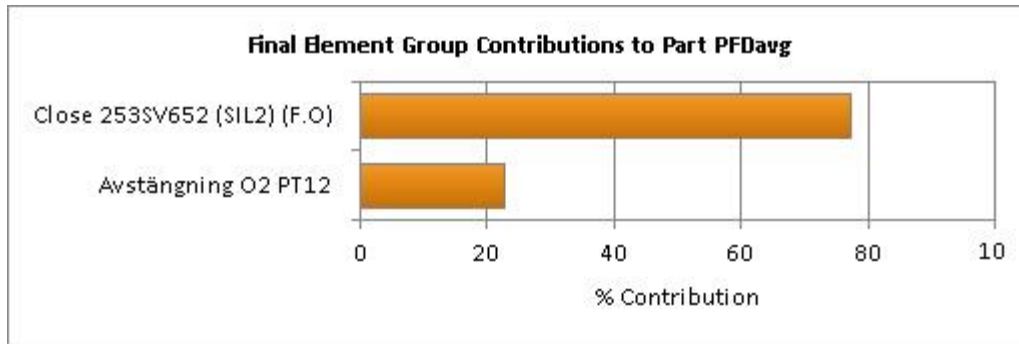
28.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-036 SIL1 Gastryck N2 (253PZ362, 253PZ363 & 253PZ364|L2) Safety Instrumented Function is quantified as follows.

Table 211 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
7.98E-03	2	N/A	842.62	0	2

Number of Final Element group(s): 2
Voting between groups: 2oo2
 β -factor [%]:



28.3.3.1 Final Element Group 1: Avstängning O2 PT12

The information and reliability data underneath describe the Avstängning O2 PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2 PT12.

Table 212 Reliability Data Final Element Group Avstängning O2 PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	5.9
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

28.3.3.2 Final Element Group 2: Close 253SV652 (SIL2) (F.O)

The information and reliability data underneath describe the Close 253SV652 (SIL2) (F.O) final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Transworld Mars		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close 253SV652 (SIL2) (F.O).

Table 213 Reliability Data Final Element Group Close 253SV652 (SIL2) (F.O)

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Transworld Mars				3.73E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	13.2
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

29 SIF-037 SIL2 Tryckvakt efter N2 bufferttank 253TA005 (253PZ323, 253PZ324, 253PZ325|L2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-037 SIL2 Tryckvakt efter N2 bufferttank 253TA005 (253PZ323, 253PZ324, 253PZ325|L2) Safety Instrumented Function.

29.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL2 Tryckvakt efter N2 bufferttank 253TA005 (253PZ323, 253PZ324, 253PZ325|L2)
 SIF Tag: SIF-037
 SIF Description: SIL2 Tryckvakt efter N2 bufferttank 253TA005 (253PZ323, 253PZ324, 253PZ325|L2)
 =>

Close N2 till kylkrets (254SV594 SIL2)
 Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)
 Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)
 Close avstängningsventil N2 till reaktortopp (253SV607 SIL2)
 Open Toptryck nödventil eft. våtfilter red. gas (345SV599)

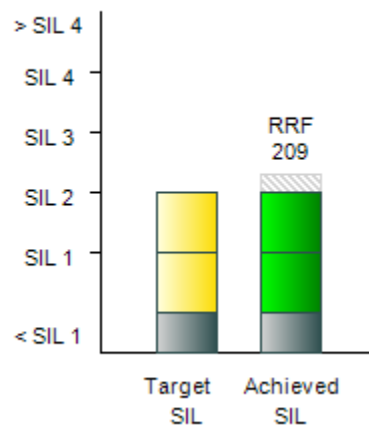
29.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 209.2



29.3 SIL Verification

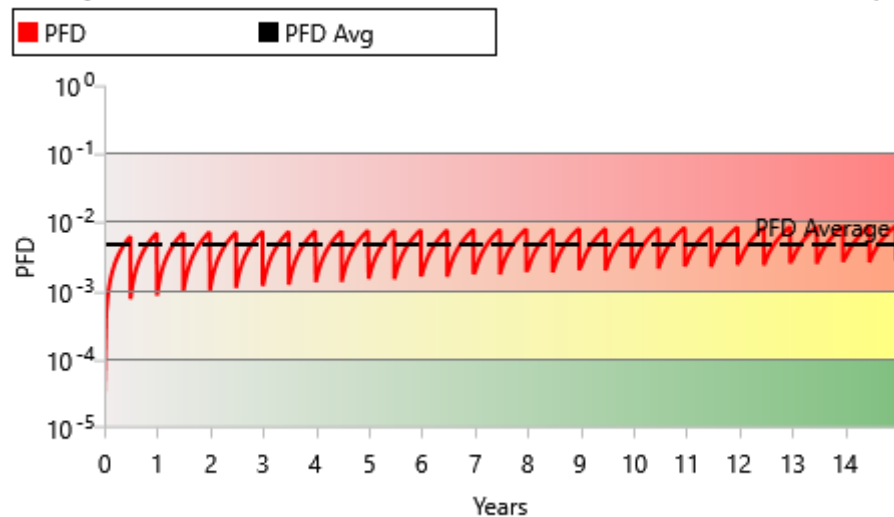
This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-037 SIL2 Tryckvakt efter N2 bufferttank 253TA005 (253PZ323, 253PZ324, 253PZ325|L2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:

SIL2 Tryckvakt efter N2 bufferttank 253TA005 (253PZ323, 253PZ324, 253PZ325|L2)



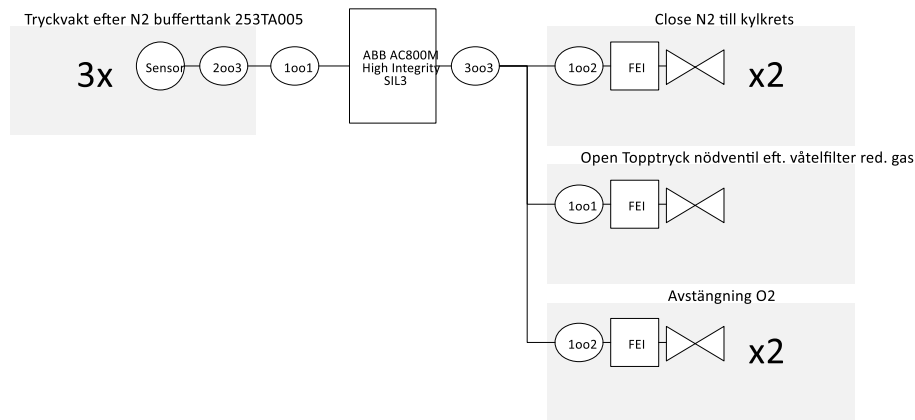
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-037 SIL2 Tryckvakt efter N2 bufferttank 253TA005 (253PZ323, 253PZ324, 253PZ325|L2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 214 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
4.78E-03	209.2	2	2	N/A	62.63



The SIF Conceptual Design is modeled as shown in the picture below.



29.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-037 SIL2 Tryckvakt efter N2 bufferttank 253TA005 (253PZ323, 253PZ324, 253PZ325|L2) Safety Instrumented Function is quantified as follows.

Table 215 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.20E-04	3	N/A	1857.58	1	2

Number of Sensor group(s): 1

Voting between groups: 1oo1

β-factor [%]: N/A

29.3.1.1 Sensor Group 1: Tryckvakt efter N2 bufferttank 253TA005

The information and reliability data underneath describe the Tryckvakt efter N2 bufferttank 253TA005 sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical

HFT: 1 β -factor [%]: 10

Equipment Leg (each): „

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: Low, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
---	-----------	---	---	---	---

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Tryckvakt efter N2 bufferttank 253TA005.

Table 216 Reliability Data Sensor Group Tryckvakt efter N2 bufferttank 253TA005

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.34E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	75.5
									ROUTE 2 _H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

29.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-037 SIL2 Tryckvakt efter N2 bufferttank 253TA005 (253PZ323, 253PZ324, 253PZ325|L2) Safety Instrumented Function is quantified as follows.

Table 217 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
3.73E-08	3	N/A	69.92	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β -factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 218 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

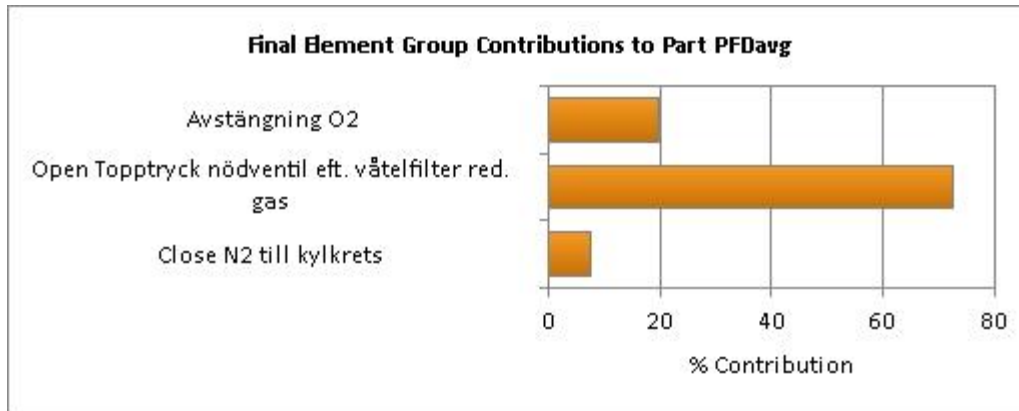
29.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-037 SIL2 Tryckvakt efter N2 bufferttank 253TA005 (253PZ323, 253PZ324, 253PZ325|L2) Safety Instrumented Function is quantified as follows.

Table 219 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.66E-03	2	N/A	886.38	0	2

Number of Final Element group(s): 3
Voting between groups: 3oo3
 β -factor [%]: 10



29.3.3.1 Final Element Group 1: Close N2 till kylkrets

The information and reliability data underneath describe the Close N2 till kylkrets final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	1oo2	Voting type:	Identical
HFT:	1	β -factor [%]:	10
Equipment Leg (each):	,	SERH VERSION	2H
Parker N		PIU	AC TYPE
Rotork			SIL CAP
Transworld Mars			
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02		
Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes			
MRT [Hours]:	24		
Proof Test Interval [Months]:	6	Proof Test Coverage [%]:	99
Proof Test Execution:	Duration 1 hours		

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close N2 till kylkrets.

Table 220 Reliability Data Final Element Group Close N2 till kylkrets

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Transworld Mars				3.73E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	13.2
						ROUTE 2H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

29.3.3.2 Final Element Group 2: Open Toptryck nödventil eft. våtfilter red. gas

The information and reliability data underneath describe the Open Toptryck nödventil eft. våtfilter red. gas final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Open Toptryck nödventil eft. våtfilter red. gas.

Table 221 Reliability Data Final Element Group Open Toptryck nödventil eft. våtfilter red. gas

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				3.70E-07			2.20E-06
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
SFF [%]							13.3
ROUTE 2 _H COMPLIANT							-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

29.3.3.3 Final Element Group 3: Avstängning O2

The information and reliability data underneath describe the Avstängning O2 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg
(each): ,

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3

Rotork	-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]:	24		
Proof Test Interval [Months]:	6	Proof Test Coverage [%]:	99
Proof Test Execution:	Leak Test; Duration 1 hours		

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2.

Table 222 Reliability Data Final Element Group Avstängning O2

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	28.4
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

30 SIF-038 SIL2 Hög vätskenivå WESP (345LI354/5/6|H2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-038 SIL2 Hög vätskenivå WESP (345LI354/5/6|H2) Safety Instrumented Function.

30.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Hög vätskenivå WESP (345LI354/5/6 H2)
SIF Tag	SIF-038
SIF Description:	SIL2 Hög vätskenivå WESP (345LI354/5/6 H2) => Trip Reduktionsgaskompressor (346PU101 SIL2) Close Avstängningsventil 1 O2 inj. (251SV555 SIL2) Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

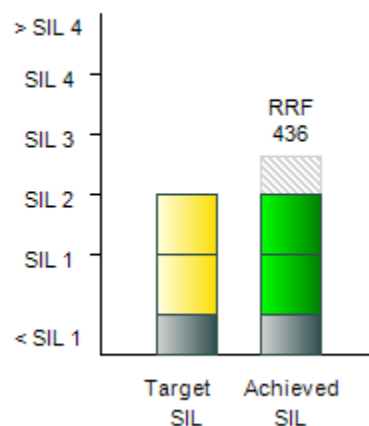
30.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 436.2



30.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-038 SIL2 Hög vätskenivå WESP (345LI354/5/6|H2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time:	15 years
Startup time:	24 hours

The SIF operates in the Low demand mode.

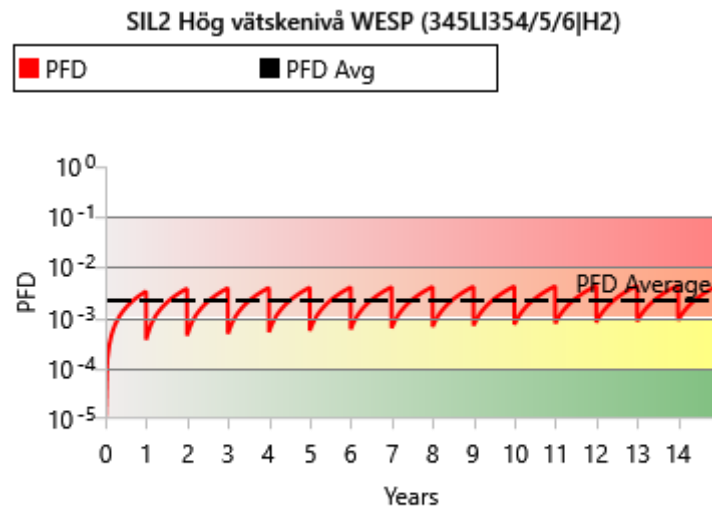
Analyst:

Analysis Date: 10/4/2019

Design References

IEC 61511 Functional safety: Safety Instrumented Systems for the Process Industry Sector

Comments:

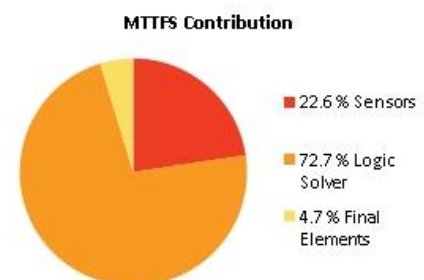
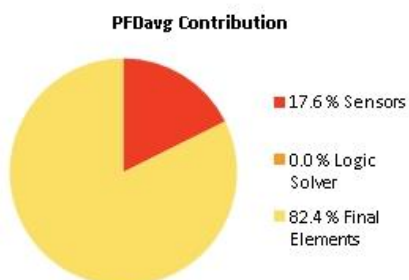


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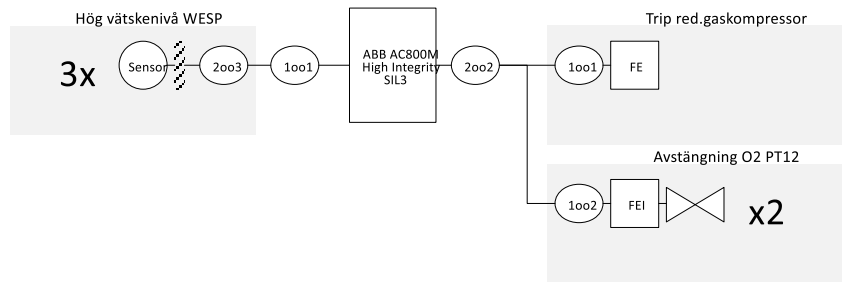
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-038 SIL2 Hög vätskenivå WESP (345LI354/5/6|H2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 223 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
2.29E-03	436.2	2	2	N/A	51.71



The SIF Conceptual Design is modeled as shown in the picture below.



30.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-038 SIL2 Hög vätskenivå WESP (345LI354/5/6|H2) Safety Instrumented Function is quantified as follows.

Table 224 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.05E-04	3	N/A	228.67	1	2

Number of Sensor group(s): 1
Voting between groups: 1001
β-factor [%]: N/A

30.3.1.1 Sensor Group 1: Hög vätskenivå WESP

The information and reliability data underneath describe the Hög vätskenivå WESP sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2003
Voting type: Identical
HFT: 1
β-factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Enderess Hauser FMR50-57		-	-	B	3
Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off					
PR Electronics 9106, 1 active in/1 active out	2014.4.03	-	-	A	2
ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
MRT [Hours]: 24					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 99		

Proof Test Execution:

Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Hög vätskenivå WESP.

Table 225 Reliability Data Sensor Group Hög vätskenivå WESP

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Enderess Hauser FMR50-57	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	2.33E-06 <i>2.33E-06</i>	0.00E+00 <i>0.00E+00</i>	2.49E-07 <i>2.49E-07</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>
PR Electronics 9106, 1 active in/1 active out	1.46E-07 <i>0.00E+00</i>	2.70E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>2.70E-08</i>	0.00E+00 <i>0.00E+00</i>	4.10E-08 <i>1.87E-07</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.77E-07 <i>1.77E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		84.5
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

30.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-038 SIL2 Hög vätskenivå WESP (345LI354/5/6|H2) Safety Instrumented Function is quantified as follows.

Table 226 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.86E-08	3	N/A	71.17	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:

ABB AC800M High Integrity SIL3

SERH VERSION	PIU	SIL Cap
2013.1.02	-	N/A

Logic Solver Voting:

Architecture Type:

B

HFT:

1

β-factor [%]:

1

MRT [Hours]:

24

Proof Test Interval [Months]:

12

Proof Test Coverage [%]:

99

Proof Test Execution:

Duration 1 hours

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 227 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

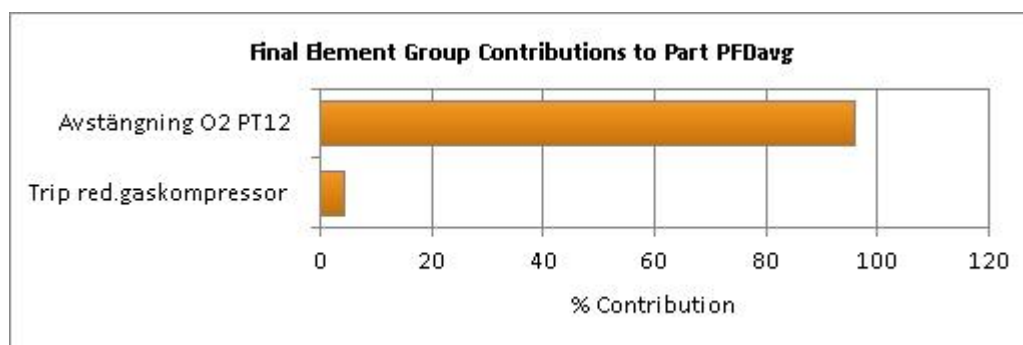
30.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-038 SIL2 Hög vätskenivå WESP (345LI354/5/6|H2) Safety Instrumented Function is quantified as follows.

Table 228 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.89E-03	2	N/A	1094.89	0	2

Number of Final Element group(s): 2
Voting between groups: 2oo2
β-factor [%]:



30.3.3.1 Final Element Group 1: Trip red.gaskompressor

The information and reliability data underneath describe the Trip red.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
ASC880 R2 STO		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip red.gaskompressor.

Table 229 Reliability Data Final Element Group Trip red.gaskompressor

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
ASC880 R2 STO		1.08E-08		1.35E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		89.5
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

30.3.3.2 Final Element Group 2: Avstängning O2 PT12

The information and reliability data underneath describe the Avstängning O2 PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg
(each): ,

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2 PT12.

Table 230 Reliability Data Final Element Group Avstängning O2 PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		5.9
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

31 SIF-039 SIL2 Tryck efter WESP red. gaskrets (345PI361|H2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-039 SIL2 Tryck efter WESP red. gaskrets (345PI361|H2) Safety Instrumented Function.

31.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL2 Tryck efter WESP red. gaskrets (345PI361|H2)
 SIF Tag: SIF-039
 SIF Description: SIL2 Tryck efter WESP red. gaskrets (345PI361|H2) =>
 Open Avstängningsventiler venturi reduktionsgas (345SV520/21/22 SIL2)
 Open Toptryck nödventil eft. våtelfilter red. gas (345SV599)

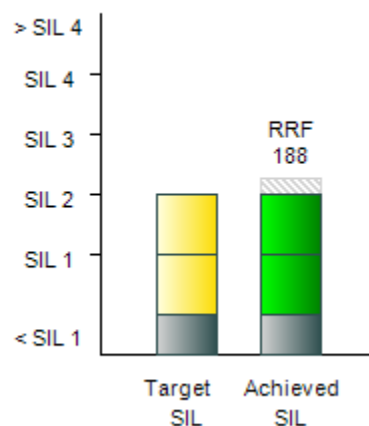
31.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 188.1



31.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-039 SIL2 Tryck efter WESP red. gaskrets (345PI361|H2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
 Startup time: 24 hours

The SIF operates in the Low demand mode.

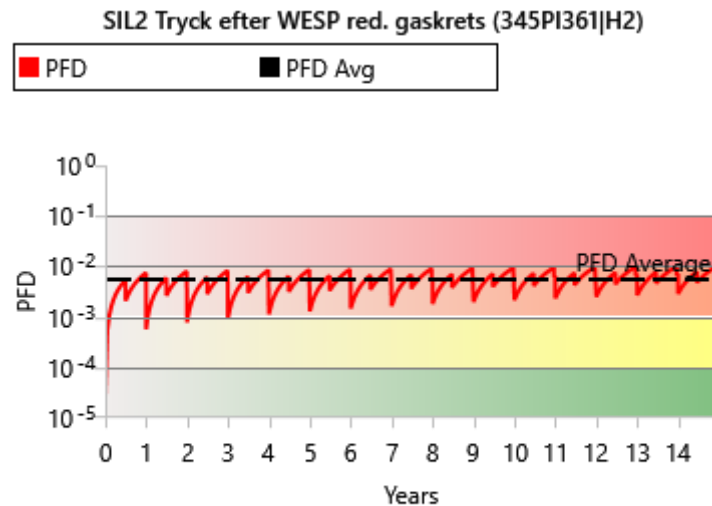
Analyst:

Analysis Date:

10/4/2019

Design References

Comments:

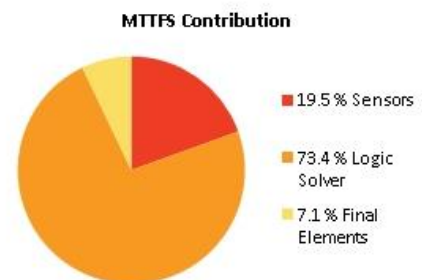
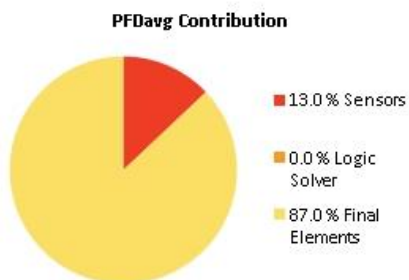


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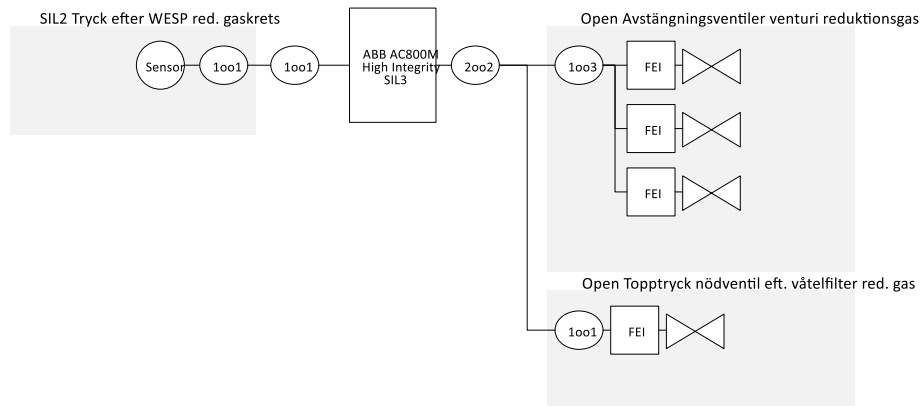
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-039 SIL2 Tryck efter WESP red. gaskrets (345PI361|H2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 231 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
5.31E-03	188.1	2	2	N/A	53.21



The SIF Conceptual Design is modeled as shown in the picture below.



31.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-039 SIL2 Tryck efter WESP red. gaskrets (345PI361|H2) Safety Instrumented Function is quantified as follows.

Table 232 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.89E-04	2	N/A	273.24	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

31.3.1.1 Sensor Group 1: SIL2 Tryck efter WESP red. gaskrets

The information and reliability data underneath describe the SIL2 Tryck efter WESP red. gaskrets sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm

Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group SIL2 Tryck efter WESP red. gaskrets.

Table 233 Reliability Data Sensor Group SIL2 Tryck efter WESP red. gaskrets

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.96E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.17E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	78.5
									ROUTE 2 _H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

31.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-039 SIL2 Tryck efter WESP red. gaskrets (345PI361|H2) Safety Instrumented Function is quantified as follows.

Table 234 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment: ABB AC800M High Integrity SIL3 SERH VERSION 2013.1.02 PIU - SIL Cap N/A

Logic Solver Voting: Architecture Type: B

HFT: 1 β-factor [%]: 1

MRT [Hours]: 24

Proof Test Interval [Months]: 12

Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 235 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

31.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-039 SIL2 Tryck efter WESP red. gaskrets (345PI361|H2) Safety Instrumented Function is quantified as follows.

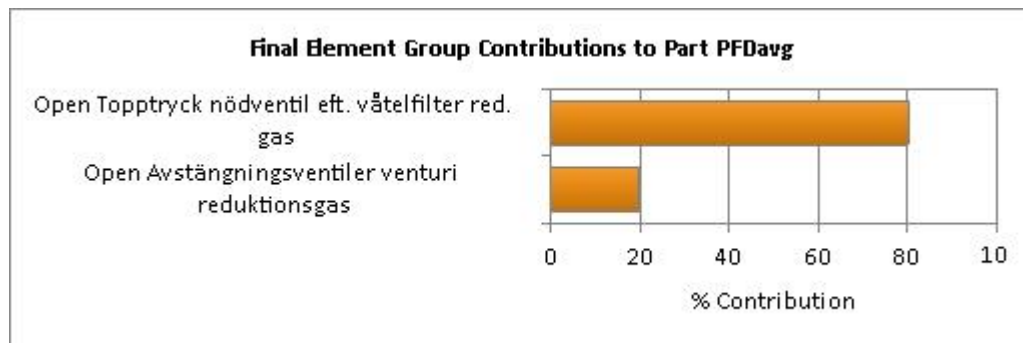
Table 236 Final Element Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.63E-03	2	N/A	750.88	0	2

Number of Final Element group(s): 2

Voting between groups: 2oo2

β-factor [%]:



31.3.3.1 Final Element Group 1: Open Avstängningsventiler venturi reduktionsgas

The information and reliability data underneath describe the Open Avstängningsventiler venturi reduktionsgas final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo3 Voting type: Diverse

HFT: 2 β -factor [%]: 10

Equipment Leg (1):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
GEFA HG1		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

Equipment Leg (2):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
GEFA HG1		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

Equipment Leg (3):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
Flowserve Worcester Series 44/59/459/599	2013.3.03	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

MRT [Hours]: 24
 Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90
 Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Open Avstängningsventiler venturi reduktionsgas.

Table 237 Reliability Data Final Element Group Open Avstängningsventiler venturi reduktionsgas

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
345SV520							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
GEFA HG1				1.56E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	16.7
						ROUTE 2 _H COMPLIANT	-
345SV521							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			

Component	Failure Rates [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
GEFA HG1				1.56E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		16.7
					Route 2 _H Compliant		-
345SV522							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
Flowserve Worcester Series 44/59/459/599		1.46E-07		3.22E-07			8.80E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		24.0
					Route 2 _H Compliant		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

31.3.3.2 Final Element Group 2: Open Topstryck nödventil eft. våtelfilter red. gas

The information and reliability data underneath describe the Open Topstryck nödventil eft. våtelfilter red. gas final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1

Voting type: Identical

HFT: 0

β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _h	PIU	ACTYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 96.5

Proof Test Execution: Duration 1 hours

Partial Valve Stroke Test Interval 6M Partial Valve Stroke Test Coverage[%] 73.3

Partial Valve Stroke Test Execution Manual

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Open Topstryck nödventil eft. våtelfilter red. gas.

Table 238 Reliability Data Final Element Group Open Topstryck nödventil eft. våtelfilter red. gas

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				3.70E-07			2.20E-06
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		13.3
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

32 SIF-040 SIL2 Tryck efter WESP processgaskrets (353PI361|H2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-040 SIL2 Tryck efter WESP processgaskrets (353PI361|H2) Safety Instrumented Function.

32.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL2 Tryck efter WESP processgaskrets (353PI361|H2)
 SIF Tag: SIF-040
 SIF Description: SIL2 Tryck efter WESP processgaskrets (353PI361|H2) =>
 Open Avstängningsventiler venturi processgas (353SV520/21/22 SIL2)
 Open Toptryck nödventil eft. våtelfilter red. gas (353SV599)

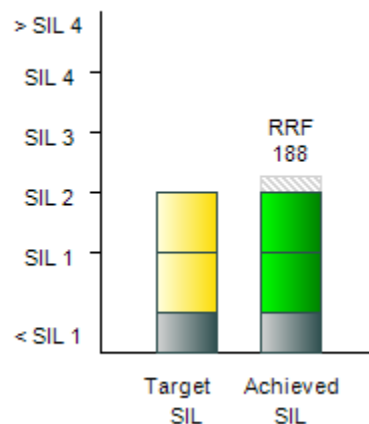
32.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 188.1



32.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-040 SIL2 Tryck efter WESP processgaskrets (353PI361|H2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
 Startup time: 24 hours

The SIF operates in the Low demand mode.

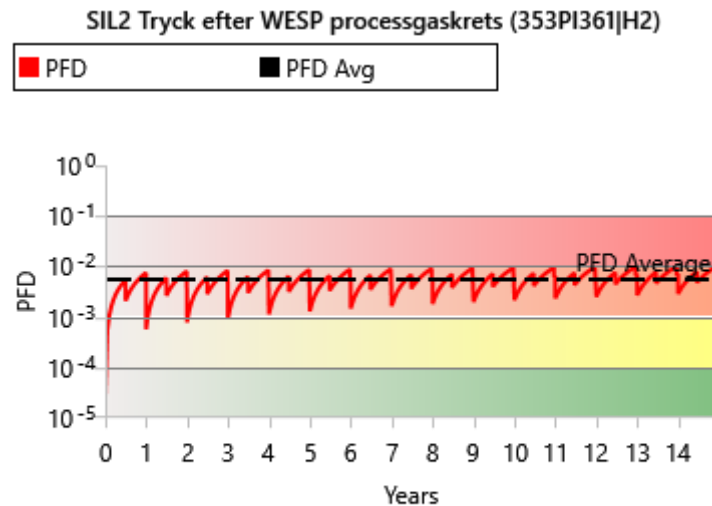
Analyst:

Analysis Date:

10/4/2019

Design References

Comments:

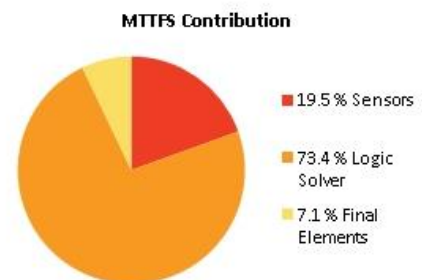
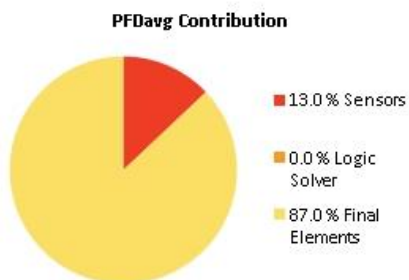


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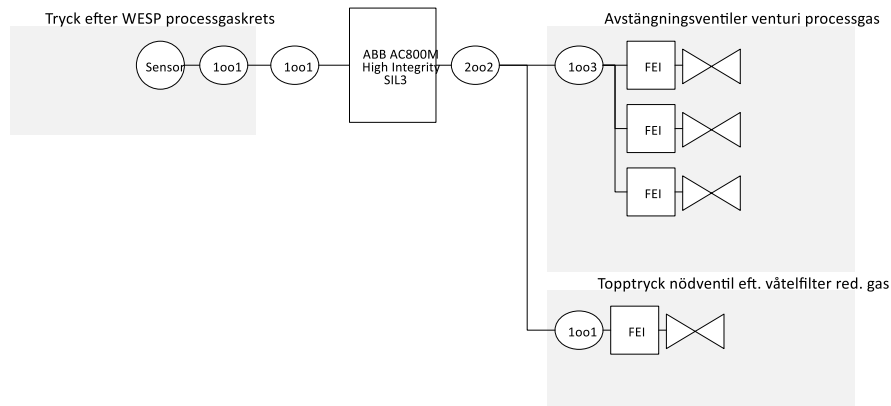
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-040 SIL2 Tryck efter WESP processgaskrets (353PI361|H2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 239 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
5.31E-03	188.1	2	2	N/A	53.21



The SIF Conceptual Design is modeled as shown in the picture below.



32.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-040 SIL2 Tryck efter WESP processgaskrets (353PI361|H2) Safety Instrumented Function is quantified as follows.

Table 240 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.89E-04	2	N/A	273.24	0	2

Number of Sensor group(s): 1
Voting between groups: 1001
 β -factor [%]: N/A

32.3.1.1 Sensor Group 1: Tryck efter WESP processgaskrets

The information and reliability data underneath describe the Tryck efter WESP processgaskrets sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm

Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3
MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Tryck efter WESP processgaskrets.

Table 241 Reliability Data Sensor Group Tryck efter WESP processgaskrets

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.96E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.17E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	78.5
									ROUTE 2 _H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

32.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-040 SIL2 Tryck efter WESP processgaskrets (353PI361|H2) Safety Instrumented Function is quantified as follows.

Table 242 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment: ABB AC800M High Integrity SIL3 SERH VERSION 2013.1.02 PIU - SIL Cap N/A

Logic Solver Voting: Architecture Type: B

HFT: 1 β-factor [%]: 1

MRT [Hours]: 24

Proof Test Interval [Months]: 12

Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 243 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

32.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-040 SIL2 Tryck efter WESP processgaskrets (353PI361|H2) Safety Instrumented Function is quantified as follows.

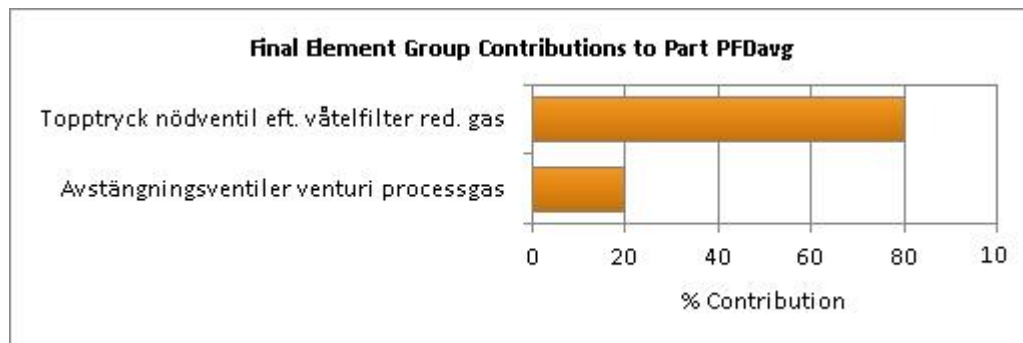
Table 244 Final Element Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.63E-03	2	N/A	750.88	0	2

Number of Final Element group(s): 2

Voting between groups: 2oo2

β-factor [%]:



32.3.3.1 Final Element Group 1: Avstängningsventiler venturi processgas

The information and reliability data underneath describe the Avstängningsventiler venturi processgas final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo3 Voting type: Diverse

HFT: 2 β -factor [%]: 10

Equipment Leg (1):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
GEFA HG1		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

Equipment Leg (2):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
GEFA HG1		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

Equipment Leg (3):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Sirca AP-APM		-	-	A	3
Flowserve Worcester Series 44/59/459/599	2013.3.03	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängningsventiler venturi processgas.

Table 245 Reliability Data Final Element Group Avstängningsventiler venturi processgas

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
353SV520							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
GEFA HG1				1.56E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	16.7
						ROUTE 2 _H COMPLIANT	-
353SV521							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			

Component	Failure Rates [1/hr]						
	SD	SU	DD	DU	AD	AU	NE
GEFA HG1				1.56E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		16.7
					Route 2 _H Compliant		-
353SV522							
Parker N		5.70E-08		6.54E-07			
Sirca AP-APM	3.00E-10	3.32E-10		3.48E-10			
Flowserve Worcester Series 44/59/459/599		1.46E-07		3.22E-07			8.80E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		24.0
					Route 2 _H Compliant		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

32.3.3.2 Final Element Group 2: Topstryck nödventil eft. våtelfilter red. gas

The information and reliability data underneath describe the Topstryck nödventil eft. våtelfilter red. gas final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	ACTYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: No, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 96.5
Proof Test Execution: Duration 1 hours
Partial Valve Stroke Test Interval 6M Partial Valve Stroke Test Coverage[%] 73.3
Partial Valve Stroke Test Execution Manual

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Topstryck nödventil eft. våtelfilter red. gas.

Table 246 Reliability Data Final Element Group Topstryck nödventil eft. våtelfilter red. gas

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				3.70E-07			2.20E-06
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		13.3
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

33 SIF-041 SIL1 Temp kondensor reduktionsgas (345TI301|H2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-041 SIL1 Temp kondensor reduktionsgas (345TI301|H2) Safety Instrumented Function.

33.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL1 Temp kondensor reduktionsgas (345TI301 H2)
SIF Tag	SIF-041
SIF Description:	SIL1 Temp kondensor reduktionsgas (345TI301 H2) => Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2) Trip Reduktionsgaskompressor (346PU101 SIL2) Close Avstängningsventil 1 O2 inj. (251SV555 SIL2) Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

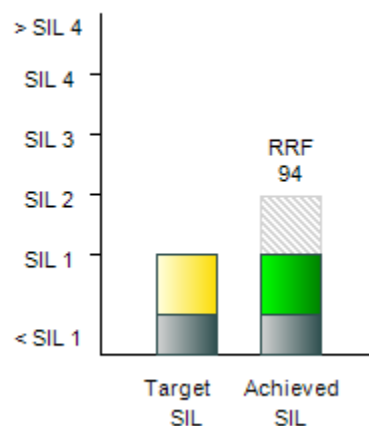
33.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 1 with RRF = 94.5



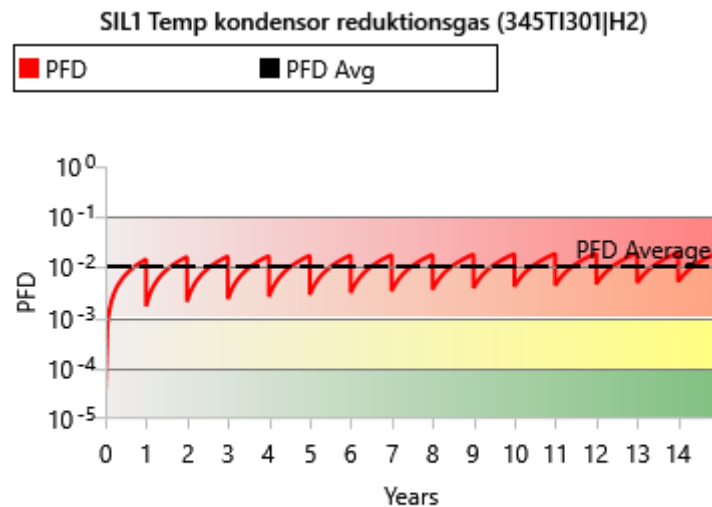
33.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-041 SIL1 Temp kondensor reduktionsgas (345TI301|H2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:

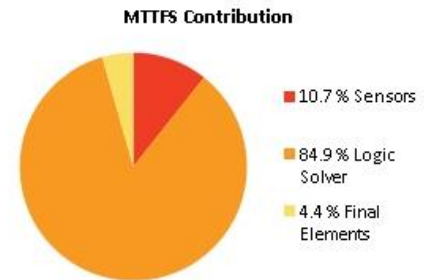
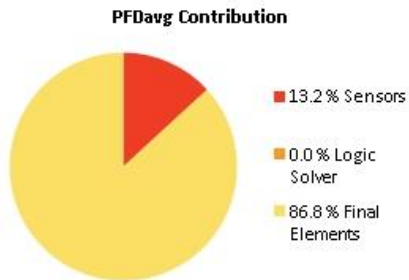


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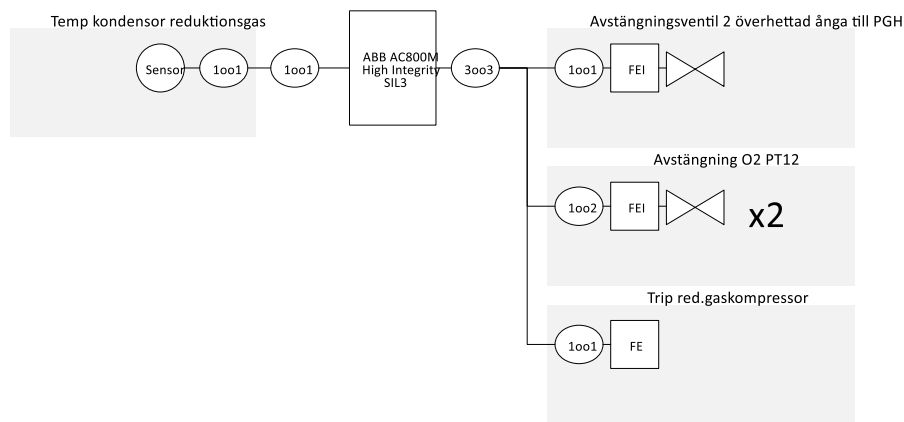
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-041 SIL1 Temp kondensor reduktionsgas (345TI301|H2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 247 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
1.06E-02	94.5	1	2	N/A	59.36



The SIF Conceptual Design is modeled as shown in the picture below.



33.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-041 SIL1 Temp kondensor reduktionsgas (345TI301|H2) Safety Instrumented Function is quantified as follows.

Table 248 Sensor Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.40E-03	2	N/A	554.98	0	2

Number of Sensor group(s): 1
Voting between groups: 1001
 β -factor [%]: N/A

33.3.1.1 Sensor Group 1: Temp kondensor reduktionsgas

The information and reliability data underneath describe the Temp kondensor reduktionsgas sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β -factor [%]: N/A

Equipment Leg (each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser iTemp TMT 182	2006.2.02	-	-	B	-

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
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MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Temp kondensor reduktionsgas.

Table 249 Reliability Data Sensor Group Temp kondensor reduktionsgas

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser iTemp TMT 182	1.66E-08 <i>0.00E+00</i>	9.16E-08 <i>0.00E+00</i>	9.18E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>1.83E-07</i>	0.00E+00 <i>0.00E+00</i>	1.07E-07 <i>1.24E-07</i>	0.00E+00 <i>0.00E+00</i>	1.22E-09 <i>1.22E-09</i>	9.78E-08 <i>9.78E-08</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	62.6
									ROUTE 2 _H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

33.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-041 SIL1 Temp kondensor reduktionsgas (345TI301|H2) Safety Instrumented Function is quantified as follows.

Table 250 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
3.73E-08	3	N/A	69.92	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:		SERH VERSION	PIU	SIL Cap
	ABB AC800M High Integrity SIL3	2013.1.02	-	N/A
Logic Solver Voting:		Architecture Type:	B	
HFT:	1	β -factor [%]:	1	
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 251 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
					SFF [%]		100.0

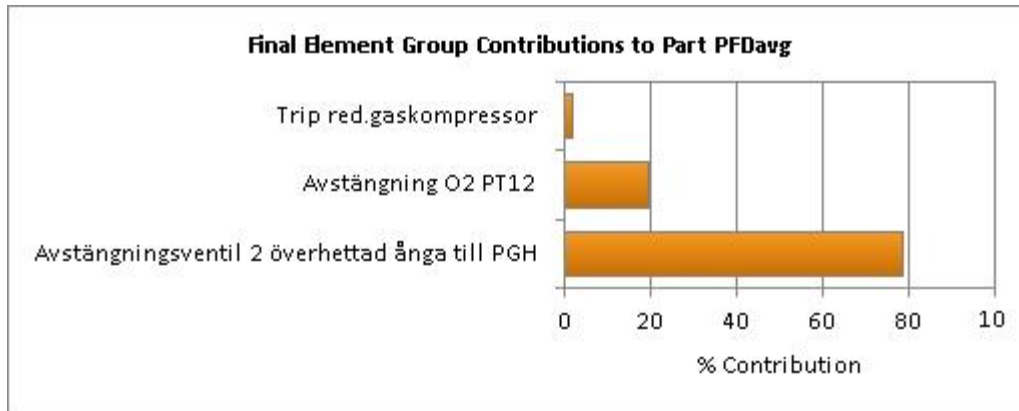
33.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-041 SIL1 Temp kondensor reduktionsgas (345TI301|H2) Safety Instrumented Function is quantified as follows.

Table 252 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
9.19E-03	2	N/A	1343.75	0	2

Number of Final Element group(s): 3
Voting between groups: 3oo3
 β -factor [%]:



33.3.3.1 Final Element Group 1: Avstängningsventil 2 överhettad ånga till PGH

The information and reliability data underneath describe the Avstängningsventil 2 överhettad ånga till PGH final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängningsventil 2 överhettad ånga till PGH.

Table 253 Reliability Data Final Element Group Avstängningsventil 2 överhettad ånga till PGH

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				5.55E-07			2.02E-06
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
SFF [%]							11.5
ROUTE 2h COMPLIANT							-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

33.3.3.2 Final Element Group 2: Avstängning O2 PT12

The information and reliability data underneath describe the Avstängning O2 PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical

HFT: 1 β -factor [%]: 10

Equipment Leg
(each): ,

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2 PT12.

Table 254 Reliability Data Final Element Group Avstängning O2 PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		5.9
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

33.3.3.3 Final Element Group 3: Trip red.gaskompressor

The information and reliability data underneath describe the Trip red.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
ASC880 R2 STO		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3
MRT [Hours]:	24				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	90		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip red.gaskompressor.

Table 255 Reliability Data Final Element Group Trip red.gaskompressor

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
ASC880 R2 STO		1.08E-08		1.35E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	89.5
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

34 SIF-042 SIL1 Temp kondensor processgas (353TI301|H2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-042 SIL1 Temp kondensor processgas (353TI301|H2) Safety Instrumented Function.

34.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL1 Temp kondensor processgas (353TI301 H2)
SIF Tag	SIF-042
SIF Description:	SIL1 Temp kondensor processgas (353TI301 H2) => Trip Processgaskompressor (354PU101 SIL2) Close NG till kylkrets (252SV614SIL2)

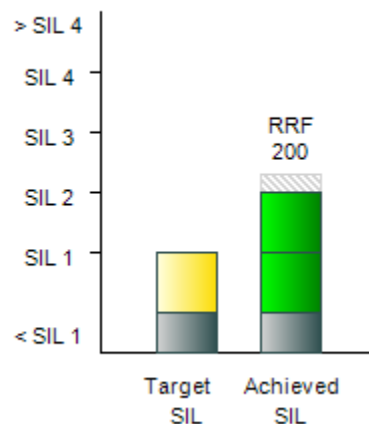
34.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 200.2



34.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-042 SIL1 Temp kondensor processgas (353TI301|H2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time:	15 years
Startup time:	24 hours

The SIF operates in the Low demand mode.

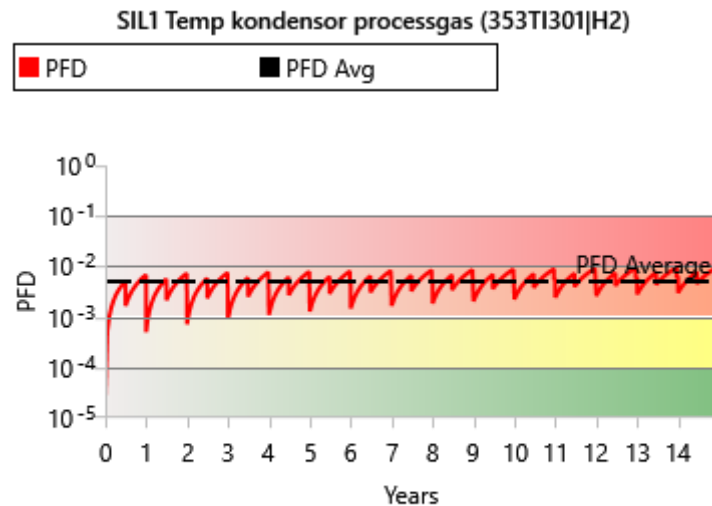
Analyst:

Analysis Date:

10/4/2019

Design References

Comments:

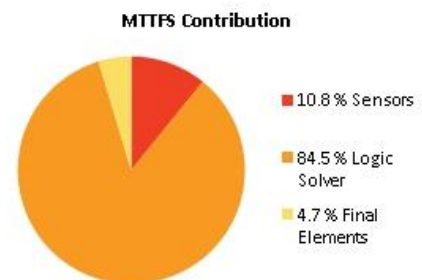
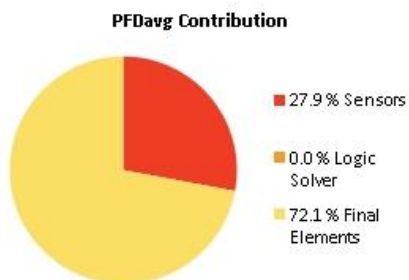


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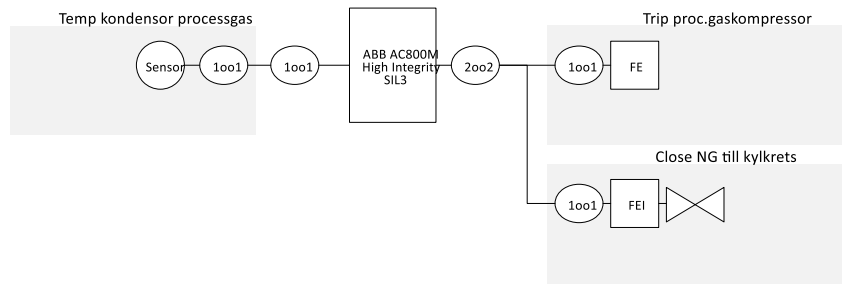
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-042 SIL1 Temp kondensor processgas (353TI301|H2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 256 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
4.99E-03	200.2	2	2	N/A	60.13



The SIF Conceptual Design is modeled as shown in the picture below.



34.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-042 SIL1 Temp kondensor processgas (353TI301|H2) Safety Instrumented Function is quantified as follows.

Table 257 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.40E-03	2	N/A	554.98	0	2

Number of Sensor group(s): 1
Voting between groups: 1001
β-factor [%]: N/A

34.3.1.1 Sensor Group 1: Temp kondensor processgas

The information and reliability data underneath describe the Temp kondensor processgas sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser iTemp TMT 182	2006.2.02	-	-	B	-
Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off					
ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
MRT [Hours]: 24					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 90		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Temp kondensor processgas.

Table 258 Reliability Data Sensor Group Temp kondensor processgas

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser iTemp TMT 182	1.66E-08 <i>0.00E+00</i>	9.16E-08 <i>0.00E+00</i>	9.18E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>1.83E-07</i>	0.00E+00 <i>0.00E+00</i>	1.07E-07 <i>1.24E-07</i>	0.00E+00 <i>0.00E+00</i>	1.22E-09 <i>1.22E-09</i>	9.78E-08 <i>9.78E-08</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		62.6
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

34.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-042 SIL1 Temp kondensor processgas (353TI301|H2) Safety Instrumented Function is quantified as follows.

Table 259 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.86E-08	3	N/A	71.17	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β-factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 260 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

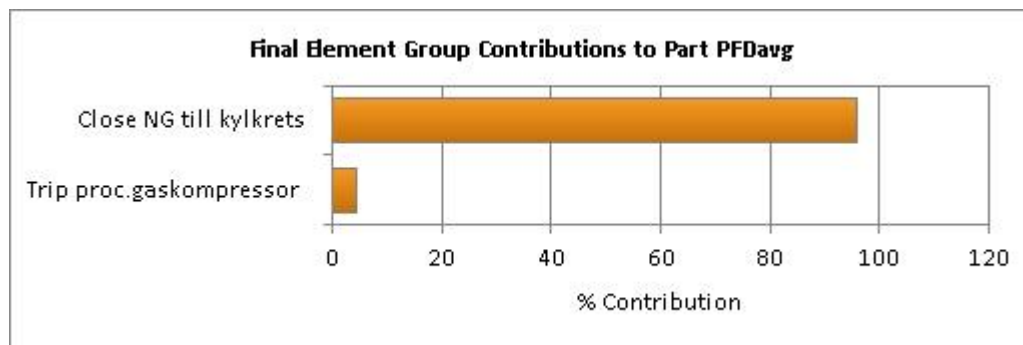
34.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-042 SIL1 Temp kondensor processgas (353TI301|H2) Safety Instrumented Function is quantified as follows.

Table 261 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
3.60E-03	2	N/A	1286.95	0	2

Number of Final Element group(s): 2
Voting between groups: 2oo2
β-factor [%]:



34.3.3.1 Final Element Group 1: Trip proc.gaskompressor

The information and reliability data underneath describe the Trip proc.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
ASC880 R2 STO		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3
MRT [Hours]:	24				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	90		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip proc.gaskompressor.

Table 262 Reliability Data Final Element Group Trip proc.gaskompressor

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
ASC880 R2 STO		1.08E-08		1.35E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
SFF [%]							89.5
ROUTE 2 _H COMPLIANT							-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

34.3.3.2 Final Element Group 2: Close NG till kylkrets

The information and reliability data underneath describe the Close NG till kylkrets final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	1oo1	Voting type:	Identical
HFT:	0	β-factor [%]:	N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Transworld Mars		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]:	24				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	97		
Proof Test Execution:	Duration 1 hours				
Partial Valve Stroke Test Interval	6M	Partial Valve Stroke Test Coverage[%]	77		
Partial Valve Stroke Test Execution	Manual				

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close NG till kylkrets.

Table 263 Reliability Data Final Element Group Close NG till kylkrets

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Transworld Mars				3.73E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		49.2
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

35 SIF-043 SIL2 Skintemp förquench reduktionsgas (345TI307|H2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-043 SIL2 Skintemp förquench reduktionsgas (345TI307|H2) Safety Instrumented Function.

35.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Skintemp förquench reduktionsgas (345TI307 H2)
SIF Tag	SIF-043
SIF Description:	SIL2 Skintemp förquench reduktionsgas (345TI307 H2) => Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2) Trip Reduktionsgaskompressor (346PU101 SIL2) Close Avstängningsventil 1 O2 inj. (251SV555 SIL2) Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

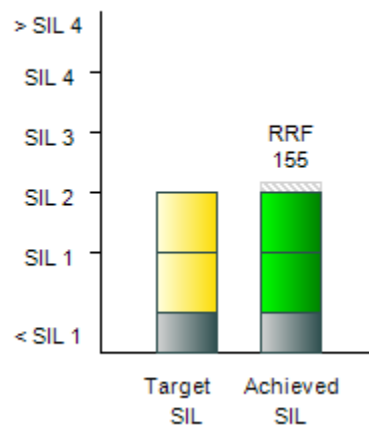
35.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 155.6



35.3 SIL Verification

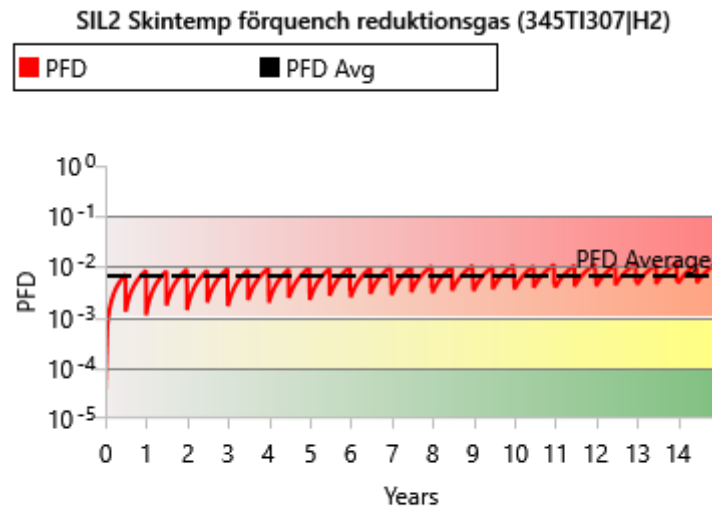
This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-043 SIL2 Skintemp förquench reduktionsgas (345TI307|H2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst:
Design References
Comments:

Analysis Date: 10/4/2019

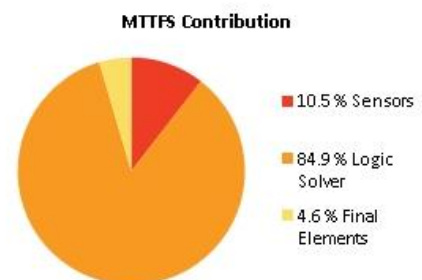
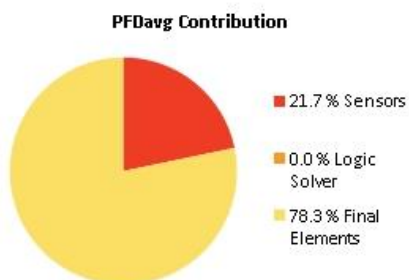


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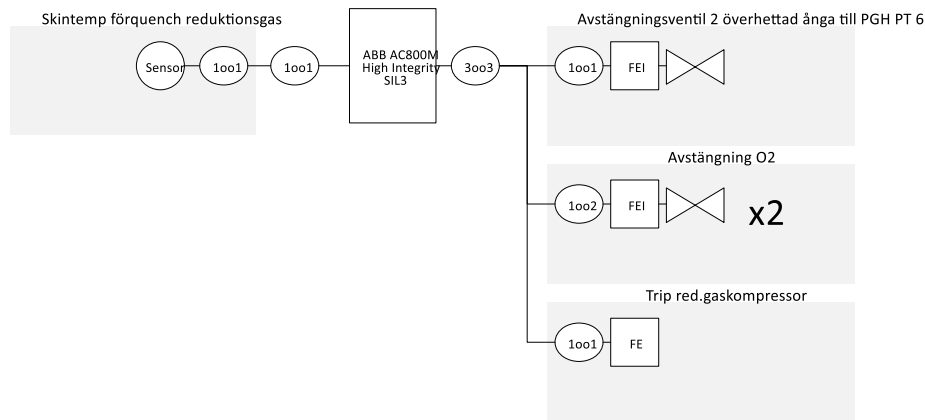
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-043 SIL2 Skintemp förquench reduktionsgas (345TI307|H2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 264 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
6.42E-03	155.6	2	2	N/A	58.35



The SIF Conceptual Design is modeled as shown in the picture below.



35.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-043 SIL2 Skintemp förquench reduktionsgas (345TI307|H2) Safety Instrumented Function is quantified as follows.

Table 265 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.40E-03	2	N/A	554.98	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

35.3.1.1 Sensor Group 1: Skintemp förquench reduktionsgas

The information and reliability data underneath describe the Skintemp förquench reduktionsgas sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg (each):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser iTemp TMT 182	2006.2.02	-	-	B	-

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3
MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90
Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Skintemp förquench reduktionsgas.

Table 266 Reliability Data Sensor Group Skintemp förquench reduktionsgas

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser iTemp TMT 182	1.66E-08 <i>0.00E+00</i>	9.16E-08 <i>0.00E+00</i>	9.18E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>1.83E-07</i>	0.00E+00 <i>0.00E+00</i>	1.07E-07 <i>1.24E-07</i>	0.00E+00 <i>0.00E+00</i>	1.22E-09 <i>1.22E-09</i>	9.78E-08 <i>9.78E-08</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		62.6
								ROUTE 2H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

35.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-043 SIL2 Skintemp förquench reduktionsgas (345TI307|H2) Safety Instrumented Function is quantified as follows.

Table 267 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.60E-08	3	N/A	68.72	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment: ABB AC800M High Integrity SIL3 SERH VERSION 2013.1.02 PIU - SIL Cap N/A

Logic Solver Voting: Architecture Type: B

HFT: 1 β-factor [%]: 1

MRT [Hours]: 24
 Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
 Proof Test Execution: Duration 1 hours

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 268 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

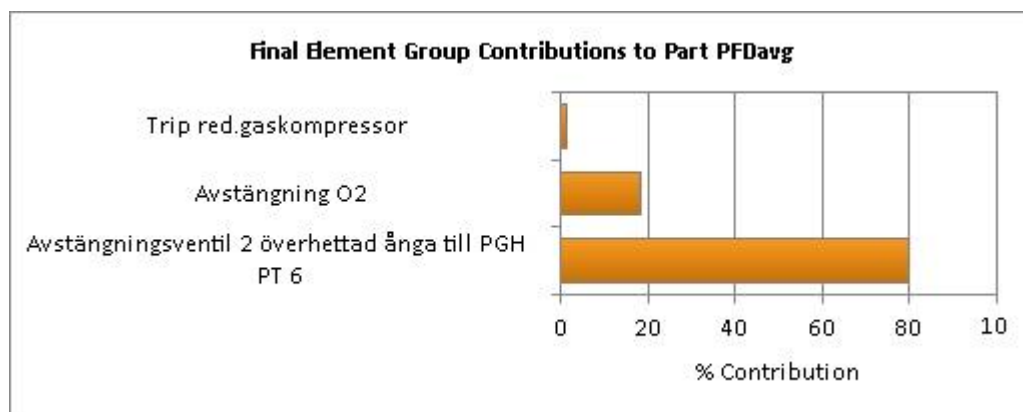
35.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-043 SIL2 Skintemp förquench reduktionsgas (345TI307|H2) Safety Instrumented Function is quantified as follows.

Table 269 Final Element Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
5.03E-03	2	N/A	1278.03	0	2

Number of Final Element group(s): 3
 Voting between groups: 3oo3
 β -factor [%]: 10



Rotork	-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]:	24		
Proof Test Interval [Months]:	6	Proof Test Coverage [%]:	99
Proof Test Execution:	Leak Test; Duration 1 hours		

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2.

Table 271 Reliability Data Final Element Group Avstängning O2

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		28.4
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

35.3.3.3 Final Element Group 3: Trip red.gaskompressor

The information and reliability data underneath describe the Trip red.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	1oo1	Voting type:	Identical
HFT:	0	β-factor [%]:	N/A

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
ASC880 R2 STO		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3
MRT [Hours]:	24				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip red.gaskompressor.

Table 272 Reliability Data Final Element Group Trip red.gaskompressor

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
ASC880 R2 STO		1.08E-08		1.35E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		89.5
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

36 SIF-044 SIL2 Skintemp quench reduktionsgas (345TI308|H2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-044 SIL2 Skintemp quench reduktionsgas (345TI308|H2) Safety Instrumented Function.

36.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Skintemp quench reduktionsgas (345TI308 H2)
SIF Tag	SIF-044
SIF Description:	SIL2 Skintemp förquench reduktionsgas (345TI307 H2) => Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2) Trip Reduktionsgaskompressor (346PU101 SIL2) Close Avstängningsventil 1 O2 inj. (251SV555 SIL2) Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

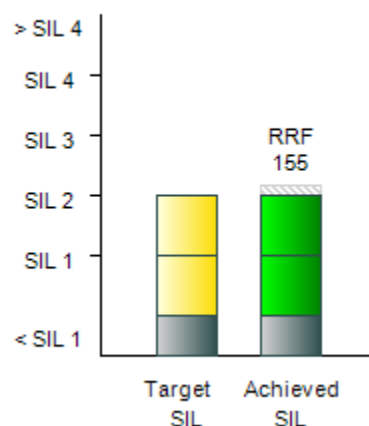
36.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 155.1



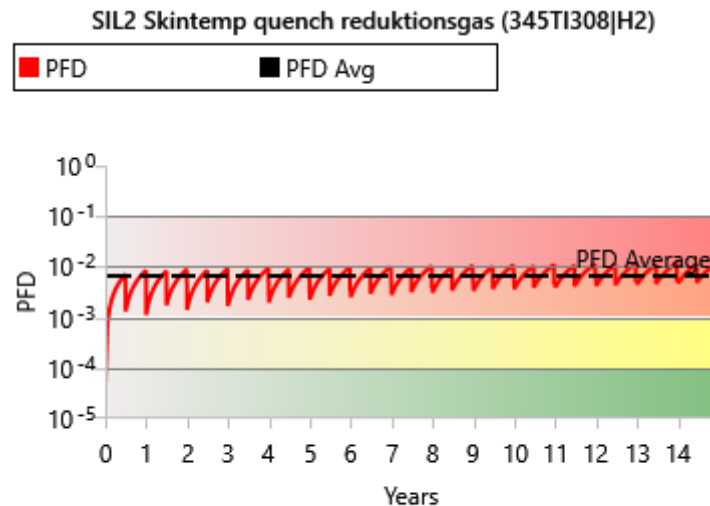
36.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-044 SIL2 Skintemp quench reduktionsgas (345TI308|H2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:

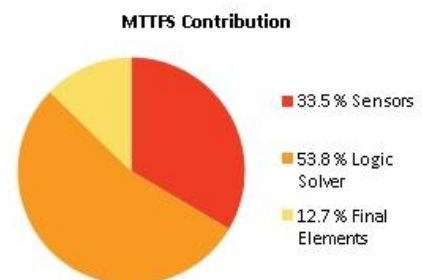
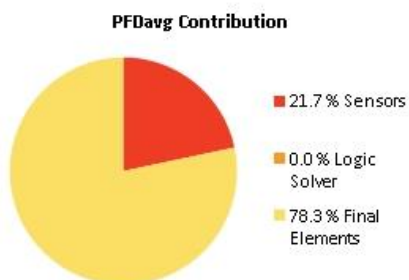


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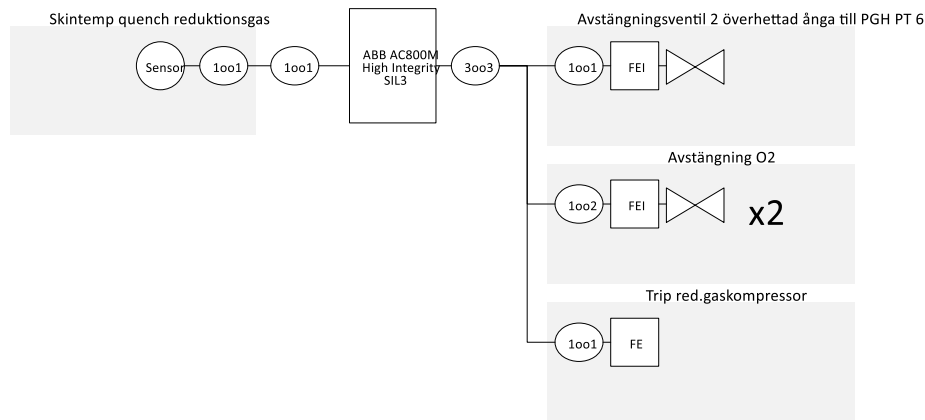
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-044 SIL2 Skintemp quench reduktionsgas (345TI308|H2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 273 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
6.45E-03	155.1	2	2	N/A	39.92



The SIF Conceptual Design is modeled as shown in the picture below.



36.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-044 SIL2 Skintemp quench reduktionsgas (345TI308|H2) Safety Instrumented Function is quantified as follows.

Table 274 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.40E-03	2	N/A	119.15	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

36.3.1.1 Sensor Group 1: Skintemp quench reduktionsgas

The information and reliability data underneath describe the Skintemp quench reduktionsgas sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg

(each):	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser iTemp TMT 182	2006.2.02	-	-	B	-

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3
 Analog In Module: AI880A/TU844/845/TY801 2013.1.02 - - B 3
 MRT [Hours]: 24
 Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 90
 Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Skintemp quench reduktionsgas.

Table 275 Reliability Data Sensor Group Skintemp quench reduktionsgas

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser iTemp TMT 182	1.66E-08 <i>0.00E+00</i>	9.16E-08 <i>0.00E+00</i>	9.18E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>1.83E-07</i>	0.00E+00 <i>0.00E+00</i>	1.07E-07 <i>1.24E-07</i>	0.00E+00 <i>0.00E+00</i>	1.22E-09 <i>1.22E-09</i>	9.78E-08 <i>9.78E-08</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
Analog In Module: AI880A/TU844/845/TY801	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	3.69E-07 <i>7.53E-07</i>	5.00E-10 <i>5.00E-10</i>	3.84E-07 <i>0.00E+00</i>	1.32E-10 <i>1.32E-10</i>	3.89E-08 <i>3.89E-08</i>	3.93E-10 <i>3.93E-10</i>	2.11E-07 <i>2.11E-07</i>
								SFF [%]		89.0
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

36.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-044 SIL2 Skintemp quench reduktionsgas (345TI308|H2) Safety Instrumented Function is quantified as follows.

Table 276 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
3.59E-09	3	N/A	74.24	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:

SERH VERSION PIU SIL Cap

ABB AC800M High Integrity SIL3		2013.1.02	-	N/A
Logic Solver Voting:		Architecture Type:	B	
HFT:	1	β -factor [%]:	1	
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 277 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		100.0

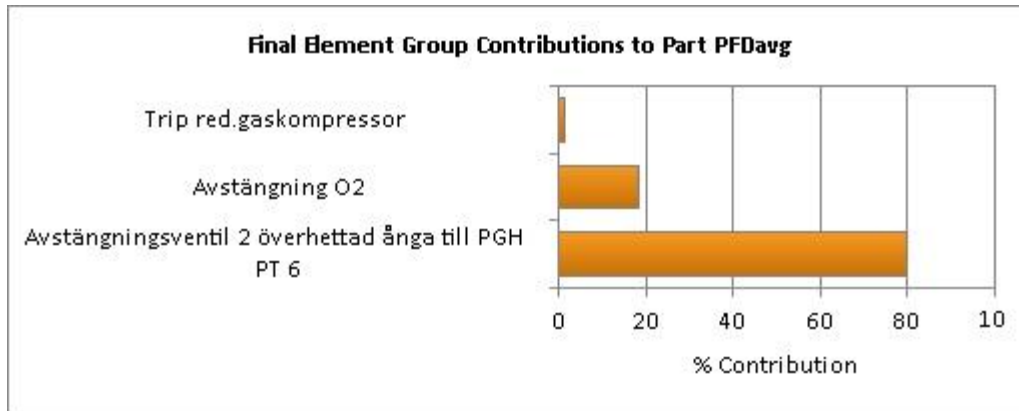
36.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-044 SIL2 Skintemp quench reduktionsgas (345TI308|H2) Safety Instrumented Function is quantified as follows.

Table 278 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
5.06E-03	2	N/A	313.71	0	2

Number of Final Element group(s):	3
Voting between groups:	3oo3
β -factor [%]:	10



36.3.3.1 Final Element Group 1: Avstängningsventil 2 överhettad ånga till PGH PT 6

The information and reliability data underneath describe the Avstängningsventil 2 överhettad ånga till PGH PT 6 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3
Digital Out Module: DO880/TU842/843	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängningsventil 2 överhettad ånga till PGH PT 6.

Table 279 Reliability Data Final Element Group Avstängningsventil 2 överhettad ånga till PGH PT 6

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				5.55E-07			2.02E-06
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
SFF [%]							45.3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

36.3.3.2 Final Element Group 2: Avstängning O2

The information and reliability data underneath describe the Avstängning O2 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3
Digital Out Module: DO880/TU842/843	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 99
Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2.

Table 280 Reliability Data Final Element Group Avstängning O2

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
					SFF [%]		28.4
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

36.3.3.3 Final Element Group 3: Trip red.gaskompressor

The information and reliability data underneath describe the Trip red.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
ASC880 R2 STO		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3
Digital Out Module: DO880/TU842/843	2013.1.02	-	-	B	3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip red.gaskompressor.

Table 281 Reliability Data Final Element Group Trip red.gaskompressor

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
ASC880 R2 STO		1.08E-08		1.35E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
						SFF [%]	89.5
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

37 SIF-045 SIL2 Skintemp förquench processgas (353TI307|H2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-045 SIL2 Skintemp förquench processgas (353TI307|H2) Safety Instrumented Function.

37.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL2 Skintemp förquench processgas (353TI307|H2)
 SIF Tag: SIF-045
 SIF Description: SIL2 Skintemp förquench processgas (353TI307|H2) =>
 Trip Processgaskompressor (354PU101 SIL2)
 Close NG till kylkrets (252SV614 SIL2)

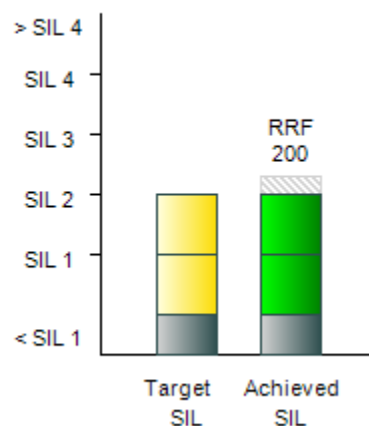
37.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 200.2



37.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-045 SIL2 Skintemp förquench processgas (353TI307|H2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
 Startup time: 24 hours

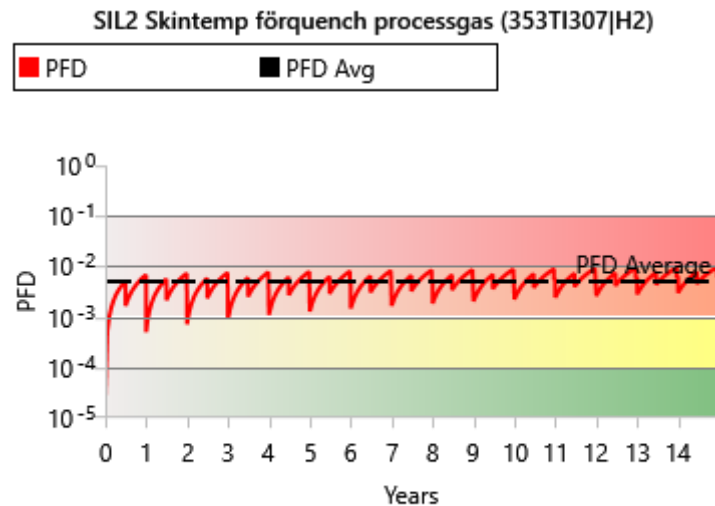
The SIF operates in the Low demand mode.

Analyst:

Analysis Date: 10/4/2019

Design References

Comments:

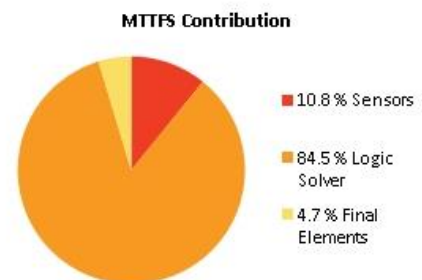
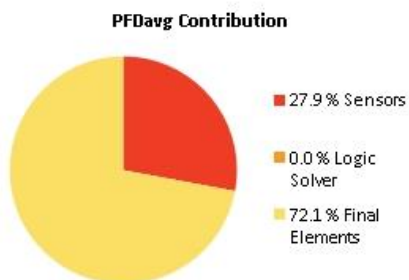


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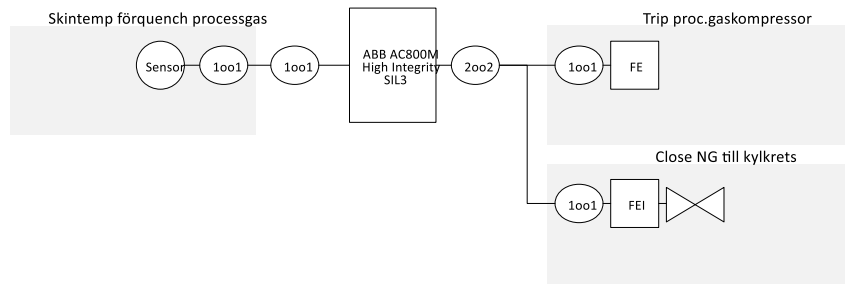
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-045 SIL2 Skintemp förquench processgas (353TI307|H2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 282 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
4.99E-03	200.2	2	2	N/A	60.13



The SIF Conceptual Design is modeled as shown in the picture below.



37.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-045 SIL2 Skintemp förquench processgas (353TI307|H2) Safety Instrumented Function is quantified as follows.

Table 283 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.40E-03	2	N/A	554.98	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

37.3.1.1 Sensor Group 1: Skintemp förquench processgas

The information and reliability data underneath describe the Skintemp förquench processgas sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser iTemp TMT 182	2006.2.02	-	-	B	-
Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off					
ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
MRT [Hours]: 24					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 90		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Skintemp förquench processgas.

Table 284 Reliability Data Sensor Group Skintemp förquench processgas

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser iTemp TMT 182	1.66E-08 <i>0.00E+00</i>	9.16E-08 <i>0.00E+00</i>	9.18E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>1.83E-07</i>	0.00E+00 <i>0.00E+00</i>	1.07E-07 <i>1.24E-07</i>	0.00E+00 <i>0.00E+00</i>	1.22E-09 <i>1.22E-09</i>	9.78E-08 <i>9.78E-08</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
								SFF [%]		62.6
								ROUTE 2 _H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

37.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-045 SIL2 Skintemp förquench processgas (353TI307|H2) Safety Instrumented Function is quantified as follows.

Table 285 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.86E-08	3	N/A	71.17	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β-factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 286 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

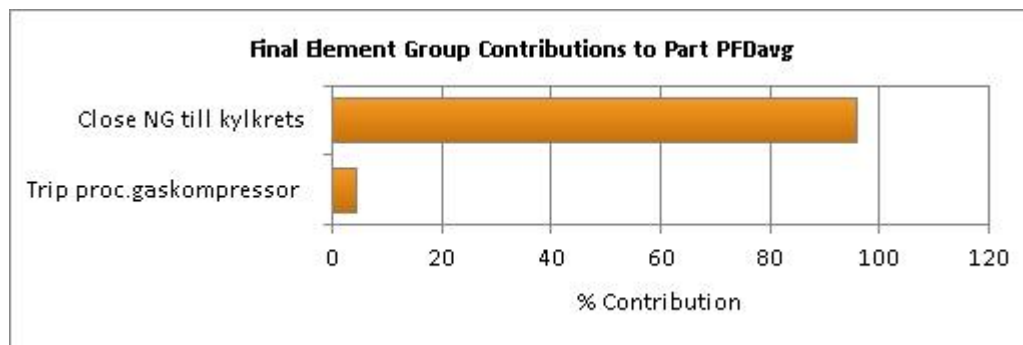
37.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-045 SIL2 Skintemp förqrench processgas (353TI307|H2) Safety Instrumented Function is quantified as follows.

Table 287 Final Element Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
3.60E-03	2	N/A	1286.95	0	2

Number of Final Element group(s): 2
Voting between groups: 2oo2
 β -factor [%]:



37.3.3.1 Final Element Group 1: Trip proc.gaskompressor

The information and reliability data underneath describe the Trip proc.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
ASC880 R2 STO		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3
MRT [Hours]:	24				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	90		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip proc.gaskompressor.

Table 288 Reliability Data Final Element Group Trip proc.gaskompressor

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
ASC880 R2 STO		1.08E-08		1.35E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
SFF [%]							89.5
ROUTE 2 _H COMPLIANT							-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

37.3.3.2 Final Element Group 2: Close NG till kylkrets

The information and reliability data underneath describe the Close NG till kylkrets final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Transworld Mars		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 97
Proof Test Execution: Duration 1 hours
Partial Valve Stroke Test Interval 6M Partial Valve Stroke Test Coverage[%] 77
Partial Valve Stroke Test Execution Manual

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close NG till kylkrets.

Table 289 Reliability Data Final Element Group Close NG till kylkrets

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Transworld Mars				3.73E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		49.2
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

38 SIF-046 SIL2 Skintemp quench processgas (353TI308|H2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-046 SIL2 Skintemp quench processgas (353TI308|H2) Safety Instrumented Function.

38.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL2 Skintemp quench processgas (353TI308|H2)
 SIF Tag: SIF-046
 SIF Description: SIL2 Skintemp quench processgas (353TI308|H2) =>
 Trip Processgaskompressor (354PU101 SIL2)
 Close NG till kylkrets (252SV614/252RV618 SIL2)

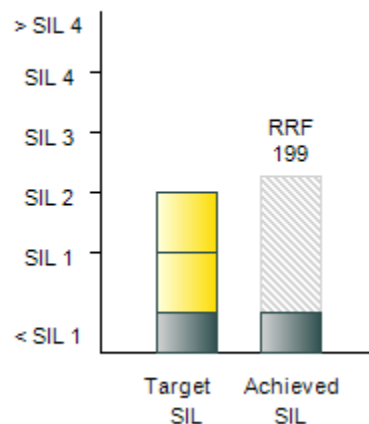
38.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 0 with RRF = 199.1



38.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-046 SIL2 Skintemp quench processgas (353TI308|H2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
 Startup time: 24 hours

The SIF operates in the Low demand mode.

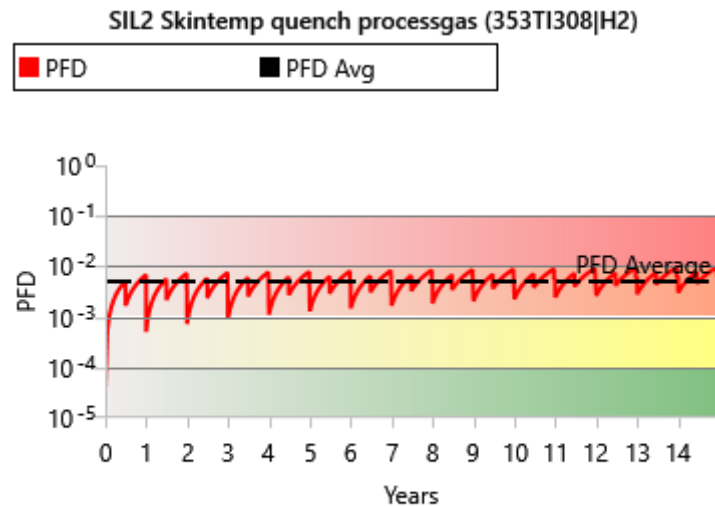
Analyst:

Analysis Date:

10/4/2019

Design References

Comments:

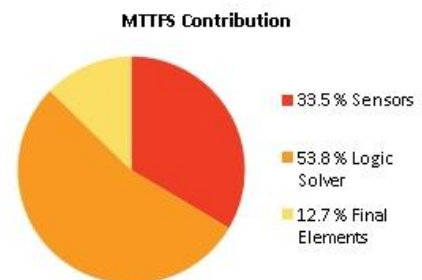
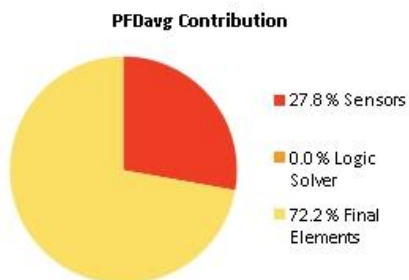


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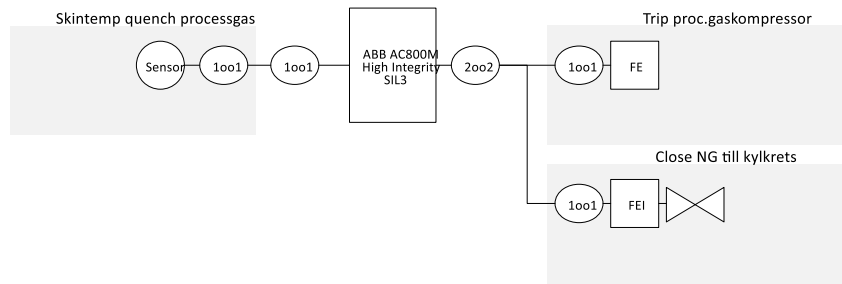
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-046 SIL2 Skintemp quench processgas (353TI308|H2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 290 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61508	SYSTEMATIC CAPABILITY	
5.02E-03	199.1	2	1	0	39.94



The SIF Conceptual Design is modeled as shown in the picture below.



38.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-046 SIL2 Skintemp quench processgas (353TI308|H2) Safety Instrumented Function is quantified as follows.

Table 291 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61508	SYSTEMATIC CAPABILITY			
1.40E-03	1	0	119.15	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

38.3.1.1 Sensor Group 1: Skintemp quench processgas

The information and reliability data underneath describe the Skintemp quench processgas sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser iTemp TMT 182	2006.2.02	-	-	B	-
Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off					
ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
Analog In Module: AI880A/TU844/845/TY801	2013.1.02	-	-	B	3
MRT [Hours]: 24					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 90		

Proof Test Execution:

Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Skintemp quench processgas.

Table 292 Reliability Data Sensor Group Skintemp quench processgas

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser iTemp TMT 182	1.66E-08 <i>0.00E+00</i>	9.16E-08 <i>0.00E+00</i>	9.18E-08 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>1.83E-07</i>	0.00E+00 <i>0.00E+00</i>	1.07E-07 <i>1.24E-07</i>	0.00E+00 <i>0.00E+00</i>	1.22E-09 <i>1.22E-09</i>	9.78E-08 <i>9.78E-08</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
Analog In Module: AI880A/TU844/845/TY80 1	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	3.69E-07 <i>7.53E-07</i>	5.00E-10 <i>5.00E-10</i>	3.84E-07 <i>0.00E+00</i>	1.32E-10 <i>1.32E-10</i>	3.89E-08 <i>3.89E-08</i>	3.93E-10 <i>3.93E-10</i>	2.11E-07 <i>2.11E-07</i>
								SFF [%]		89.0
								ROUTE 2H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

38.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-046 SIL2 Skintemp quench processgas (353TI308|H2) Safety Instrumented Function is quantified as follows.

Table 293 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61508	SYSTEMATIC CAPABILITY			
3.59E-09	3	3	74.24	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	3
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β-factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 294 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	100.0

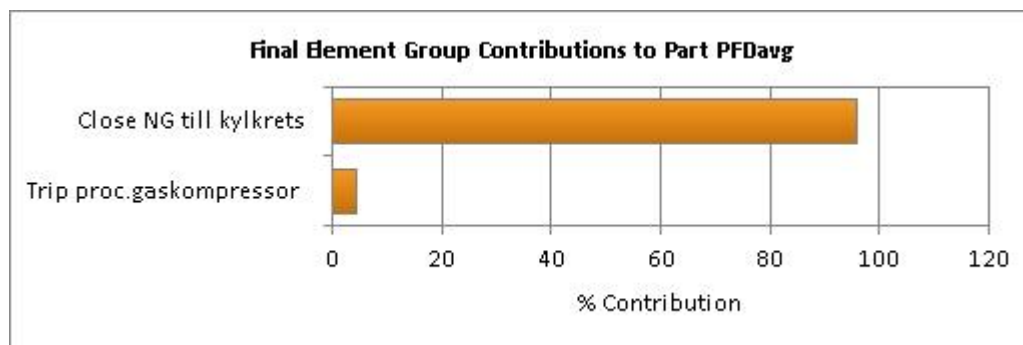
38.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-046 SIL2 Skintemp quench processgas (353TI308|H2) Safety Instrumented Function is quantified as follows.

Table 295 Final Element Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61508	SYSTEMATIC CAPABILITY			
3.63E-03	1	3	314.84	0	2

Number of Final Element group(s): 2
Voting between groups: 2oo2
β-factor [%]:



38.3.3.1 Final Element Group 1: Trip proc.gaskompressor

The information and reliability data underneath describe the Trip proc.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	1oo1	Voting type:	Identical		
HFT:	0	β-factor [%]:	N/A		
Equipment Leg (each):		SERH VERSION	2H	PIU	ACTY
ASC880 R2 STO			-	-	A
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02		-	-	B
Digital Out Module: DO880/TU842/843	2013.1.02		-	-	B
MRT [Hours]:	24				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	90		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip proc.gaskompressor.

Table 296 Reliability Data Final Element Group Trip proc.gaskompressor

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
ASC880 R2 STO		1.08E-08		1.35E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
						SFF [%]	89.5
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

38.3.3.2 Final Element Group 2: Close NG till kylkrets

The information and reliability data underneath describe the Close NG till kylkrets final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	1oo1	Voting type:	Identical		
HFT:	0	β-factor [%]:	N/A		
Equipment Leg (each):		SERH VERSION	2H	PIU	ACTY
Parker N			-	-	A
Rotork			-	-	A
Transworld Mars			-	-	A
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02		-	-	B
Digital Out Module: DO880/TU842/843	2013.1.02		-	-	B
Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes					
MRT [Hours]:	24				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	97		

Proof Test Execution: Duration 1 hours
 Partial Valve Stroke Test Interval 6M Partial Valve Stroke Test 77
 Coverage[%]
 Partial Valve Stroke Test Execution Manual

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close NG till kylkrets.

Table 297 Reliability Data Final Element Group Close NG till kylkrets

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Transworld Mars				3.73E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
						SFF [%]	49.2
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

39 SIF-047 SIL1 Flödesmätare till förquench reduktionsgaskrets (345FI305|L2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-047 SIL1 Flödesmätare till förquench reduktionsgaskrets (345FI305|L2) Safety Instrumented Function.

39.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL1 Flödesmätare till förquench reduktionsgaskrets (345FI305 L2)
SIF Tag	SIF-047
SIF Description:	SIL1 Flödesmätare till förquench reduktionsgaskrets (345FI305 L2) => Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2) Trip Reduktionsgaskompressor (346PU101 SIL2) Close Avstängningsventil 1 O2 inj. (251SV555 SIL2) Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

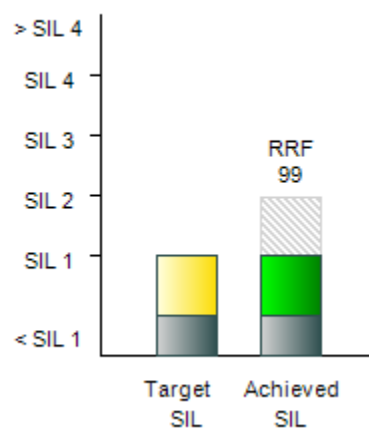
39.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 1 with RRF = 99



39.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-047 SIL1 Flödesmätare till förquench reduktionsgaskrets (345FI305|L2) Safety Instrumented Function. In order to perform

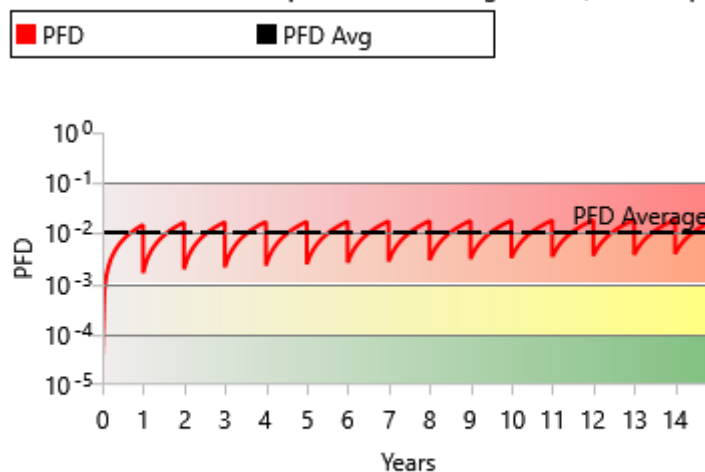
the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019
Design References
Comments:

SIL1 Flödesmätare till förquench reduktionsgaskrets (345FI305|L2)



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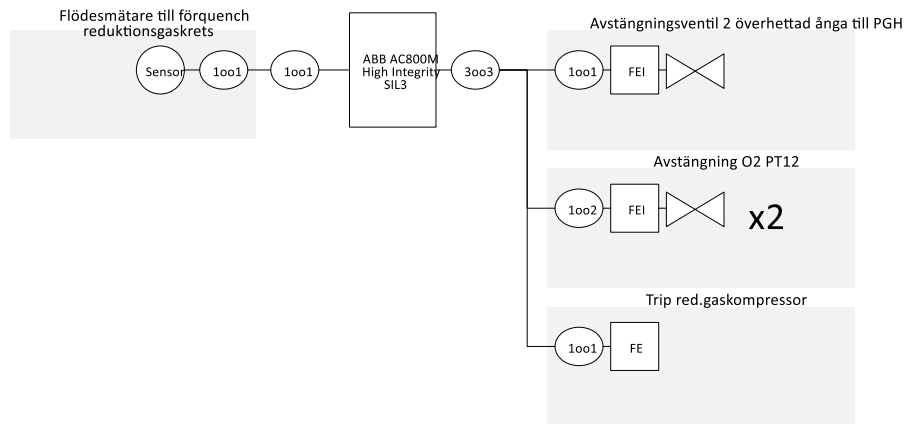
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-047 SIL1 Flödesmätare till förquench reduktionsgaskrets (345FI305|L2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 298 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
1.01E-02	99	1	2	N/A	22.26



The SIF Conceptual Design is modeled as shown in the picture below.



39.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-047 SIL1 Flödesmätare till förquench reduktionsgaskrets (345FI305 | L2) Safety Instrumented Function is quantified as follows.

Table 299 Sensor Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.01E-03	2	N/A	33.51	0	2

Number of Sensor group(s): 1

Voting between groups: 1oo1

β-factor [%]: N/A

39.3.1.1 Sensor Group 1: Flödesmätare till förquench reduktionsgaskrets

The information and reliability data underneath describe the Flödesmätare till förquench reduktionsgaskrets sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β -factor [%]: N/A

Equipment Leg (each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Promag P200		-	-	B	2

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
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MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 98

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Flödesmätare till förquench reduktionsgaskrets.

Table 300 Reliability Data Sensor Group Flödesmätare till förquench reduktionsgaskrets

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Promag P200	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	3.39E-06 <i>3.39E-06</i>	0.00E+00 <i>0.00E+00</i>	1.55E-07 <i>1.55E-07</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	95.6
									ROUTE 2 _H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

39.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-047 SIL1 Flödesmätare till förquench reduktionsgaskrets (345FI305|L2) Safety Instrumented Function is quantified as follows.

Table 301 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
3.73E-08	3	N/A	69.92	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:		SERH VERSION	PIU	SIL Cap
	ABB AC800M High Integrity SIL3	2013.1.02	-	N/A
Logic Solver Voting:		Architecture Type:	B	
HFT:	1	β -factor [%]:	1	
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 302 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

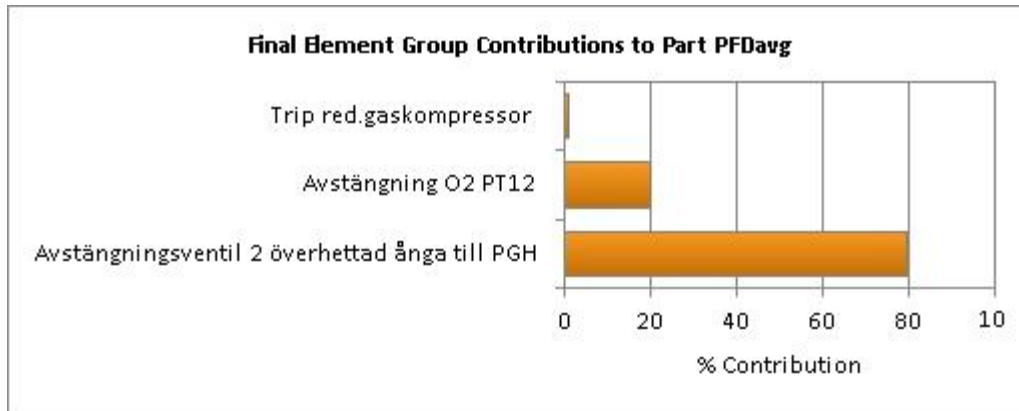
39.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-047 SIL1 Flödesmätare till förqrench reduktionsgaskrets (345FI305|L2) Safety Instrumented Function is quantified as follows.

Table 303 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
9.10E-03	2	N/A	1279.87	0	2

Number of Final Element group(s): 3
Voting between groups: 3oo3
 β -factor [%]: 10



39.3.3.1 Final Element Group 1: Avstängningsventil 2 överhettad ånga till PGH

The information and reliability data underneath describe the Avstängningsventil 2 överhettad ånga till PGH final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängningsventil 2 överhettad ånga till PGH.

Table 304 Reliability Data Final Element Group Avstängningsventil 2 överhettad ånga till PGH

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				5.55E-07			2.02E-06
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
SFF [%]							11.5
ROUTE 2H COMPLIANT							-

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
ASC880 R2 STO		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3
MRT [Hours]:	24				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip red.gaskompressor.

Table 306 Reliability Data Final Element Group Trip red.gaskompressor

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
ASC880 R2 STO		1.08E-08		1.35E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	89.5
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

40 SIF-048 SIL1 Flödesmätare till förquench processgaskrets (353FI305|L2)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-048 SIL1 Flödesmätare till förquench processgaskrets (353FI305|L2) Safety Instrumented Function.

40.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL1 Flödesmätare till förquench processgaskrets (353FI305 L2)
SIF Tag	SIF-048
SIF Description:	SIL1 Flödesmätare till förquench processgaskrets (353FI305 L2) => Trip Processgaskompressor (354PU101 SIL2) Close NG till kylkrets (252SV614 SIL2)

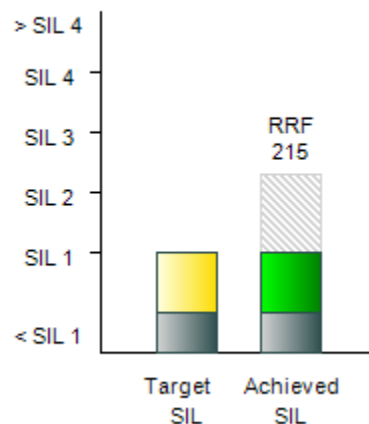
40.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 1 with RRF = 215.9



40.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-048 SIL1 Flödesmätare till förquench processgaskrets (353FI305|L2) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time:	15 years
Startup time:	24 hours

The SIF operates in the Low demand mode.

Analyst:

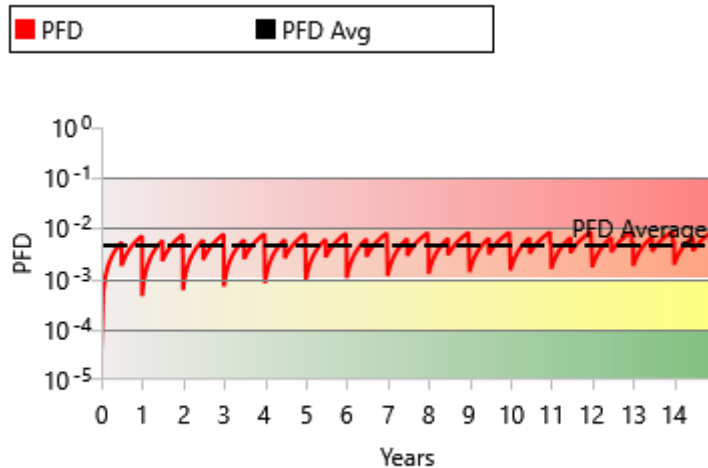
Analysis Date:

10/4/2019

Design References

Comments:

SIL1 Flödesmätare till förquench processgaskrets (353FI305|L2)



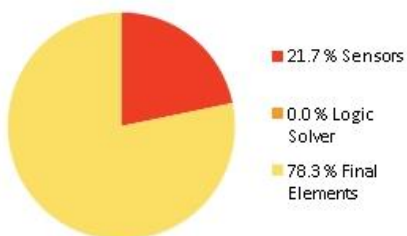
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Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-048 SIL1 Flödesmätare till förquench processgaskrets (353FI305|L2) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

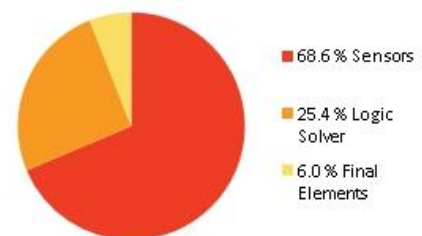
Table 307 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61508	SYSTEMATIC CAPABILITY	
4.63E-03	215.9	2	1	2	18.84

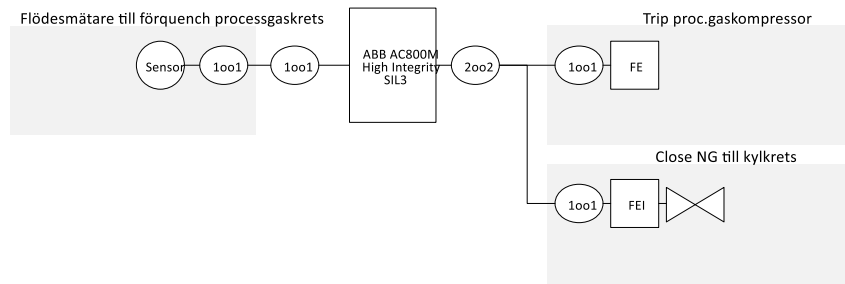
PFD_{avg} Contribution



MTTFS Contribution



The SIF Conceptual Design is modeled as shown in the picture below.



40.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-048 SIL1 Flödesmätare till förquench processgaskrets (353FI305 | L2) Safety Instrumented Function is quantified as follows.

Table 308 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61508	SYSTEMATIC CAPABILITY			
1.01E-03	2	2	27.45	0	2

Number of Sensor group(s): 1
Voting between groups: 1001
β-factor [%]: N/A

40.3.1.1 Sensor Group 1: Flödesmätare till förquench processgaskrets

The information and reliability data underneath describe the Flödesmätare till förquench processgaskrets sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001 Voting type: Identical
HFT: 0 β-factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Promag P200		-	-	B	2
Trip Direction: Low, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off					
ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
Analog In Module: AI880A/TU844/845/TY801	2013.1.02	-	-	B	3
MRT [Hours]: 24					
Proof Test Interval [Months]: 12			Proof Test Coverage [%]: 98		

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Flödesmätare till förquench processgaskrets.

Table 309 Reliability Data Sensor Group Flödesmätare till förquench processgaskrets

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Promag P200	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	3.39E-06 <i>3.39E-06</i>	0.00E+00 <i>0.00E+00</i>	1.55E-07 <i>1.55E-07</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
Analog In Module: AI880A/TU844/845/TY80 1	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	3.69E-07 <i>7.53E-07</i>	5.00E-10 <i>5.00E-10</i>	3.84E-07 <i>0.00E+00</i>	1.32E-10 <i>1.32E-10</i>	3.89E-08 <i>3.89E-08</i>	3.93E-10 <i>3.93E-10</i>	2.11E-07 <i>2.11E-07</i>
								SFF [%]		96.4
								ROUTE 2H COMPLIANT		-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

40.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-048 SIL1 Flödesmätare till förquench processgaskrets (353FI305 | L2) Safety Instrumented Function is quantified as follows.

Table 310 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61508	SYSTEMATIC CAPABILITY			
3.59E-09	3	3	74.24	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	3
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β-factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 311 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	100.0

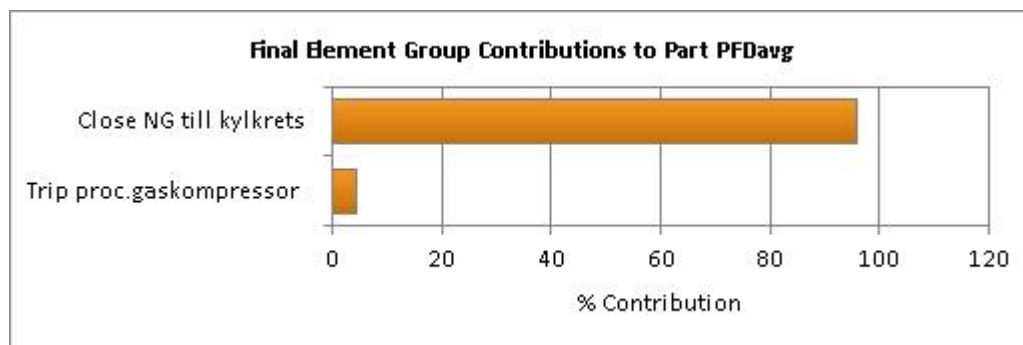
40.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-048 SIL1 Flödesmätare till förquench processgaskrets (353FI305 | L2) Safety Instrumented Function is quantified as follows.

Table 312 Final Element Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61508	SYSTEMATIC CAPABILITY			
3.63E-03	1	3	314.84	0	2

Number of Final Element group(s): 2
Voting between groups: 2oo2
β-factor [%]:



40.3.3.1 Final Element Group 1: Trip proc.gaskompressor

The information and reliability data underneath describe the Trip proc.gaskompressor final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	1oo1	Voting type:	Identical		
HFT:	0	β-factor [%]:	N/A		
Equipment Leg (each):		SERH VERSION	2H	PIU	ACTY
ASC880 R2 STO			-	-	A
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02		-	-	B
Digital Out Module: DO880/TU842/843	2013.1.02		-	-	B
MRT [Hours]:	24				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	90		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Trip proc.gaskompressor.

Table 313 Reliability Data Final Element Group Trip proc.gaskompressor

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
ASC880 R2 STO		1.08E-08		1.35E-08			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
						SFF [%]	89.5
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

40.3.3.2 Final Element Group 2: Close NG till kylkrets

The information and reliability data underneath describe the Close NG till kylkrets final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	1oo1	Voting type:	Identical		
HFT:	0	β-factor [%]:	N/A		
Equipment Leg (each):		SERH VERSION	2H	PIU	ACTY
Parker N			-	-	A
Rotork			-	-	A
Transworld Mars			-	-	A
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02		-	-	B
Digital Out Module: DO880/TU842/843	2013.1.02		-	-	B
Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes					
MRT [Hours]:	24				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	97		

Proof Test Execution: Duration 1 hours
 Partial Valve Stroke Test Interval 6M Partial Valve Stroke Test 77
 Coverage[%]
 Partial Valve Stroke Test Execution Manual

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close NG till kylkrets.

Table 314 Reliability Data Final Element Group Close NG till kylkrets

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Transworld Mars				3.73E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
					SFF [%]		49.2
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

41 SIF-075 SIL2 Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck ((253PZ323, 253PZ324, 253PZ325) - (343PZ305, 343PZ307, 343PZ308) < 0,5) => Close N2 till reduktionskrets (254SV582)

This chapter details the SIL Verification results, selections, and assumptions for the SIF-075 SIL2 Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck ((253PZ323, 253PZ324, 253PZ325) - (343PZ305, 343PZ307, 343PZ308) < 0,5) => Close N2 till reduktionskrets (254SV582) Safety Instrumented Function.

41.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name:	SIL2 Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck ((253PZ323, 253PZ324, 253PZ325) - (343PZ305, 343PZ307, 343PZ308) < 0,5) => Close N2 till reduktionskrets (254SV582)
SIF Tag	SIF-075
SIF Description:	SIL2 Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck ((253PZ323, 253PZ324, 253PZ325) - (343PZ305, 343PZ307, 343PZ308) < 0,5) => Close N2 till reduktionskrets (254SV582)

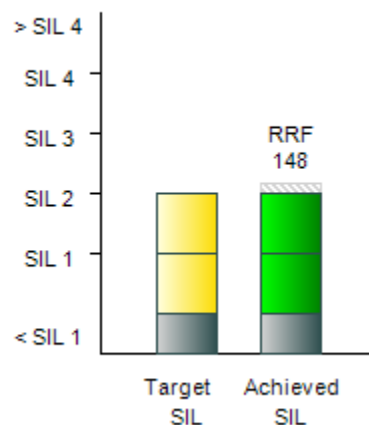
41.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 148.7



41.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-075 SIL2 Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck ((253PZ323, 253PZ324, 253PZ325) - (343PZ305, 343PZ307, 343PZ308) < 0,5) => Close N2 till reduktionskrets (254SV582) Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years
Startup time: 24 hours

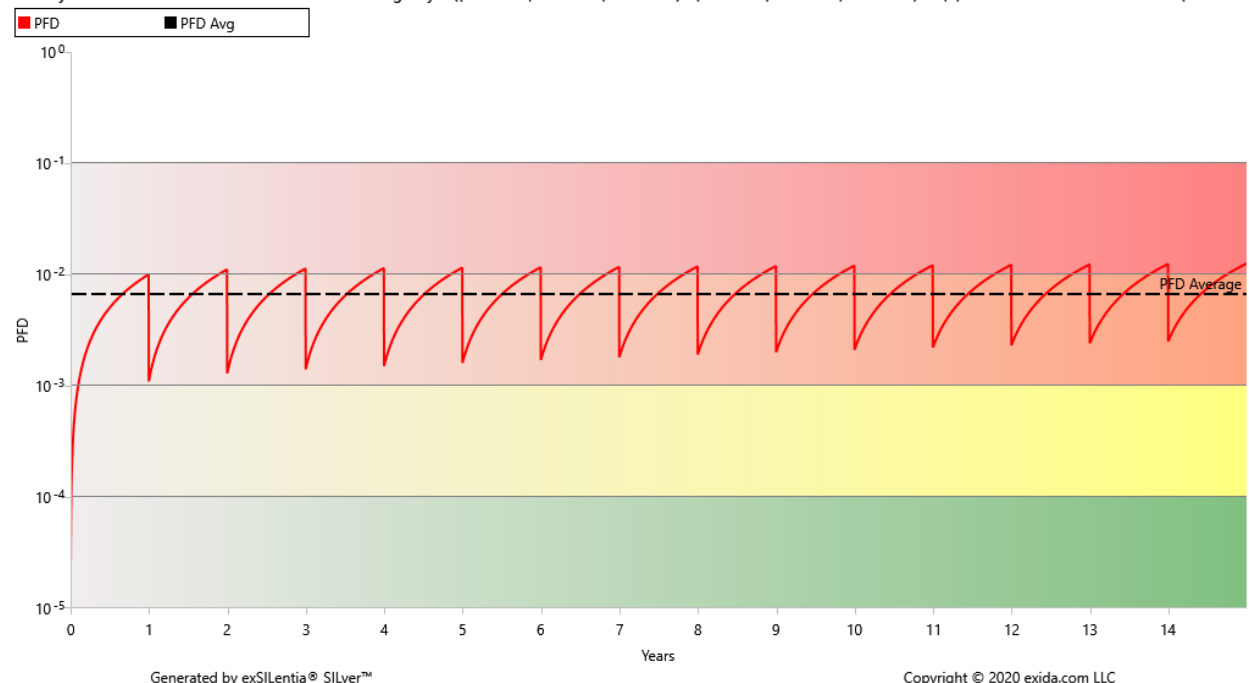
The SIF operates in the Low demand mode.

Analyst: Analysis Date: 10/4/2019

Design References

Comments:

SIL2 Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck ((253PZ323, 253PZ324, 253PZ325) - (343PZ305, 343PZ307, 343PZ308) < 0,5) => Close N2 till reduktionskrets (254SV582)



Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-075 SIL2 Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck ((253PZ323, 253PZ324, 253PZ325) - (343PZ305, 343PZ307, 343PZ308) < 0,5) => Close N2 till reduktionskrets (254SV582) Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

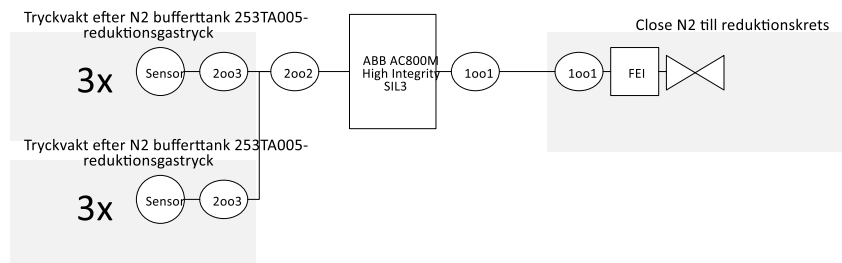
Table 315 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
6.72E-03	148.7	2	2	N/A	65.02



The SIF Conceptual Design is modeled as shown in the picture below.



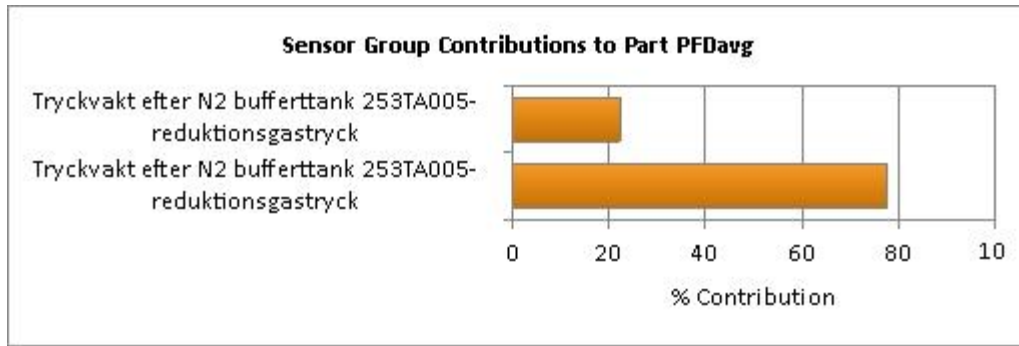
41.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-075 SIL2 Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck ((253PZ323, 253PZ324, 253PZ325) - (343PZ305, 343PZ307, 343PZ308) < 0,5) => Close N2 till reduktionskrets (254SV582) Safety Instrumented Function is quantified as follows.

Table 316 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
5.37E-04	3	N/A	10637.24	1	2

Number of Sensor group(s): 2
Voting between groups: 2oo2
β-factor [%]:



41.3.1.1 Sensor Group 1: Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck

The information and reliability data underneath describe the Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo3 Voting type: Identical

HFT: 1 β -factor: 10
[%]:

Equipment Leg (each): „

	SERH VERSION	2H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: High, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
---	-----------	---	---	---	---

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck.

Table 317 Reliability Data Sensor Group Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>6.85E-08</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>4.44E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	18.5
									ROUTE 2H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

41.3.1.2 Sensor Group 2: Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck

The information and reliability data underneath describe the Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group:	2oo3	Voting type:	Identical				
HFT:	1	β-factor [%]:	10				
Equipment Leg (each):	„						
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75			SERH VERSION	2H	PIU	AC TYPE	SIL CAP
			2006.2.02	-	-	B	2
	Trip Direction: Low, Trip On Alarm: Off, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off						
ABB AC800M High Integrity SIL3 [AnalogIn]			2013.1.02	-	-	B	3
MRT [Hours]:	24						
Proof Test Interval [Months]:		12		Proof Test Coverage [%]:		99	
Proof Test Execution:		Duration 1 hours					

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck.

Table 318 Reliability Data Sensor Group Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.79E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.34E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>4.28E-09</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>1.83E-08</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	75.5
									ROUTE 2H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

41.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-075 SIL2 Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck ((253PZ323, 253PZ324, 253PZ325) - (343PZ305, 343PZ307, 343PZ308) < 0,5) => Close N2 till reduktionskrets (254SV582) Safety Instrumented Function is quantified as follows.

Table 319 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.56E-08	3	N/A	72.04	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A

Logic Solver Voting: Architecture Type: B

HFT: 1 β -factor [%]: 1

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 320 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

41.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-075 SIL2 Tryckvakt efter N2 bufferttank 253TA005-reduktionsgastryck ((253PZ323, 253PZ324, 253PZ325) - (343PZ305, 343PZ307, 343PZ308) < 0,5) => Close N2 till reduktionskrets (254SV582) Safety Instrumented Function is quantified as follows.

Table 321 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.19E-03	2	N/A	711.69	0	2

Number of Final Element group(s): 1
Voting between groups: 1001
 β -factor [%]: N/A

41.3.3.1 Final Element Group 1: Close N2 till reduktionskrets

The information and reliability data underneath describe the Close N2 till reduktionskrets final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1001 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Transworld Mars		-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 95
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Close N2 till reduktionskrets.

Table 322 Reliability Data Final Element Group Close N2 till reduktionskrets

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Transworld Mars				3.73E-07			
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
SFF [%]							49.2
ROUTE 2 _H COMPLIANT							-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

42 SIF-079 SIL2 252PZ337 (H2/L2) => close ånga, H2, O2

This chapter details the SIL Verification results, selections, and assumptions for the SIF-079 SIL2 252PZ337 (H2/L2) => close ånga, H2, O2 Safety Instrumented Function.

42.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL2 252PZ337 (H2/L2) => close ånga, H2, O2
 SIF Tag: SIF-079
 SIF Description: SIL2 252PZ337 (H2/L2) =>

Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)
 Close Avstängningsventil 1 H2 till brännare PGH (341SV665 SIL2)
 Close Avstängningsventil 2 H2 till brännare PGH (341SV667 SIL2)

Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)
 Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

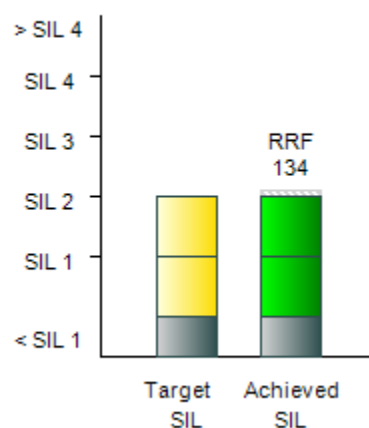
42.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 134.4



42.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-079 SIL2 252PZ337 (H2/L2) => close ånga, H2, O2 Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

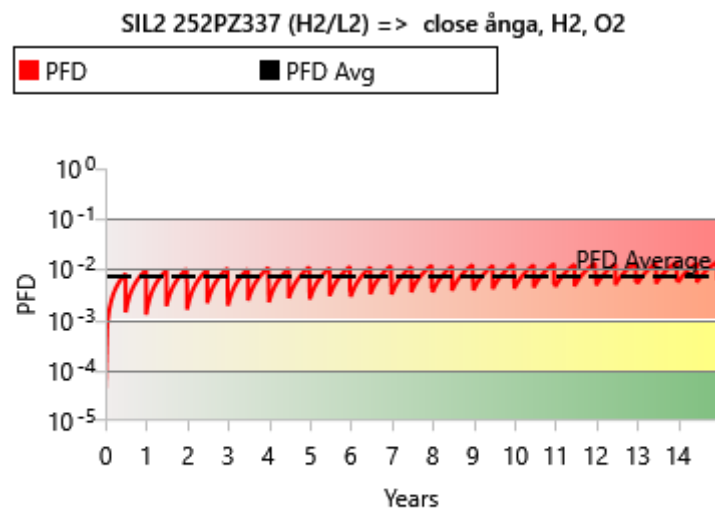
The SIF operates in the Low demand mode.

Analyst:

Analysis Date: 10/4/2019

Design References

Comments:



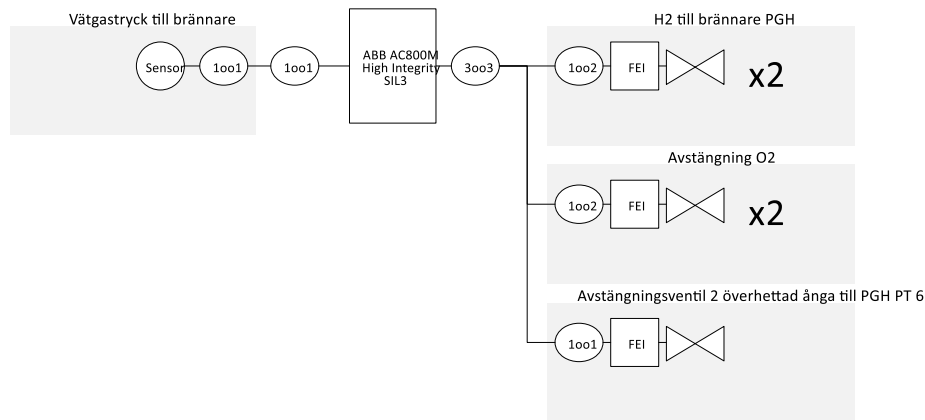
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-079 SIL2 252PZ337 (H2/L2) => close ånga, H2, O2 Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 323 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
7.44E-03	134.4	2	2	N/A	51.18



The SIF Conceptual Design is modeled as shown in the picture below.



42.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-079 SIL2 252PZ337 (H2/L2) => close ånga, H2, O2 Safety Instrumented Function is quantified as follows.

Table 324 Sensor Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.89E-04	2	N/A	273.24	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

42.3.1.1 Sensor Group 1: Vätgastryck till brännare

The information and reliability data underneath describe the Vätgastryck till brännare sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β -factor [%]: N/A

Equipment Leg (each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Vätgastryck till brännare.

Table 325 Reliability Data Sensor Group Vätgastryck till brännare

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.96E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.17E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
									SFF [%]	78.5
									ROUTE 2 _H COMPLIANT	-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

42.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-079 SIL2 252PZ337 (H2/L2) => close ånga, H2, O2 Safety Instrumented Function is quantified as follows.

Table 326 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
5.49E-08	3	N/A	67.55	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	2013.1.02	PIU	-	SIL Cap	N/A
Logic Solver Voting:		Architecture Type:	B				
HFT:	1	β -factor [%]:	1				
MRT [Hours]:	24						
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99				
Proof Test Execution:	Duration 1 hours						

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 327 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

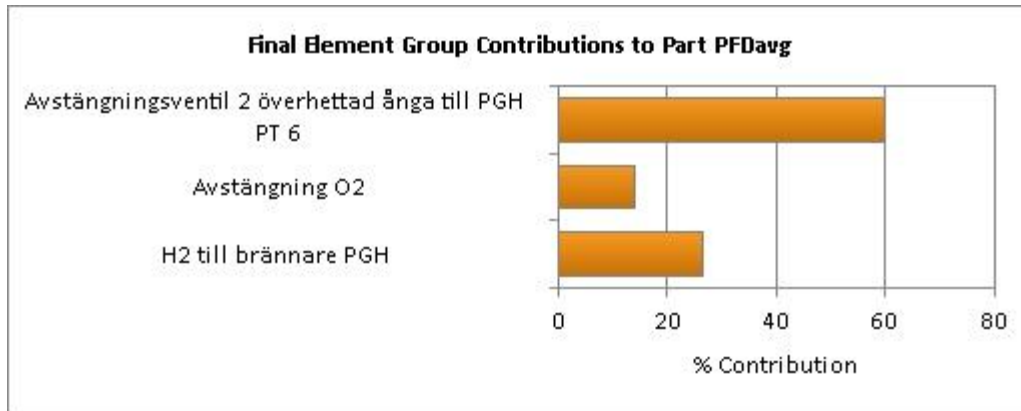
42.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-079 SIL2 252PZ337 (H2/L2) => close ånga, H2, O2 Safety Instrumented Function is quantified as follows.

Table 328 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.75E-03	2	N/A	931.52	0	2

Number of Final Element group(s): 3
Voting between groups: 3oo3
 β -factor [%]:



42.3.3.1 Final Element Group 1: H2 till brännare PGH

The information and reliability data underneath describe the H2 till brännare PGH final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2

Voting type: Identical

HFT: 1

β -factor [%]: 10

Equipment Leg
(each): ,

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24

Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 95

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group H2 till brännare PGH.

Table 329 Reliability Data Final Element Group H2 till brännare PGH

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	5.9
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

42.3.3.2 Final Element Group 2: Avstängning O2

The information and reliability data underneath describe the Avstängning O2 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 99
Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2.

Table 330 Reliability Data Final Element Group Avstängning O2

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
SFF [%]							28.4
ROUTE 2 _H COMPLIANT							-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

42.3.3.3 Final Element Group 3: Avstängningsventil 2 överhettad ånga till PGH PT 6

The information and reliability data underneath describe the Avstängningsventil 2 överhettad ånga till PGH PT 6 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24

Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängningsventil 2 överhettad ånga till PGH PT 6.

Table 331 Reliability Data Final Element Group Avstängningsventil 2 överhettad ånga till PGH PT 6

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				5.55E-07			2.02E-06
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
SFF [%]							45.3
ROUTE 2 _H COMPLIANT							-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

43 SIF-080 SIL1 271XI323 Ånga ej tillräckligt överhettad => permissive 271SV556

This chapter details the SIL Verification results, selections, and assumptions for the SIF-080 SIL1 271XI323 Ånga ej tillräckligt överhettad => permissive 271SV556

Safety Instrumented Function.

43.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL1 271XI323 Ånga ej tillräckligt överhettad => permissive 271SV556

SIF Tag SIF-080

SIF Description: SIL1 271XI323 Ånga ej tillräckligt överhettad => permissive 271SV556

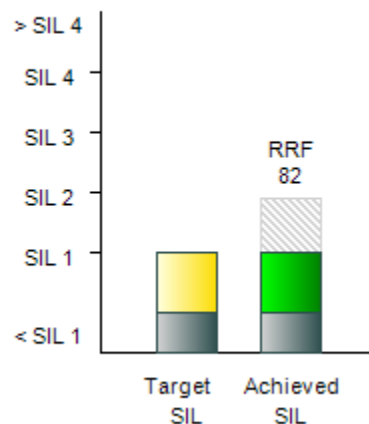
43.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 1 with RRF ≥ 10

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 1 with RRF = 82



43.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-080 SIL1 271XI323 Ånga ej tillräckligt överhettad => permissive 271SV556

Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

The SIF operates in the Low demand mode.

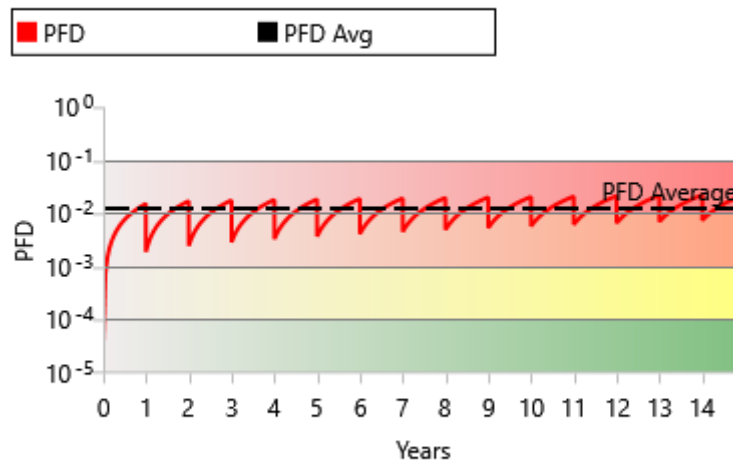
Analyst:

Analysis Date: 4/28/2020

Design References

Comments:

SIL1 271XI323 Ånga ej tillräckligt överhettad => permissive 271SV556



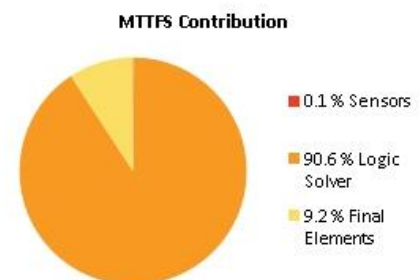
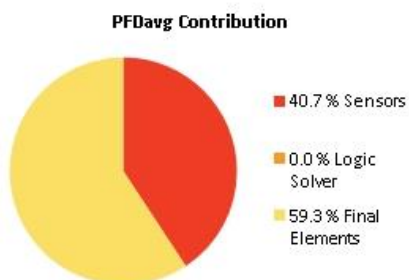
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Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-080 SIL1 271XI323 Ånga ej tillräckligt överhettad => permissive 271SV556

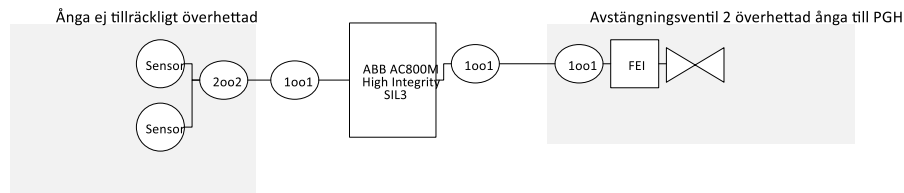
Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 332 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
1.22E-02	82	1	2	N/A	65.68



The SIF Conceptual Design is modeled as shown in the picture below.



43.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-080 SIL1 271XI323 Ånga ej tillräckligt överhettad => permissive 271SV556

Safety Instrumented Function is quantified as follows.

Table 333 Sensor Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
4.98E-03	2	N/A	47842.09	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

43.3.1.1 Sensor Group 1: Ånga ej tillräckligt överhettad

The information and reliability data underneath describe the Ånga ej tillräckligt överhettad sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 2oo2 Voting type: Diverse
HFT: 0 β-factor [%]: 10

Equipment Leg (1):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser iTemp TMT 182	2006.2.02	-	-	B	-
Trip Direction: None, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off					
ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
Equipment Leg (2):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2
Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off					
ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3

MRT [Hours]: 24
 Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 93.4
 Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Ånga ej tillräckligt överhettad.

Table 334 Reliability Data Sensor Group Ånga ej tillräckligt överhettad

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
271TI322										
Endress+Hauser iTemp TMT 182	1.66E-08 <i>0.00E+00</i>	9.16E-08 <i>0.00E+00</i>	9.18E-08 <i>0.00E+00</i>	0.00E+00 <i>2.00E-07</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.07E-07 <i>1.07E-07</i>	0.00E+00 <i>0.00E+00</i>	1.22E-09 <i>1.22E-09</i>	9.78E-08 <i>9.78E-08</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]		67.7	
							ROUTE 2 _H COMPLIANT		-	
271PI320										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>6.85E-08</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>4.44E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]		18.5	
							ROUTE 2 _H COMPLIANT		-	

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

43.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-080 SIL1 271XI323 Ånga ej tillräckligt överhettad => permissive 271SV556

Safety Instrumented Function is quantified as follows.

Table 335 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.01E-08	3	N/A	72.46	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High

Integrity SIL3

Equipment:		SERH VERSION	PIU	SIL Cap
	ABB AC800M High Integrity SIL3	2013.1.02	-	N/A
Logic Solver Voting:		Architecture Type:	B	
HFT:	1	β -factor [%]:	1	
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 336 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
SFF [%]							100.0

43.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-080 SIL1 271XI323 Ånga ej tillräckligt överhettad => permissive 271SV556

Safety Instrumented Function is quantified as follows.

Table 337 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
7.25E-03	2	N/A	712.45	0	2

Number of Final Element group(s): 1
Voting between groups: 1001
 β -factor [%]: N/A

43.3.3.1 Final Element Group 1: Avstängningsventil 2 överhettad ånga till PGH

The information and reliability data underneath describe the Avstängningsventil 2 överhettad ånga till PGH final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1

Voting type:

Identical

HFT: 0

β -factor [%]:

N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24

Proof Test Interval [Months]:

12

Proof Test Coverage [%]:

99

Proof Test Execution:

Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängningsventil 2 överhettad ånga till PGH.

Table 338 Reliability Data Final Element Group Avstängningsventil 2 överhettad ånga till PGH

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				5.55E-07			2.02E-06
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
					SFF [%]		11.5
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

44 SIF-081 SIL2 Mätvärde på tillsatt flöde naturgas 252XI565*K

This chapter details the SIL Verification results, selections, and assumptions for the SIF-081 SIL2 Mätvärde på tillsatt flöde naturgas 252XI565*K Safety Instrumented Function.

44.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL2 Mätvärde på tillsatt flöde naturgas 252XI565*K
 SIF Tag: SIF-081
 SIF Description: SIL2 Mätvärde på tillsatt flöde naturgas 252XI565*K

Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)
 Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

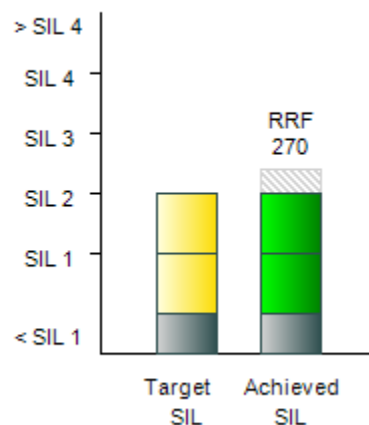
44.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 270.7



44.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-081 SIL2 Mätvärde på tillsatt flöde naturgas 252XI565*K Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

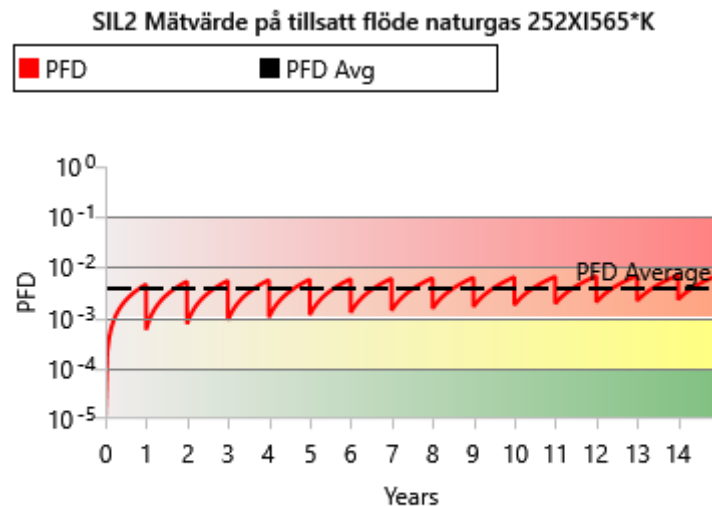
The SIF operates in the Low demand mode.

Analyst:

Analysis Date: 10/4/2019

Design References

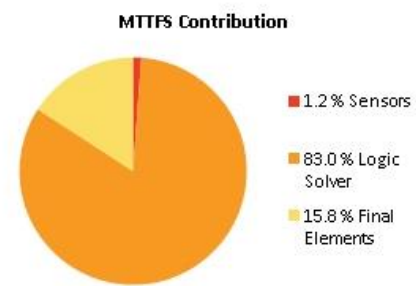
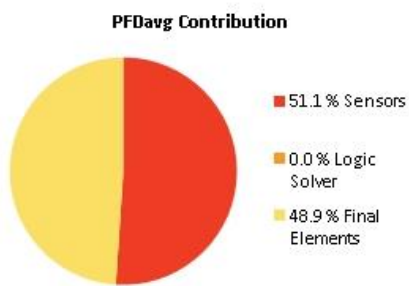
Comments:



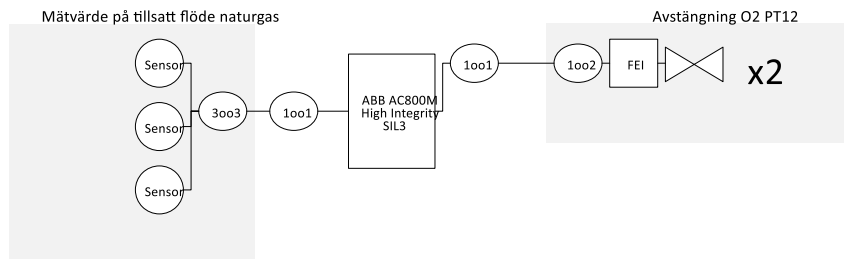
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-081 SIL2 Mätvärde på tillsatt flöde naturgas 252XI565*K Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 339 Functional Safety Performance

PFDavg	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFDavg	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
3.69E-03	270.7	2	2	N/A	59.79



The SIF Conceptual Design is modeled as shown in the picture below.



44.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-081 SIL2 Mätvärde på tillsatt flöde naturgas 252XI565*K Safety Instrumented Function is quantified as follows.

Table 340 Sensor Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.89E-03	2	N/A	5138.39	0	2

Number of Sensor group(s): 1
Voting between groups: 1oo1
β-factor [%]: N/A

44.3.1.1 Sensor Group 1: Mätvärde på tillsatt flöde naturgas

The information and reliability data underneath describe the Mätvärde på tillsatt flöde naturgas sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 3oo3 Voting type: Diverse
HFT: 0 β-factor [%]: 10

Equipment Leg (1):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Deltabar S PMD75, FMD77, FMD78	2006.2.02	-	-	B	2

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: On, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
Equipment Leg (2):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser iTemp TMT 182	2006.2.02	-	-	B	-

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: On, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
Equipment Leg (3):	SERH VERSION	2_H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: On, Alarm Setting: Under Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn]	2013.1.02	-	-	B	3
MRT [Hours]:	24				
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	95		
Proof Test Execution:	Duration 1 hours				

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Mätvärde på tillsatt flöde naturgas.

Table 341 Reliability Data Sensor Group Mätvärde på tillsatt flöde naturgas

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
252PD565										
Endress+Hauser Deltabar S PMD75, FMD77, FMD78	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>4.48E-07</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	6.89E-08 <i>6.89E-08</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	4.30E-07 <i>4.30E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]		87.5	
							ROUTE 2 _H COMPLIANT		-	
252TI565										
Endress+Hauser iTemp TMT 182	1.66E-08 <i>0.00E+00</i>	9.16E-08 <i>0.00E+00</i>	9.18E-08 <i>0.00E+00</i>	0.00E+00 <i>2.00E-07</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.07E-07 <i>1.07E-07</i>	0.00E+00 <i>0.00E+00</i>	1.22E-09 <i>1.22E-09</i>	9.78E-08 <i>9.78E-08</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]		67.7	
							ROUTE 2 _H COMPLIANT		-	
252PI565										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>4.48E-07</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>6.53E-08</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
							SFF [%]		88.0	
							ROUTE 2 _H COMPLIANT		-	

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

44.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-081 SIL2 Mätvärde på tillsatt flöde naturgas 252XI565*K Safety Instrumented Function is quantified as follows.

Table 342 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
2.84E-08	3	N/A	72.03	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A
Logic Solver Voting:	Architecture Type:	B		
HFT:	β-factor [%]:	1		
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 343 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
						SFF [%]	100.0

44.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-081 SIL2 Mätvärde på tillsatt flöde naturgas 252XI565*K Safety Instrumented Function is quantified as follows.

Table 344 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
1.81E-03	3	N/A	377.66	1	2

Number of Final Element group(s): 1
 Voting between groups: 1001
 β-factor [%]: N/A

44.3.3.1 Final Element Group 1: Avstängning O2 PT12

The information and reliability data underneath describe the Avstängning O2 PT12 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1002 Voting type: Identical
 HFT: 1 β-factor [%]: 10

Equipment Leg (each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
 Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99
 Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2 PT12.

Table 345 Reliability Data Final Element Group Avstängning O2 PT12

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	5.9
						ROUTE 2 _H COMPLIANT	-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

45 SIF-082 SIL2 252PZ337 (H2/L2) => close ånga, NG, O2

This chapter details the SIL Verification results, selections, and assumptions for the SIF-082 SIL2 252PZ337 (H2/L2) => close ånga, NG, O2 Safety Instrumented Function.

45.1 General SIF Information

The following characterizes the Safety Instrumented Function.

SIF Name: SIL2 252PZ337 (H2/L2) => close ånga, NG, O2
 SIF Tag: SIF-082
 SIF Description: SIL2 252PZ337 (H2/L2) =>

Close Avstängningsventil 2 överhettad ånga till PGH (271SV556 SIL2)
 Close Avstängningsventil 1 NG till brännare PGH (252SV544 SIL2)
 Close Avstängningsventil 2 NG till brännare PGH (252SV546 SIL2)

Close Avstängningsventil 1 O2 inj. (251SV555 SIL2)
 Close Avstängningsventil 2 O2 inj. (251SV556 SIL2)

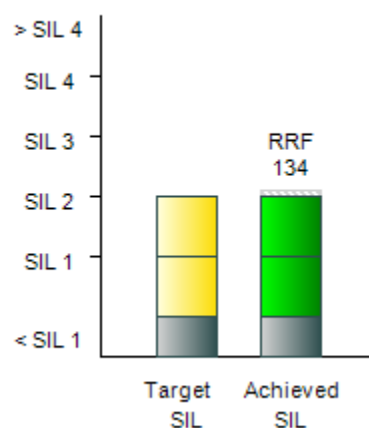
45.2 Safety Integrity Levels

The target Safety Integrity Level determined for this Safety Instrumented Function is:

SIL 2 with RRF ≥ 100

The calculated achieved Safety Integrity Level for this Safety Instrumented Function is:

SIL 2 with RRF = 134.2



45.3 SIL Verification

This section provides a detailed overview of the Safety Integrity Level verification performed for the SIF-082 SIL2 252PZ337 (H2/L2) => close ånga, NG, O2 Safety Instrumented Function. In order to perform the reliability calculations part of the Safety Integrity Level verification, the following assumptions have been made.

Mission Time: 15 years

Startup time: 24 hours

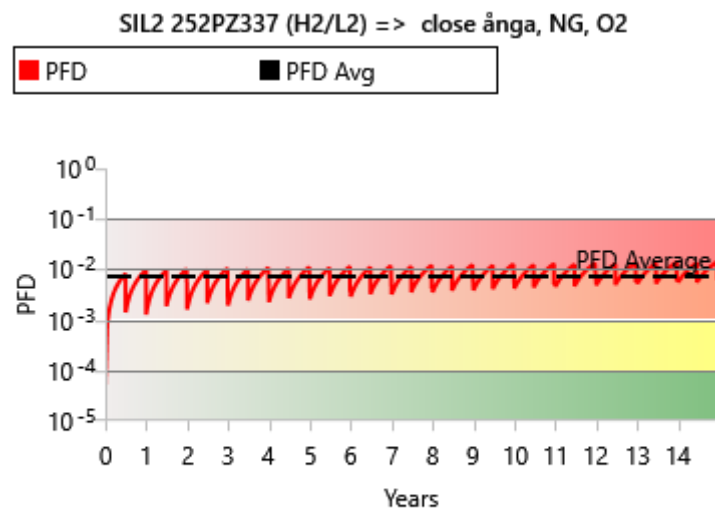
The SIF operates in the Low demand mode.

Analyst:

Analysis Date: 10/4/2019

Design References

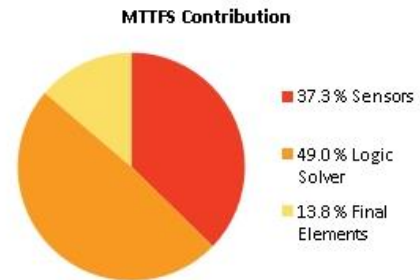
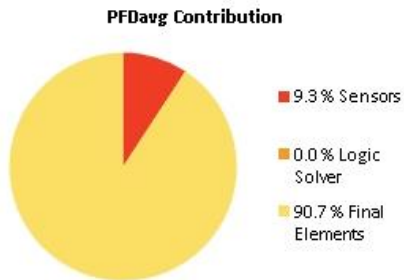
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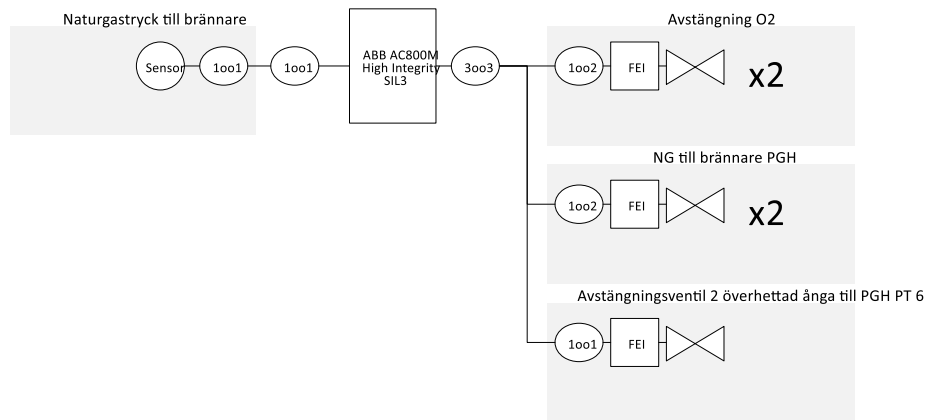
Given the reliability data and SIL verification selections and assumptions described in the subsequent subsections in this report the SIF-082 SIL2 252PZ337 (H2/L2) => close ånga, NG, O2 Safety Instrumented Function achieves the functional safety performance as displayed in the Functional Safety Performance table.

Table 346 Functional Safety Performance

PFD _{AVG}	RRF	ACHIEVED SIL			MTTFS (YEARS)
		PFD _{AVG}	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY	
7.45E-03	134.2	2	2	N/A	36.35



The SIF Conceptual Design is modeled as shown in the picture below.



45.3.1 Sensor Part Configuration

The functional safety and spurious trip behavior of the sensor part of the SIF-082 SIL2 252PZ337 (H2/L2) => close ånga, NG, O2 Safety Instrumented Function is quantified as follows.

Table 347 Sensor Part Functional Safety Performance

PFDavg	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.90E-04	2	N/A	97.51	0	2

Number of Sensor group(s): 1
Voting between groups: 1001
 β -factor [%]: N/A

45.3.1.1 Sensor Group 1: Naturgastryck till brännare

The information and reliability data underneath describe the Naturgastryck till brännare sensor group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical

HFT: 0 β -factor [%]: N/A

Equipment Leg (each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	2006.2.02	-	-	B	2

Trip Direction: High, Trip On Alarm: On, Over / Under Range Detection: Off, Alarm Setting: Over Range, Alarm Filter: Off, External Comparison: Off

ABB AC800M High Integrity SIL3 [AnalogIn] 2013.1.02 - - B 3

Analog In Module: AI880A/TU844/845/TY801 2013.1.02 - - B 3

MRT [Hours]: 24

Proof Test Interval [Months]: 12 Proof Test Coverage [%]: 99

Proof Test Execution: Duration 1 hours

The Reliability Data Sensor Group table shows the reliability data used during the SIL verification of sensor group Naturgastryck till brännare.

Table 348 Reliability Data Sensor Group Naturgastryck till brännare

COMPONENT	FAILURE RATES [1/HR]									
	FAIL LOW	FAIL HIGH	FAIL DET.	SD	SU	DD	DU	AD	AU	NE
Each Leg										
Endress+Hauser Cerabar S PMP71, PMP72 HT, PMP75	5.21E-08 <i>0.00E+00</i>	6.85E-08 <i>0.00E+00</i>	3.27E-07 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>3.96E-07</i>	0.00E+00 <i>0.00E+00</i>	6.53E-08 <i>1.17E-07</i>	0.00E+00 <i>0.00E+00</i>	1.00E-08 <i>1.00E-08</i>	3.03E-07 <i>3.03E-07</i>
ABB AC800M High Integrity SIL3 [AnalogIn]	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	4.28E-09 <i>2.26E-08</i>	4.28E-13 <i>4.28E-13</i>	1.83E-08 <i>0.00E+00</i>	1.83E-12 <i>1.83E-12</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	1.09E-08 <i>1.09E-08</i>
Analog In Module: AI880A/TU844/845/TY801	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	0.00E+00 <i>0.00E+00</i>	3.69E-07 <i>7.53E-07</i>	5.00E-10 <i>5.00E-10</i>	3.84E-07 <i>0.00E+00</i>	1.32E-10 <i>1.32E-10</i>	3.89E-08 <i>3.89E-08</i>	3.93E-10 <i>3.93E-10</i>	2.11E-07 <i>2.11E-07</i>
										SFF [%]
										91.2
										ROUTE 2 _H COMPLIANT
										-

The data shown in blue and italic indicates the effect of the PLC detection configuration performed by the logic solver on the sensor group data.

The failure rates displayed in red & italic font show the adjusted failure rates due to the External Comparison.

45.3.2 Logic Solver Part Configuration

The functional safety and spurious trip behavior of the logic solver part of the SIF-082 SIL2 252PZ337 (H2/L2) => close ånga, NG, O2 Safety Instrumented Function is quantified as follows.

Table 349 Logic Solver Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
3.59E-09	3	N/A	74.24	1	2

The information and reliability data underneath describe the ABB AC800M High Integrity SIL3 logic solver group as it has been analyzed in this Safety Integrity Level verification.

Logic Solver Name: ABB AC800M High Integrity SIL3

Equipment:	ABB AC800M High Integrity SIL3	SERH VERSION	PIU	SIL Cap
		2013.1.02	-	N/A
Logic Solver Voting:		Architecture Type:	B	
HFT:	1	β-factor [%]:	1	
MRT [Hours]:	24			
Proof Test Interval [Months]:	12	Proof Test Coverage [%]:	99	
Proof Test Execution:	Duration 1 hours			

The Reliability Data Logic Solver Group table shows the reliability data used during the SIL verification of logic solver group ABB AC800M High Integrity SIL3.

Table 350 Reliability Data Logic Solver Group ABB AC800M High Integrity SIL3

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Main Processor: PM865/TP830 SM811/TP868 (SM810/P855)	4.51E-07	7.54E-07	1.46E-06	5.56E-11	3.63E-09	1.19E-08	6.79E-07
Power Supply: 1*SD821 + 1*SS823 + 2*MCB	1.53E-06	9.06E-10			2.16E-08	4.52E-10	1.47E-07
Analog In Module: AI880A/TU844/845/TY801	3.69E-07	5.00E-10	3.84E-07	1.32E-10	3.89E-08	3.93E-10	2.11E-07
Channel: Analog In	4.28E-09	4.28E-13	1.83E-08	1.83E-12			1.09E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
Channel: Digital Out	8.53E-08	1.91E-08					4.07E-08
						SFF [%]	100.0

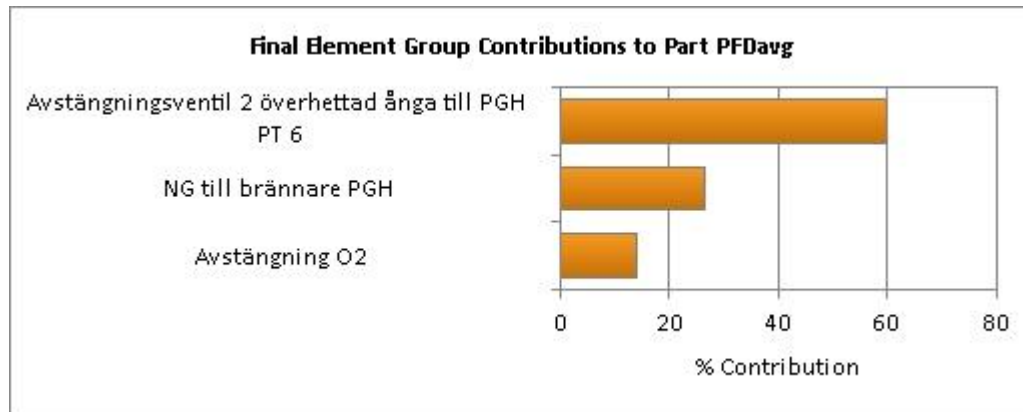
45.3.3 Final Element Part Configuration

The functional safety and spurious trip behavior of the final element part of the SIF-082 SIL2 252PZ337 (H2/L2) => close ånga, NG, O2 Safety Instrumented Function is quantified as follows.

Table 351 Final Element Part Functional Safety Performance

PFD _{AVG}	SIL LIMITS		MTTFS (YEARS)	HFT	SSI
	ARCH. CONSTRAINTS IEC 61511	SYSTEMATIC CAPABILITY			
6.76E-03	2	N/A	264.15	0	2

Number of Final Element group(s): 3
Voting between groups: 3oo3
 β -factor [%]:



45.3.3.1 Final Element Group 1: Avstängning O2

The information and reliability data underneath describe the Avstängning O2 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical
HFT: 1 β -factor [%]: 10

Equipment Leg
(each): ,

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3
Digital Out Module: DO880/TU842/843	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24
Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 99
Proof Test Execution: Leak Test; Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängning O2.

Table 352 Reliability Data Final Element Group Avstängning O2

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
							SFF [%]
							28.4
							ROUTE 2 _H COMPLIANT
							-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

45.3.3.2 Final Element Group 2: NG till brännare PGH

The information and reliability data underneath describe the NG till brännare PGH final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo2 Voting type: Identical

HFT: 1 β -factor [%]: 10

Equipment Leg

(each):	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3
Digital Out Module: DO880/TU842/843	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: No

MRT [Hours]: 24

Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 95

Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group NG till brännare PGH.

Table 353 Reliability Data Final Element Group NG till brännare PGH

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				1.90E-06			6.72E-07
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
							SFF [%]
							28.4
							ROUTE 2 _H COMPLIANT
							-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

45.3.3.3 Final Element Group 3: Avstängningsventil 2 överhettad ånga till PGH PT 6

The information and reliability data underneath describe the Avstängningsventil 2 överhettad ånga till PGH PT 6 final element group as it has been analyzed in this Safety Integrity Level verification.

Voting within group: 1oo1 Voting type: Identical
HFT: 0 β -factor [%]: N/A

Equipment Leg
(each):

	SERH VERSION	2 _H	PIU	AC TYPE	SIL CAP
Parker N		-	-	A	3
Rotork		-	-	A	3
Pekos Floating Ball Valve, metal seat, < 8"	2010.1.01	-	-	A	3
ABB AC800M High Integrity SIL3 [DigitalOut]	2013.1.02	-	-	B	3
Digital Out Module: DO880/TU842/843	2013.1.02	-	-	B	3

Clean Service: Yes, Close on Trip: Yes, Full Stroke: Yes

MRT [Hours]: 24
Proof Test Interval [Months]: 6 Proof Test Coverage [%]: 99
Proof Test Execution: Duration 1 hours

The Reliability Data Final Element Group table shows the reliability data used during the SIL verification of final element group Avstängningsventil 2 överhettad ånga till PGH PT 6.

Table 354 Reliability Data Final Element Group Avstängningsventil 2 överhettad ånga till PGH PT 6

COMPONENT	FAILURE RATES [1/HR]						
	SD	SU	DD	DU	AD	AU	NE
Each Leg							
Parker N		5.70E-08		6.54E-07			
Rotork				3.15E-08			
Pekos Floating Ball Valve, metal seat, < 8"				5.55E-07			2.02E-06
ABB AC800M High Integrity SIL3 [DigitalOut]	8.53E-08	1.91E-08					4.07E-08
Digital Out Module: DO880/TU842/843	3.98E-07	9.79E-09	4.17E-07	1.36E-10	3.89E-08	3.93E-10	3.68E-07
					SFF [%]		45.3
					ROUTE 2 _H COMPLIANT		-

The failure rates displayed in blue & italic font show the adjusted failure rates due to the Partial Valve Stroke Test performed (if any). Diagnostic coverage factor(s) for the Partial Stroke Test is (are) provided on a per leg basis

46 SILver™ Group Reuse Overview

This chapter provides an overview of the various groups that were defined during the SIL verification analyses of the Hybrit Pilot Plant project. The reuse of groups is available between Safety Instrumented Functions to account for identical hardware used in those different Safety Instrumented Functions. The group reuse allows users to specify a specific group once, reuse it, and with all future changes made to a single group cascading to all Safety Instrumented Functions it is used in. It is instrumental for the user to review all SIFs a single group is used in when changes are made to that group.

This overview indicates the groups that were reused, the number of times they were reused, and the Safety Instrumented Functions that they were used in.

46.1 Sensor Groups Reused

Gastrück toppgas (used 4 times)	SIF-013, SIF-014, SIF-011, SIF-012
Gastrück transferline (used 3 times)	SIF-021, SIF-024, SIF-020
Tryckvakt O2-inj (used 3 times)	SIF-022, SIF-023, SIF-020
Temperatur konvektionsdel (used 2 times)	SIF-027, SIF-028

46.2 Logic Solver Groups Reused

ABB AC800M High Integrity SIL3 (used 72 times)	SIF-996, SIF-997, SIF-998, SIF-026, SIF-001, SIF-034, SIF-002, SIF-003, SIF-004, SIF-005, SIF-051, SIF-052, SIF-053, SIF-054, SIF-056, SIF-058, SIF-059, SIF-060, SIF-061, SIF-062, SIF-063, SIF-076, SIF-077, SIF-064, SIF-065, SIF-067, SIF-068, SIF-069, SIF-070, SIF-071, SIF-074, SIF-073, SIF-072, SIF-075, SIF-019, SIF-013, SIF-014, SIF-017, SIF-018, SIF-081, SIF-039, SIF-040, SIF-042, SIF-045, SIF-046, SIF-048, SIF-011, SIF-021, SIF-024, SIF-022, SIF-023, SIF-020, SIF-025, SIF-027, SIF-028, SIF-029, SIF-015, SIF-012, SIF-031, SIF-032, SIF-033, SIF-035, SIF-036, SIF-037, SIF-038, SIF-041, SIF-043, SIF-044, SIF-047, SIF-080, SIF-079, SIF-082
ABB AC800M High Integrity SIL3 (used 9 times)	SIF-006, SIF-007, SIF-008, SIF-009, SIF-049, SIF-050, SIF-055, SIF-057, SIF-066

46.3 Final Element Groups Reused

Avstängning O2 (used 6 times)	SIF-026, SIF-037, SIF-043, SIF-044, SIF-079, SIF-082
Trip red.gaskompressor (used 6 times)	SIF-050, SIF-025, SIF-043, SIF-044, SIF-047, SIF-038
Trip proc.gaskompressor (used 5 times)	SIF-055, SIF-042, SIF-045, SIF-046, SIF-048
Trip red.gaskompressor (used 2 times)	SIF-049, SIF-041
Avstängningsventil 2 överhettad ånga till PGH (used 8 times)	SIF-076, SIF-016, SIF-017, SIF-018, SIF-028, SIF-041, SIF-047, SIF-080
H2 till brännare PGH (used 2 times)	SIF-019, SIF-079
NG till brännare PGH (used 2 times)	SIF-019, SIF-082
Close NG till kylkrets	SIF-042, SIF-045, SIF-046, SIF-048

(used 4 times)

Avstängning O2 PT12

(used 18 times)

NG till brännare PGH PT12

(used 5 times)

Avstängningsventil 2 överhettad ånga
till PGH PT 6

(used 5 times)

SIF-021, SIF-022, SIF-024, SIF-023, SIF-020, SIF-025, SIF-027, SIF-029,
SIF-015, SIF-031, SIF-032, SIF-033, SIF-035, SIF-036, SIF-041, SIF-047,
SIF-081, SIF-038

SIF-025, SIF-027, SIF-029, SIF-031, SIF-032

SIF-012, SIF-043, SIF-044, SIF-079, SIF-082

47 Abbreviations and Definitions

47.1 Abbreviations

HFT	Hardware Fault Tolerance
IPL	Independent Protection Layers
MoonN	"M" out of "N"
MTTFS	Mean Time To Fail Spurious
MRT	Mean Repair Time
PFD	Probability of Failure on Demand
PFDavg	Average Probability of Failure on Demand
PFH	Probability of a Dangerous Failure per Hour
PIU	Proven In Use
PTC	Proof Test Coverage
PTI	Proof Test Interval
RRF	Risk Reduction Factor
SERH	Safety Equipment Reliability Handbook
SFF	Safe Failure Fraction
SIF	Safety Instrumented Function
SIL	Safety Integrity Level
SIS	Safety Instrumented System
SSI	Site Safety Index
SRS	Safety Requirements Specification
β -factor	Beta factor, indicating common cause susceptibility
SD	Safe Detected
SU	Safe Undetected
DD	Dangerous Detected
DU	Dangerous Undetected
AD	Annunciation Detected
AU	Annunciation Undetected
NE	No Effect

47.2 Definitions

Architectural Constraints	Indication of maximum SIL achievable of a SIF's subsystems architecture based on minimum hardware fault tolerance requirements. This is determined solely on the basis of Type A or Type B device selection, redundancy (hardware fault tolerance), and the safe failure fraction (calculated or conservatively assumed if no data is provided). It does not pertain to Systematic Capability or certification levels. Architectural Constraint limits are determined based on IEC 61508 or IEC 61511 tables.
Mission Time	This is the time period that the SIF is expected to be operational. Alternatively this is the time period between points in time where the SIF can be considered as new. Note that this is not the proof test interval because of imperfect proof tests considerations. The mission time could, for example, correspond to the major turnaround period of a unit.
Startup Time	This represents the time it will take to perform maintenance and repair on a unit after a spurious trip plus the time it will take to restart the unit. This is typically longer than just the average repair time.

Systematic Capability

Indication of the maximum SIL achievable based on the SIF's equipment items systematic integrity alone. This is determined on the basis of the IEC 61508 certification level of each equipment item and/or the systematic integrity claim made by the user for a specific equipment item as part of its Proven In use justification. This addresses the requirement in IEC 61511 that all equipment used in a SIF must either be assessed per IEC 61508 or proven in use.

48 Disclaimer, Assumptions, Equipment Data

48.1 Disclaimer

The user of the exSILentia® software is responsible for verification of all results obtained and their applicability to any particular situation. Calculations are performed per guidelines in applicable international standards. *exida.com L.L.C.* accepts no responsibility for the correctness of the regulations or standards on which the tool is based. In particular, *exida.com L.L.C.* accepts no liability for decisions based on the results of this software. The *exida.com L.L.C.* guarantee is restricted to the correction of errors or deficiencies within a reasonable period when such errors or deficiencies are brought to our attention in writing. *exida.com L.L.C.* accepts no responsibility for adjustments made to this automatically generated report made by the user.

48.2 Assumptions SILver™

SIL Capability

If the user disables the **Consider IEC SIL Capability** option, it is assumed that the justification of equipment used is performed outside of the SILver tool as in that case the results of the SIL verification only address the quantitative requirements of IEC 61511.

A list of all other assumptions on which the SILver™ tool is based can be found in the exSILentia® user guide.

48.3 Equipment Data

exida has compiled a proprietary equipment failure database. This database is a compilation of failure data collected from a variety of public and confidential sources and presents an industry average. The database is published as the "Safety Equipment Reliability Handbook" ISBN-13: 978-1-934977-15-6. The reliability data collection process as described in this book applies to the SILver equipment data collection process.

The user is responsible for determining the applicability of the failure data to any particular environment. The stress levels assumed to determine the equipment failure rate are average worst case for an industrial environment. Descriptions of the environmental profiles used are included in the Safety Equipment Reliability Handbook. Accurate plant specific data is preferable to general industry average data. Industrial plant sites with high levels of stress must use failure rate data that is adjusted to a higher value to account for the specific conditions of the plant.